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**Optimal Power Sharing in DC Grid-
Connected Microgrids for Rural
Electrification**

FINAL REPORT

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ELECTRICAL
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ENGINEERING
DEPARTMENT

Statement of Academic Honesty

Appendix C: Statement of Academic Honesty

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FACULTY OF ENGINEERING
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Abstract

DC microgrids have gained increasing attention as a viable solution for rural electrification due to their compatibility with renewable energy sources and improved efficiency compared to traditional AC systems. The integration of distributed generation and storage enables localized energy management and enhances reliability in regions with limited grid access. This project aims to develop and evaluate an OPF framework for analysing power sharing in grid-connected DC microgrids. The study investigates how variations in PV generation, battery storage, and grid interaction affect system performance, efficiency, and power distribution. A MATLAB-based optimization model was implemented using a branch flow formulation and solved using nonlinear programming techniques. The framework incorporates PV generation, battery energy storage systems, converter loss modelling, and bidirectional grid interaction, and is applied to three microgrid configurations: a 7-bus radial system, a modified IEEE 14-bus system, and a clustered 20-bus network over a 24-hour simulation period. The results show consistent trends across all systems, where full PV generation leads to lower system losses, while reduced PV increases reliance on battery storage. In the absence of PV, grid support becomes necessary, resulting in the highest losses. Additionally, smaller networks exhibit higher converter activity due to limited power flow paths, while larger networks enable more localized power sharing. Collectively, the study demonstrates that coordinated optimization of renewable generation, storage, and grid interaction is critical for achieving efficient, reliable, and scalable DC microgrid operation for rural electrification.

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List of Abbreviations

AC	– Alternating Current
BESS	– Battery Energy Storage System
BFM	– Branch Flow Model
CGCS	– Centralized Generation Centralized Storage
CGDS	– Centralized Generation Distributed Storage
DC	– Direct Current
DG	– Distributed Generation
DGDS	– Distributed Generation Distributed Storage
DGDSA	– Distributed Generation Distributed Storage Architecture
DGS	– Distributed Generation Systems
DHI	– Diffuse Horizontal Irradiance
DNI	– Direct Normal Irradiance
DPS	– Distributed Power Sharing
GHI	– Global Horizontal Irradiance
IEEE	– Institute of Electrical and Electronics Engineers
KCL	– Kirchhoff’s Current Law
LCOE	– Levelized Cost of Electricity
LP	– Linear Programming
MATLAB	– Matrix Laboratory
MILP	– Mixed-Integer Linear Programming
MPPT	– Maximum Power Point Tracking
NREL	– National Renewable Energy Laboratory
NSRDB	– National Solar Radiation Database
OPF	– Optimal Power Flow
P2P	– Peer-to-Peer
PCC	– Point of Common Coupling
PV	– Photovoltaic
QP	– Quadratic Programming
RES	– Renewable Energy Sources
SOCP	– Second-Order Cone Programming

1 Introduction

Access to reliable and affordable electricity is one of the most pressing challenges for rural communities worldwide. Despite global progress in expanding energy access, more than 770 million people, most of whom live in Sub-Saharan Africa and South Asia, still remain without electricity, suffering from energy poverty (Rabia Khan et al.). This severe energy deficit significantly hinders economic development, education outcomes, and public health, with many relying on inefficient and polluting fossil fuels like kerosene and wood for basic needs (Parimalram Achintya Madduri December 2016). In response to the United Nations Sustainable Development Goals (SDG 7), which aims to ensure universal access to modern electric services by 2030, a strong research interest has emerged in designing microgrids specifically adapted for these underserved rural communities (Lucas Richard et al., 2022). In response to these challenges, renewable energy technologies, particularly direct current (DC) microgrids, have emerged as promising solutions for rural electrification. These systems harness locally available resources such as solar photovoltaic (PV), wind, or biomass, and distribute electricity through modular and scalable networks that are inherently compatible with modern DC appliances.

While DC microgrids offer clear advantages in terms of efficiency and compatibility with renewable energy sources, many real-world energy systems operate within an existing alternating current (AC) infrastructure. As a result, there has been increasing interest in hybrid AC/DC microgrids, which integrate both AC and DC subsystems through power electronic converters (Chaudhari 2023). These hybrid configurations enable flexible operation, allowing microgrids to function either in islanded mode or connected to the

utility grid. Grid-connected operation is particularly important in rural electrification contexts, as it enhances system reliability by providing backup support during periods of low renewable generation, while also enabling excess locally generated power to be exported back to the grid (Dewan Mahnaaz Mahmud 2022). Work done by Mohamed El-Hendawi et al. and Nandini K. Krishnamurthy et al. has highlighted that integrating DC microgrids with the utility grid can improve overall energy efficiency, reduce reliance on centralized generation, and support more stable and resilient power systems (Mohamed El-Hendawi 2018) (Nandini K. Krishnamurthy 2025). Additionally, limited attention has been given to how grid connection influences peer-to-peer energy sharing strategies within distributed nanogrid architectures. To address these gaps, this study extends existing DC microgrid energy sharing models by incorporating grid-connected operation through a bidirectional AC/DC interface. This allows the microgrid to dynamically transition between local energy exchange and grid support, providing a more realistic representation of practical deployment scenarios. By integrating grid interaction directly into the optimal power flow formulation, the proposed approach enables a more comprehensive analysis of system performance, including the trade-offs between local energy sharing, grid dependency, and converter-related losses.

The justification for this project stems from both the technical and socioeconomic benefits of grid-connected DC microgrids. From a technical perspective, DC architectures offer several advantages over conventional AC systems, including fewer conversion stages, reduced distribution losses, and greater compatibility with renewable sources and storage devices (Babangida Modu et al., 2023). Studies consistently show that distributed generation and storage systems are more efficient, scalable, and cost-effective than centralized approaches, particularly in remote areas where grid extension is impractical

(Mashood Nasir et al., 2020). However, incorporating a grid connection introduces additional operational flexibility, allowing the microgrid to draw power during generation shortages and export excess renewable energy when available. This enhances system reliability and reduces dependence on local generation alone. Economically, DC microgrids have demonstrated lower levelized costs of electricity (LCOE), making them affordable for low-income households. The ability to interact with the grid further improves economic performance by optimizing when energy is imported or exported. These systems also support the transition away from costly fossil fuels while aligning with the United Nations Sustainable Development Goals, which call for universal access to affordable and clean energy by 2030. Importantly, the novelty of this project lies in its integration of grid-connected operation within an advanced optimization framework. The proposed model incorporates both local peer-to-peer energy sharing and grid interaction, while accounting for distribution and converter losses within the optimal power flow formulation. This approach not only ensures cost-effectiveness and reliability but also provides new insights into the coordinated optimization of energy exchange between interconnected nanogrids and the utility grid.

The objectives of this project are multi-fold. First, the project involves reviewing various DC microgrid topologies and features used in rural electrification scenarios, with particular attention to aspects relevant to the optimal power-sharing problem. Second, existing optimization techniques applied to power dispatch in microgrids will be studied, and an appropriate optimization method will be selected and justified for implementation. Third, the objective function and associated constraints for the optimization problem will be formulated to represent the operational characteristics of the microgrid. Fourth, a modular and well-documented set of functions will be developed to read a system

description file, execute the optimization routine, and generate the corresponding results. As part of this stage, a testing and validation document will also be produced. Finally, the developed framework will be evaluated using at least three microgrid test cases of different sizes, and the results and limitations of the approach will be analyzed and discussed.

The scope of the project is focused on community-level DC microgrids, both in islanded and grid-connected modes, with a particular emphasis on renewable integration and distributed energy resources, and their interaction with the utility grid. Solar PV will be central to the analysis, given its abundance in rural areas of Asia and Africa (Rabia Khan et al.), combined with the use of household battery energy storage systems (BESS). The study specifically examines system-level aspects such as network topology design, energy management strategies, and optimal power dispatch, with particular attention to how grid-connected operation influences power sharing, reliability, and overall system performance.

The report is organized into six main sections. The Introduction sets the context of the study by outlining the objectives, scope, and significance of the work. The Literature Review critically examines existing research on rural electrification, DC and hybrid AC/DC microgrid architectures, optimization techniques, and peer-to-peer energy sharing, highlighting key gaps related to grid integration and coordinated energy management. The Method section details the modelling approach, optimization framework, and tools used to simulate a grid-connected DC microgrid and evaluate its performance under different operating conditions. The implementation and code structure section then describes how the proposed methodology was translated into a modular MATLAB framework. It outlines the overall program architecture, the main functional components, the flow of data

between modules, and the coding conventions used to ensure consistency, scalability, and maintainability across the different microgrid test cases. The Results section presents the technical outcomes of the simulations, including power-sharing efficiency, system losses, and grid interaction behaviour. These findings are then discussed in relation to their implications for rural electrification and practical microgrid deployment. Finally, the Conclusion summarizes the key contributions of the study and suggests directions for future research and implementation.

2 Literature Review

Rural electrification has been a focal point of global development efforts due to the persistent challenge of energy poverty. Studies consistently highlight that the majority of those without electricity live in rural and remote regions, where grid extension is either economically prohibitive or technically impractical (N. N. S. Rabia Khan, 2025). In response, researchers have extensively explored decentralized energy solutions, particularly renewable energy-based microgrids. This section reviews the established knowledge and recent advances in microgrid design, optimization, control strategies, and peer-to-peer power sharing, with specific emphasis on islanded and grid-connected DC microgrids for rural electrification.

2.1 Emergence of DC Microgrids

The emergence of DC microgrids is directly linked to the global challenge of achieving universal energy access, particularly in remote regions, together with rapid advancements in power technologies. Access to electricity remains uneven, with rural and remote communities disproportionately affected. Extending national grids to sparsely populated areas is frequently infeasible because of high up-front infrastructure costs and long distances (N. N. S. Rabia Khan, 2020). As a result, millions still lack reliable power, and many households rely on kerosene or biomass for basic energy services, with associated health and environmental harms (S. I. Mashood Nasir et al., 2020). Given these limitations, microgrids have emerged as a reliable and affordable solution by locating generation close to loads (Distributed Generation, DG) and leveraging Renewable Energy Sources (RES).

In many rural areas of South-East Asia and Sub-Saharan Africa, solar energy is highly abundant, often providing $5.5kWh/m^2/day$ and above of solar irradiance. This makes solar PV systems particularly promising, especially as the cost of solar panels has been rapidly decreasing (M. N. Rabia Khan et al., 2023). Microgrids incorporate various Distributed Energy Resources (DERs), and incorporating RES has enhanced their sustainability and cost-effectiveness. This renewed interest is driven by major technological advancements over the past decade, including:

- **Improvements in Power Electronics:** Developments in power electronic converters (high-efficiency DC-DC converters) have made DC systems feasible, cost-effective, and capable of meeting required specifications.
- **PV Systems and Storage:** DC microgrids are particularly compatible with major renewable sources like PV panels and batteries, which are inherently DC in nature.
- **Availability of DC Loads:** There is an extensive market availability of DC loads such as LED lights, cell phones, chargers, and computers, which are increasingly common household appliances in both urban and rural settings.

Hence, DC microgrids have emerged as a prominent solution for rural electrification due to several intrinsic advantages over AC systems:

1. **Compatibility with Renewables and Storage:** Direct integration with PV and batteries minimizes redundant conversion stages.
2. **Higher Efficiency:** DC networks avoid synchronization and many harmonics, issues inherent in AC systems, maximizing useful energy and reducing distribution losses (Rabia Khan, 2024).
3. **Cost-Effectiveness and Simplified Structure:** DC configurations can reduce complexity and cost, relative to AC systems.

4. **Distributed Generation (DGDS) Capabilities:** Distributed DC architectures enable households (prosumers) to manage their own generation and storage (DGDSA: Distributed Generation and Distributed Storage Architecture), which facilitates resource sharing and usage diversity (Rabia Khan, 2024).

Taken together, these drivers, falling PV and storage costs, appliance efficiency gains, and maturing power-electronic interfaces explain the emergence of DC microgrids as a pragmatic pathway to accelerate rural electrification while improving the technical efficiency and affordability of basic electricity services.

2.2 Grid-Connected DC Microgrids

The development of grid-connected DC microgrids builds upon the advantages of standalone DC systems by introducing the ability to interact with the existing utility network. While islanded DC microgrids are effective for localized electrification, their operation is limited by the availability of local generation and storage resources. By contrast, grid-connected configurations provide an additional layer of flexibility, allowing power to be exchanged with the upstream grid to balance fluctuations in renewable generation and load demand. In this configuration, the microgrid is no longer an isolated system but operates as part of a larger energy network. When local generation is insufficient, power can be imported from the utility grid to maintain supply continuity. Conversely, when renewable generation exceeds local demand, excess energy can be exported back to the grid (Dewan Mahnaaz Mahmud 2022). This bidirectional interaction enhances system reliability, improves utilization of renewable energy, and reduces the need for oversized storage systems. As highlighted in (Chaudhari 2023), this capability is

one of the key factors driving the adoption of grid-connected DC microgrids. The increasing integration of renewable energy sources, particularly solar PV systems and battery energy storage, has further accelerated the transition toward grid-connected microgrids.

From an operational perspective, the inclusion of grid interaction transforms the power balance of the microgrid. Instead of relying solely on local resources, the system can incorporate grid power as an additional decision variable within the energy management framework (Yazdani 2009). This formulation highlights that, in grid-connected operation, the utility grid becomes an active participant in system operation rather than a passive backup source.

The ability to interact with the grid also enables more advanced energy management strategies. Rather than simply balancing supply and demand, the system can optimize when to import or export power based on system conditions, renewable availability, and operational objectives. This has been widely explored in (Mohamed El-Hendawi 2018), where grid-connected microgrids are shown to improve economic performance, enhance operational flexibility, and support more efficient utilization of distributed energy resources. In addition to improving energy management, grid-connected operation expands the functional role of battery energy storage systems. Storage units can be used not only for local energy balancing but also for supporting grid services, smoothing renewable fluctuations, and enabling transitions between islanded and grid-connected modes. This flexibility is particularly important in real-world deployments, where microgrids must maintain stable operation under varying system conditions. Despite these advantages, the integration of grid connection also introduces additional complexity (ASADULLAH KHALID 2021). The microgrid must now satisfy interconnection requirements, maintain synchronization with the utility grid, and ensure stable operation during mode

transitions. Furthermore, the presence of power electronic interfaces introduces additional losses and control challenges that must be considered in system design and optimization.

Overall, grid-connected DC microgrids represent a significant evolution from isolated systems, combining the efficiency of DC distribution with the resilience and flexibility of grid support. They provide a scalable and practical solution for modern energy systems, particularly in rural and semi-urban environments where both reliability and adaptability are critical.

2.2.1 AC/DC Interface

The AC/DC interface forms the electrical and control boundary between the DC microgrid and the upstream AC utility system. In grid-connected configurations, this interface is typically implemented using a bidirectional power electronic converter located at the point of common coupling (PCC) (ASADULLAH KHALID 2021). Its primary function is to enable controlled two-way power transfer between the DC microgrid and the utility grid, while ensuring stable operation on both sides of the system.

In practical operation, the AC/DC interface performs multiple roles simultaneously. It allows the microgrid to import power when local renewable generation and battery resources are insufficient, and to export excess energy when generation exceeds demand. In addition, it contributes to system stability by linking the controllable DC-side resources to the AC network, thereby supporting overall power balance and voltage regulation. As discussed in (Necmi Altin 2026), the grid-side converter typically operates in rectifier mode during power import and inverter mode during export, with filtering components such as

LC filters used to reduce switching harmonics and improve power quality. From a modelling and control perspective, the AC/DC interface is not simply a static power transfer element but a dynamic system that must be actively controlled. Converter behaviour is commonly represented using hierarchical control structures consisting of inner current control loops and outer voltage or power control loops (Necmi Altin 2026). The outer loop generates reference signals based on system objectives, while the inner loop ensures fast and accurate current tracking. This separation allows the converter to respond effectively to both steady-state and transient conditions.

Beyond local control, the AC/DC interface also plays a key role in system-level energy management. In hierarchical control frameworks, the converter operates within multiple control layers. The energy management system performs high-level scheduling of power flows, the supervisory layer corrects deviations between scheduled and actual operation, and the local control layer governs converter switching and current regulation. Through this structure, the interface becomes the mechanism through which optimal operating decisions, such as when to import or export power, are physically implemented. This relationship between converter control and system-level optimization is particularly important for grid-connected microgrid studies. Because the AC/DC interface governs power exchange with the utility grid, it enables grid import and export to be treated as controllable variables within optimization problems such as optimal power flow. This allows the system to determine the most efficient combination of local generation, storage, and grid interaction, rather than relying solely on internal resources.

Another critical function of the AC/DC interface is to support transitions between grid-connected and islanded modes of operation. During disconnection from the utility grid, the converter must ensure a smooth transition to islanded operation without disrupting

supply to local loads. Similarly, reconnection requires synchronization of voltage, frequency, and phase angle at the PCC. These transition capabilities are essential for maintaining system reliability and are widely discussed in studies on grid-connected battery energy storage and microgrid control (Chaudhari 2023). Despite its benefits, the AC/DC interface also introduces additional design considerations. Power electronic conversion results in efficiency losses, and the inclusion of filters, controllers, and communication systems adds to system complexity (Necmi Altin 2026). While some optimization-based studies represent the grid interface using simplified power exchange variables, more detailed models incorporate converter losses and operational constraints to better capture real-world performance. The level of modelling detail, therefore, depends on the objectives of the study, with control-focused work emphasizing dynamic behaviour and optimization studies focusing on system-level energy flows.

Needless to say, the AC/DC interface is a fundamental component of any grid-connected DC microgrid. It enables bidirectional power exchange, maintains stability across AC and DC domains, supports coordinated energy management, and facilitates seamless transitions between operating modes. As such, it is not merely an auxiliary component but a central element that determines how effectively the microgrid interacts with the utility grid and achieves its operational objectives.

2.3 Microgrid Architectures & Topology Design

The structural design of a DC microgrid, encompassing both its architecture (where components are physically placed) and its topology (how nodes are interconnected), is

crucial as it impacts flexibility, cost, reliability, robustness, resiliency, and scalability (Babangida Modu et al., 2023).

DC microgrid architectures are generally classified based on the location of the generation resources, such as PV panels, and the energy storage, typically batteries (N. N. S. Rabia Khan, 2020). Four primary architectures exist for solar PV-based DC microgrids for rural electrification applications:

1. Centralized Generation Centralized Storage

In the centralized generation centralized storage (CGCS) architecture, all PV generation and battery storage resources are located at a single central point within the village. This configuration is structurally less complex and offers simplified system control, monitoring, and maintenance, as all major components are concentrated in one location. However, CGCS requires high-power-rated generation and storage equipment at the central node and relies heavily on the distribution network to supply all loads. As a result, the system typically experiences higher distribution losses, reduced reliability due to a single point of failure, and lower overall power availability compared to more distributed architectures.

2. Centralized Generation Distributed Storage

In the centralized generation distributed storage (CGDS) configuration, power generation is centralized, while battery storage is deployed at individual households. This architecture aims to reduce losses associated with energy storage by placing batteries closer to the loads, thereby decreasing the number of power conversion stages and reducing line losses between storage and consumption points. Additionally, CGDS introduces a level of scalability, as storage capacity can be incrementally added as demand grows. This hybrid approach has been proposed in prior rural electrification studies as a compromise between centralized simplicity and distributed efficiency.

3. Distributed Generation Centralized Storage

The distributed generation centralized storage (DGCS) architecture distributes PV generation across multiple households while maintaining a centralized energy storage system. Distributing generation closer to the loads helps reduce instantaneous distribution losses during periods of active generation. However, losses associated with stored energy are significantly increased, as energy must travel from distributed generators to the central storage during charging and then back through the network during discharging. This results in longer effective power flow paths and increased losses. Furthermore, DGCS introduces bidirectional power flows and requires more sophisticated control and communication schemes to manage resource balancing and network stability effectively.

4. Distributed Generation Distributed Storage

In the distributed generation distributed storage (DGDS) architecture, both PV generation and battery storage are installed at each household, effectively forming a network of interconnected nanogrids. Each household operates as a prosumer, capable of both producing and consuming energy. DGDS is widely regarded as the most efficient, reliable, and optimal configuration for rural microgrids due to its modularity, scalability, reduced distribution and conversion losses, and high operational efficiency. It also enables advanced functionalities such as peer-to-peer power sharing and community-level power pooling. The primary trade-off is increased structural complexity, as the high degree of distribution necessitates more advanced control strategies and greater communication requirements.

The choice of microgrid topology, however, plays a critical role in determining system performance, including reliability, flexibility, efficiency, fault tolerance, and scalability

(Babangida Modu et al., 2023). Several common DC microgrid topologies exist, each offering different trade-offs for rural electrification applications:

1. Radial Topology

In the radial DC microgrid topology, all generation sources, storage systems, and loads are connected to a single main DC bus, forming a simple feeder-like structure. This configuration is widely used due to its simplicity, ease of implementation, and low cost. It is particularly suitable for residential or small-scale systems that prefer low-voltage DC operation. However, the radial topology suffers from a major limitation in reliability, as it has a single point of failure. If a fault occurs along the main bus, it can disrupt power supply to downstream loads. Despite this, the structure allows for reduced distribution losses when loads and sources are located close to each other.

2. Multi-Bus Topology

The multi-bus topology is an extension of the radial configuration, where multiple DC buses are interconnected to improve system flexibility and reliability. This structure enables power exchange between different sections or clusters of the microgrid, allowing excess energy in one area to support deficits in another. Additionally, the presence of multiple buses introduces redundancy, allowing faulty sections to be isolated without affecting the entire system. This improves operational reliability and makes the topology suitable for larger and more complex systems. However, increased system complexity and coordination requirements are key challenges.

3. Multi-Terminal Topology

In the multi-terminal DC microgrid topology, multiple generation sources and grid interfaces are connected through several terminals, enabling multidirectional power flow. This topology is highly flexible and allows efficient power sharing among different system

components. While it enhances reliability and enables better power balancing, the control and protection of such systems become significantly more complex. Coordinating multiple converters and ensuring stable operation under varying conditions remains a major challenge.

4. Ring Topology

The ring topology connects all buses in a closed-loop structure, allowing power to flow in multiple directions. This provides an alternative path for power delivery in the event of a fault, significantly improving system reliability and resilience. Each node in the system can be connected or isolated using switches, enabling flexible operation and fault management. Due to its redundancy and fault tolerance, the ring topology is commonly used in urban and industrial applications. However, it requires more sophisticated protection and control mechanisms compared to simpler topologies.

5. Ladder Topology

The ladder topology builds upon the ring structure by introducing multiple interconnected loops, forming a ladder-like configuration. This topology offers very high redundancy and reliability, as power can be supplied through multiple alternative paths. It is capable of isolating faults effectively while maintaining system operation, making it suitable for critical applications. However, the increased structural complexity makes system design, control, and implementation more challenging.

6. Zonal Topology

In the zonal topology, the microgrid is divided into several zones, each with its own local sources, storage, and loads. Multiple DC buses supply each zone, allowing flexible and redundant power delivery. This configuration enhances system reliability and allows for localized fault isolation. Power can be supplied from different buses depending on system

conditions. However, the coordination between zones and increased system complexity requires advanced control strategies.

2.3.1 Distributed Power Sharing vs Centralized Generation Dispatch

Traditional Centralized Generation Dispatch (CGD) relies on a single system operator to schedule power output from a small number of large generation units to meet aggregate demand. Power flows unidirectionally from generation to loads, and the optimization problem primarily aims to minimize generation or fuel costs while satisfying overall demand and basic operational constraints. This approach is well-suited to conventional transmission systems but becomes less effective in microgrids with high penetration of distributed renewable energy resources.

In contrast, Distributed Power Sharing (DPS) treats the microgrid as a network of interconnected prosumers, where each node may both produce and consume energy. Power is exchanged peer-to-peer through the distribution network, and multiple distributed sources and storage units actively participate in meeting demand. Rather than dispatching a single centralized generator, the optimization determines who supplies whom, along which lines, and at what operating point, subject to network constraints. This fundamentally changes the nature of the optimization problem.

From an optimization perspective, centralized dispatch problems are relatively simple, with a small number of decision variables and constraints. Distributed power sharing significantly increases problem complexity by introducing nodal power balance constraints, branch power flow equations, voltage limits, current limits, battery state-of-charge dynamics, and converter efficiency models.

2.4 Mathematical Optimization Techniques & Energy Management

Mathematical optimization techniques are foundational to microgrid energy management, providing the framework to balance stochastic renewable energy with varying load demands while minimizing costs and losses (C. L. B. a. M. A. Chaudhari). These techniques formulate energy management as a constrained optimization problem, where an objective function is optimized while satisfying constraints. Depending on problem structure, this can involve Linear Programming (LP), Quadratic Programming (QP), or Mixed-Integer Linear Programming (MILP) when discrete decisions are present. For distributed microgrids, the underlying power-flow relationships are inherently nonlinear and nonconvex. To address this, OPF is used, with convex relaxation methods, most notably Second-Order Cone Programming (SOCP), to transform the problem into a form that guarantees convergence to a global optimum under practical conditions (M. N. Rabia Khan, Noel N. Schulz). Second-order cone programming is essentially a convex optimization technique that allows certain nonlinear relationships to be represented in a mathematically tractable form while preserving global optimality. It is particularly well-suited for power system applications because it can capture the coupling between power flow, voltage, and current through second-order cone constraints, without introducing non-convexities that make problems difficult to solve. By relaxing exact nonlinear equalities into convex cone inequalities, second-order cone programming enables efficient and reliable solution of optimal power flow problems using interior-point methods. In microgrid energy management, this approach provides a practical balance between modelling accuracy and computational efficiency, allowing complex network constraints to be handled while ensuring stable convergence to physically meaningful solutions.

3 Methodology and Approach

3.1 Research & Model Identification

The first stage of the methodology focused on identifying a suitable modelling and optimization framework for analysing energy sharing in a grid-connected DC microgrid. This stage involved reviewing existing research on DC microgrid energy management, optimal power flow (OPF), peer-to-peer (P2P) energy exchange in distributed generation systems, and grid integration. The general methodological approach adopted in this study was influenced by the work of Rabia Khan et al., who investigated optimal power sharing strategies for neighbourhood-level DC microgrids using physics-based power flow models and convex optimization techniques. Her research provides a structured framework for modelling distributed photovoltaic generation, battery energy storage systems, and peer-to-peer power exchange between interconnected nodes within a DC microgrid. This study, however, builds upon that framework by extending the model to include grid-connected operation through the implementation of utility grid extension. While original research primarily focused on islanded microgrid energy sharing between households, the present work incorporates an additional layer of operational flexibility by allowing the microgrid to interact with the external utility grid.

The microgrid architecture considered in this study follows a DGDS configuration. In this architecture, each household operates as an individual nanogrid equipped with its own PV generation unit and BESS. To represent power flow within the DC distribution network, the branch flow model (BFM) is adopted. The branch flow formulation captures the relationships between nodal power injections, branch currents, and voltage magnitudes within distribution networks. This model provides the physical representation of power

flow required for the optimization framework and forms the mathematical basis of the DC microgrid model described in the methodology. Using this network representation, an OPF problem is formulated to determine the optimal dispatch of generation, storage, and energy exchange across the microgrid. The optimization framework aims to minimize system losses while satisfying operational constraints such as power balance, voltage limits, converter efficiencies, and battery state-of-charge bounds. Through this methodology, the developed model provides a realistic representation of the operational behaviour of a distributed DC microgrid while enabling the analysis of optimal energy sharing strategies under both islanded and grid-connected operating conditions.

3.2 Build Cycle: System Design & Integration

3.2.1 Modelling of the DC Microgrid

3.2.1.1 *Nanogrid Design*

For this project, each N user is equipped with a rooftop solar PV source, a BESS, and a DC load. Within each household, the PV array is interfaced to the internal DC bus through a DC/DC converter operating under maximum power point tracking (MPPT). The battery is connected via a bidirectional DC/DC converter, enabling controlled charging/discharging to support local demand and power sharing. The DC load is supplied from the internal DC bus. For network-level energy exchange, each nanogrid is connected to an external DC bus through the inter-nanogrid lines, enabling peer-to-peer power transfer among households.

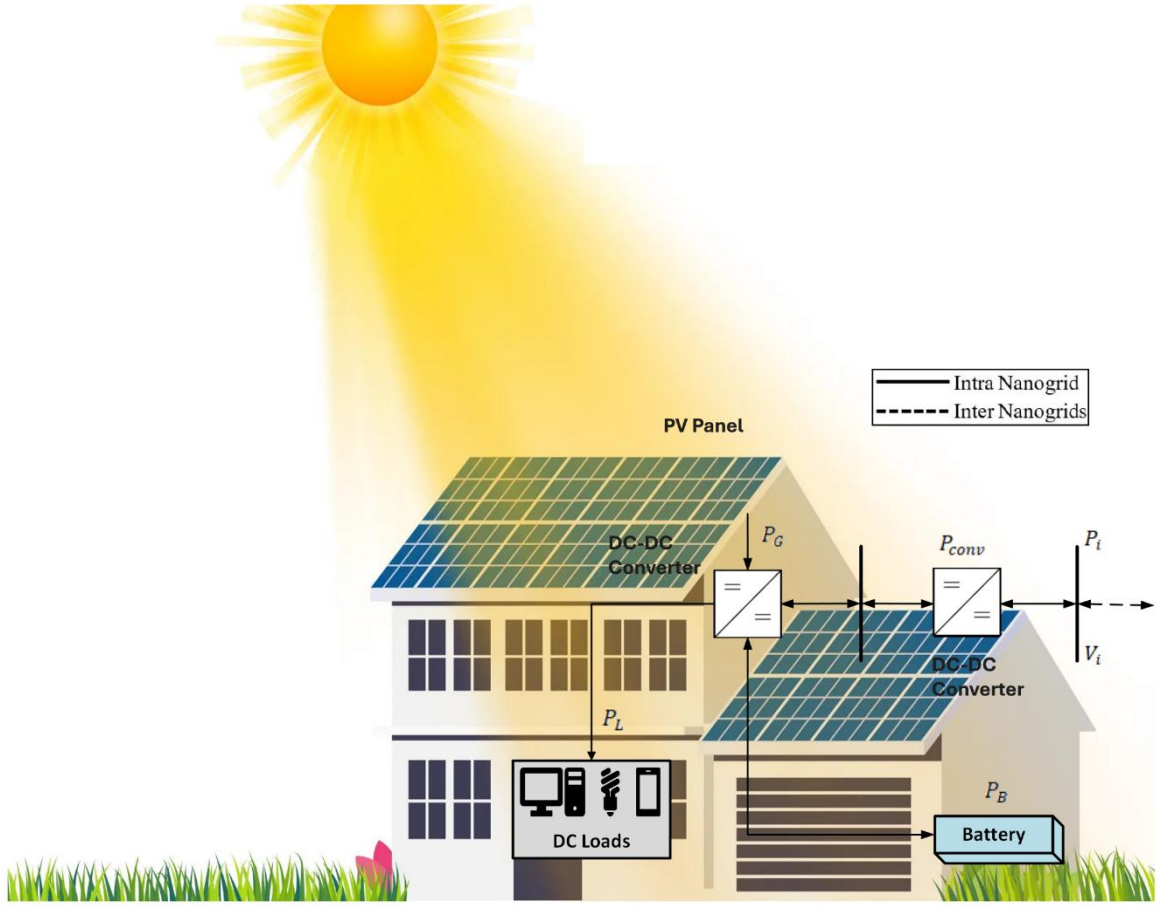


Figure 1: Nanogrid Model with components

The PV power is denoted by P_G , battery power by P_B and load demand by P_L . Each bus in the system is called a node, where two connected buses are represented by “ i ” and “ j ”, respectively. The voltage magnitude squared and power injection at nodes are V_i , V_j , P_i and P_j . The users are interconnected through resistance r_{ij} , while current and power flow from node “ i ” to node “ j ” are I_{ij} and P_{ij} , respectively.

3.2.1.2 Distribution Loss Model

In DC microgrids, power losses in the distribution network arise primarily from the resistive characteristics of the conductors connecting different nodes. These losses occur when electrical current flows through the distribution lines and are commonly referred

to as resistive or I^2R losses. The magnitude of these losses depends on the resistance of the line and the current flowing through it. For a branch connecting node i to node j , the line is characterized by a resistance r_{ij} and a current I_{ij} flowing through the conductor. While the current magnitude varies depending on operating conditions such as generation levels and load demand, the line resistance is assumed to remain constant because it is determined by the physical properties of the conductor and the fixed network topology. The distribution loss on a single line can therefore be expressed using the standard resistive loss relationship:

$$P_{loss,ij} = r_{ij}I_{ij}^2 \dots [1]$$

Where r_{ij} is the resistance of the line connecting nodes i and j , and I_{ij}^2 is the current flowing through that line.

To determine the total distribution loss within the microgrid, the losses across all interconnected lines in the network must be summed. For a system containing N nodes and a set of distribution branches E , the total distribution loss DL can be expressed as,

$$DL = \sum r_{ij}I_{ij}^2, \forall i \in N \dots [2]$$

This formulation captures the cumulative resistive losses across the entire DC distribution network. In the optimization model, the squared current term is commonly represented by the variable $l_{ij} = I_{ij}^2$, which simplifies the mathematical formulation. Using this representation, the total distribution loss can also be written as,

$$DL = \sum r_{ij}l_{ij}, \forall i \in N \dots [3]$$

Where l_{ij} represents the squared magnitude of the current flowing through the branch (i,j) .

3.2.1.3 Power Electronic Converter Loss Model (DC/DC Converter)

In a DC microgrid, power electronic converters play a critical role in regulating and controlling power flow between different components such as PV systems, batteries, and the distribution network. These converters enable functions such as voltage regulation, maximum power point tracking, and step-up or step-down power conversion. However, during this conversion process, a portion of the processed power is lost due to switching and conduction losses within the semiconductor devices and passive components of the converter. These losses are collectively referred to as power electronic converter losses. The magnitude of converter losses depends on the amount of power processed by the converter. Let P_o represent the output power processed by a converter. In general, converter losses can be represented using a polynomial approximation that relates the processed power to the associated losses. The general form of the converter loss model can be expressed as,

$$P_c = \sum_{x=0}^K k_x P_o^x \dots [4]$$

where P_c represents the converter power loss, k_x are empirical coefficients obtained from converter characteristics or manufacturer data, and K represents the order of the polynomial approximation. In practice, converter efficiency varies with its loading ratio, defined as the ratio between the output power P_o and the rated converter power P_R . This relationship can be expressed as,

$$x = \frac{P_o}{P_R} \dots [5]$$

where x represents the converter loading level. Since converter efficiency is a nonlinear function of this loading ratio, polynomial approximations are commonly used to model the efficiency characteristics.

For the purposes of this study, only the lower-order terms of the polynomial are considered, with $K = 2$. This means that converter losses are modeled as the sum of constant, linear, and quadratic components while higher-order terms are neglected. The resulting simplified converter loss model is given by:

$$P_c = \alpha + \beta P_o + \gamma P_o^2 \dots [6]$$

where the coefficients α , β , and γ represent the constant, linear, and quadratic components of the converter losses, respectively. The constant term α accounts for no-load losses that occur even when the converter processes very little power. The linear term βP_o represents conduction-related losses that increase proportionally with the processed power, while the quadratic term γP_o^2 captures additional nonlinear losses associated with switching behaviour and thermal effects.

3.2.1.4 Grid Converter Loss Model (AC/DC Converter)

In the context of the proposed DC microgrid system, the same modelling approach used for the DC/DC converter loss model is adopted for the grid-interface converter that connects the microgrid to the utility grid. Accordingly, the grid converter losses are represented using the same quadratic polynomial formulation relating the processed power to the converter losses. The grid converter loss model is therefore expressed as,

$$P_{loss_{gc}} = k_0 + k_1 P_{grid} + k_2 P_{grid}^2 \dots [7]$$

where $P_{loss_{gc}}$ represents the power loss of the grid converter, P_{grid} is the power exchanged between the microgrid and the utility grid, and k_0 , k_1 , and k_2 are empirical coefficients representing the constant, linear, and quadratic loss components of the converter. The constant coefficient, k_0 accounts for fixed converter losses that occur even when little power is processed, such as control circuitry and auxiliary electronics. The linear term, $k_1 P_{grid}$ represents conduction losses that increase approximately proportionally with the processed power, while the quadratic term, $k_2 P_{grid}^2$ captures nonlinear losses associated with switching behaviour and thermal effects within the semiconductor devices.

3.2.1.5 Implementing Grid Extension

To extend the microgrid model, the utility grid was incorporated to represent grid-connected operation scenarios where the local DC microgrid interacts with a larger power system. Rather than modelling the entire upstream transmission and distribution network, the utility grid was represented as an infinite bus equivalent connected to the microgrid through a PCC. In this configuration, the utility grid acts as both a slack source and energy sink. When local generation from renewable sources is insufficient to meet demand, power can be imported from the grid. Conversely, when the microgrid produces excess generation, such as during periods of high photovoltaic output, this surplus power can be exported back to the grid. In (Chaudhari 2023), the grid interface and power exchange are modelled in greater detail, incorporating a comprehensive representation of the grid-connected DC microgrid architecture, converter dynamics, and power management strategies. These models often include additional electrical parameters, detailed converter

operation, and extended network representations to accurately analyse system behaviour under a wide range of operating conditions. However, for this project, the grid extension was implemented using a simplified mathematical representation that focuses on the aspects most relevant to the optimal power flow formulation. Instead of modelling detailed converter switching behaviour or the full upstream network, the grid is treated as an infinite-bus system, and the AC/DC interface is represented by algebraic relationships within the OPF framework. This simplification significantly reduces computational complexity while still capturing the essential behaviour of grid power exchange and its associated losses. As a result, the model remains tractable for optimization while maintaining sufficient accuracy for analysing microgrid operation and energy management.

3.2.2 Optimal Power Flow

3.2.2.1 Branch Flow Model

The electrical behaviour of the DC microgrid network is represented using BFM. This model describes the relationships between voltage, current, and power flows within the network and provides a mathematical framework for analysing power distribution across interconnected nodes. In this formulation, the DC microgrid is represented as a connected graph $G(\mathcal{N}, \mathcal{E})$, where $\mathcal{N}$ denotes the set of buses (nodes) and $\mathcal{E}$ represents the set of distribution lines (edges) that interconnect the nodes.

Let the set of nodes be defined as $\mathcal{N} := \{1, 2, \dots, n\}$, where n represents the total number of buses in the network. The set of edges $\mathcal{E}$ describes the electrical connections between nodes, where each edge $(i, j) \in \mathcal{E}$ corresponds to a branch linking node i to node j . In

this representation, nodes correspond to buses within the microgrid while edges represent the distribution lines or branches through which power flows.

In DC distribution systems, each branch (i, j) is characterized by a purely resistive impedance $z_{ij} = r_{ij}$ since the reactive component is negligible. This implies that the reactance x_{ij} is assumed to be zero, resulting in a purely resistive line model. The corresponding branch admittance is therefore $g_{ij} = \frac{1}{r_{ij}}$, where r_{ij} is the resistance of the line connecting nodes i and j .

For each bus $i \in \mathcal{N}$, the system variables include the voltage magnitude V_i and the net real power injection P_i at that node. For each branch $(i, j) \in \mathcal{E}$, the key variables include the branch current I_{ij} flowing from node i to node j , and the corresponding power flow P_{ij} . Since the system operates in DC, all electrical quantities are real-valued and there are no reactive power components. Consequently, the apparent power reduces to real power only, and the branch flow model is governed by three fundamental physical relationships derived from basic circuit laws. The first relationship is Ohm's Law, which describes the current flowing through a branch as a function of the voltage difference between the two connected nodes:

$$I_{ij} = g_{ij}(V_i - V_j), \quad \forall (i, j) \in \mathcal{E} \dots [8]$$

This equation indicates that the current flowing through a branch is proportional to the voltage difference across the line and the branch conductance.

The second relationship is the current balance equation, derived from Kirchhoff's Current Law (KCL). It states that the net current injection at each node must equal the sum of currents flowing through all branches connected to that node:

$$I_i = \sum_{j:j \sim i} I_{ij}, \quad \forall i \in \mathcal{N} \dots [9]$$

This ensures that current entering and leaving each node is balanced, thereby maintaining conservation of charge within the network.

The third relationship defines the power injection at each bus, which relates the nodal voltage and current injection:

$$P_i = V_i I_i, \forall i \in \mathcal{N} \dots [10]$$

Together, these equations form the foundation of the branch flow model used to represent power flow in the DC microgrid. By linking nodal voltages, branch currents, and power injections, the model provides a consistent framework for analysing how electrical power is distributed throughout the network.

3.2.2.2 Optimization Framework

The optimization framework used in this work is based on the branch flow model (BFM) representation of the DC microgrid. The optimization problem is inherently non-linear and NP-hard, primarily due to the nonlinear relationships between voltage, current, and power within the network. Using the branch flow representation, the electrical behaviour of each distribution line $(i, j) \in \mathcal{E}$ can be described by the following set of equations:

$$P_{ij} + P_{ji} = r_{ij} l_{ij}, \forall (i, j) \in \mathcal{E} \dots [11]$$

$$v_i - v_j = r_{ij} (P_{ij} - P_{ji}), \forall i \rightarrow j \dots [12]$$

$$v_{ij} \cdot l_{ij} = P_{ij}^2, \forall i \sim j \dots [13]$$

Equation (11) represents the power balance within a branch. It states that the total power entering a branch from both ends is equal to the resistive power loss along that line. The variables P_{ij} and P_{ji} represent the power flow from node i to node j and from node j to node i , respectively, while r_{ij} denotes the resistance of the branch. The term

l_{ij} represents the squared magnitude of the current flowing through the line. Equation (12) describes the voltage drop across the distribution line. The voltage difference between nodes i and j is proportional to the net power flow through the branch and the line resistance. This relationship captures the resistive behaviour of DC distribution lines and links the nodal voltages with the branch power flows. Equation (13) establishes the coupling relationship between voltage, current, and power flow. It relates the squared current magnitude l_{ij} , the squared bus voltage magnitude v_i , and the branch power flow P_{ij} . This equation introduces a quadratic constraint into the formulation, making the optimal power flow problem non-convex. Because of this non-convexity, solving the optimization problem directly can be computationally challenging. Therefore, the non-convex formulation is relaxed into a convex form to enable efficient computation of the optimal solution while preserving the physical characteristics of the system.

3.2.2.3 Optimization Problem Formulation

In this work, the objective function is extended beyond conventional distribution loss minimization to include losses associated with power electronic converters and the grid interface. This allows the optimization framework to capture both resistive distribution losses and conversion losses arising from DC-DC converters and the PCC grid converter. The resulting objective function formulation is given as follows:

OPF-2; Minimize:

$$\sum_{j:j \sim i} l^{ij} r^{ij} + \sum_{i \in \mathcal{N}} P_c + \sum_{i \in \mathcal{N}} P_{loss_{gc}} \dots [14]$$

Subject to:

$$P_{G_i}(t) - P_{L_i}(t) - P_{B_i}(t) - P_{conv_i}(t) = \sum_{j:i \rightarrow j} P_{ij}(t), \forall i \in \mathcal{N} \dots [15]$$

$$P_i(t) = \sum_{j:i \rightarrow j} P_{ij}(t), \forall i \in \mathcal{N} \dots [16]$$

$$P_{ij}(t) + P_{ji}(t) = r_{ij} l_{ij}(t), \forall (i, j) \in \mathcal{E} \dots [17]$$

$$v_i(t) - v_j(t) = r_{ij} (P_{ij}(t) - P_{ji}(t)), \forall i \rightarrow j \dots [18]$$

$$v_i(t) l_{ij}(t) = (P_{ij}(t))^2, \forall i \sim j \dots [19]$$

$$l_{ij}(t) \leq (I_{ij, rated})^2, \forall (i, j) \in \mathcal{E} \dots [20]$$

$$P_{B_i}(t) - P_{L_i}(t) \leq C_B(t) [SOC_i(t) - \underline{SOC_i}], \forall i \in \mathcal{N} \dots [21]$$

$$P_{L_i}(t) - P_{B_i}(t) \leq C_B(t) [\overline{SOC_i} - SOC_i(t)], \forall i \in \mathcal{N} \dots [22]$$

$$SOC_i(t+1) = SOC_i(t) + \frac{1}{C_B(t)} P_{B_i}(t), \forall i \in \mathcal{N} \dots [23]$$

State-of-charge limits:

$$\underline{SOC_i} \leq SOC_i(t) \leq \overline{SOC_i}, \forall i \in \mathcal{N} \dots [24]$$

Generation limits at each node:

$$\underline{P_{G_i}} \leq P_{G_i}(t) \leq \overline{P_{G_i}}, \forall i \in \mathcal{N} \dots [25]$$

Converter loss constraints:

$$P_{conv_i}(t) \geq \alpha + \beta P_{o_i}(t) + \gamma P_{o_i}(t)^2 \dots [26]$$

$$P_{o_i}(t) \geq |P_{inj_i}(t)| \dots [27]$$

where P_{o_i} is an auxiliary variable representing the absolute power processed by the converter, and (α, β, γ) are the quadratic loss coefficients.

The voltage magnitudes of the branch buses must be maintained within a narrow operating range to ensure system stability and reliable performance. Specifically, the voltage at each node should not deviate by more than $\pm 5\%$ from its nominal per-unit

value. For this project, the acceptable limits are set at 0.95 p.u. and 1.05 p.u., corresponding to the lower and upper bounds, respectively.

$$\underline{v}_i \leq v_i \leq \overline{v}_i, \forall i \in \mathcal{N} \dots [28]$$

Where P_{G_i} , P_{B_i} , $P_{L_i}(t)$, P_{conv_i} , and P_i denote the generated power, battery power, load demand, converter losses, and injected power at node i , respectively. The variables $\underline{P_{G_i}}$ and $\overline{P_{G_i}}$ represent the lower and upper limits of generated power. Similarly, $\underline{v}_i$ and $\overline{v}_i$ denote the lower and upper bounds of the squared voltage magnitude. The parameters $\underline{SOC_i}$ and $\overline{SOC_i}$ define the minimum and maximum allowable state-of-charge limits of the battery.

3.2.2.4 Convex Relaxation – SOCP Formulation

Equation [19] introduces non-linearity into the model due to its dependence on the variables P , l , and v , as well as the quadratic relationship among them. Consequently, the formulated optimization problem takes the form of a Quadratically Constrained Quadratic Program (QCQP). The objective function includes a quadratic term representing converter losses, P_c and $P_{loss_{gc}}$ and the equality constraint in equation [19] further contributes to the problem's nonlinearity and nonconvexity. To obtain a globally optimal solution, these non-linear equality constraints are relaxed into convex inequalities using SOCP. The relaxed form of the constraint is expressed as:

$$v_i(t)l_{ij}(t) \geq \left(P_{ij}(t)\right)^2 \forall i \in \mathcal{N} \dots [29]$$

The resulting relaxed optimization problem (OPF-2) seeks to minimize distribution and converter losses along with the PCC grid converter losses within a DC microgrid, as

formulated in equation (14). This optimization is solved subject to the constraints defined by equations [15] – [18] and [20] – [29].

3.3 Algorithm Implementation

3.3.1 System Workflow

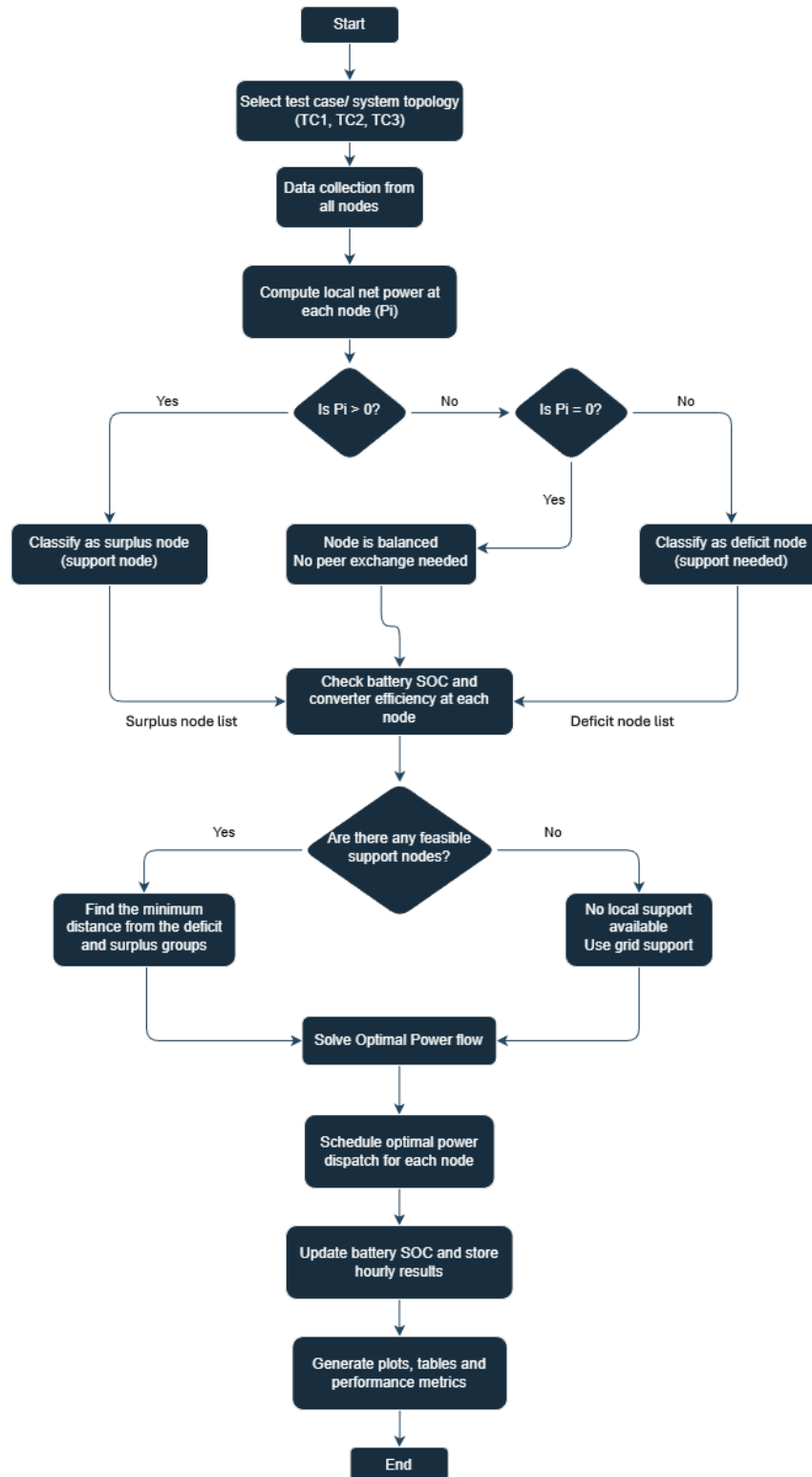


Figure 2: Flowchart of System Workflow

Figure 2 describes the procedure for simulating and optimizing power management in the DC microgrid system. It describes how data is processed, how nodes are classified based on their energy status, and how the OPF algorithm is used to determine efficient power dispatch across the network. The process begins with the selection of the test case or system topology. Three different configurations are considered in the study, labelled TC1, TC2, and TC3. Each test case represents a different microgrid structure and operating scenario, allowing the algorithm to be evaluated under varying network conditions. Once the system topology is selected, data is collected from all nodes within the microgrid. Each node represents a bus in the network and may correspond to a household (or community load) equipped with photovoltaic generation and battery storage. The collected data includes load demand, photovoltaic generation, and the state of charge of the local battery storage system. Using this information, the local net power at each node is computed. The net power, denoted as P_i , represents the difference between the locally generated power and the power demand at that node. This value determines whether the node has excess energy available for sharing or requires additional power to meet its load. A decision process is then used to classify each node according to its net power. If $P_i > 0$, the node has surplus energy and is classified as a support node capable of supplying power to other nodes in the network. If $P_i = 0$, the node is considered balanced, meaning that local generation exactly matches the load demand and no power exchange with neighbouring nodes is required. If $P_i < 0$, the node is classified as a deficit node, indicating that additional power is required to satisfy its load demand. After the nodes are classified, the algorithm evaluates the battery state of charge and converter efficiency at each node. This step ensures that nodes identified as surplus nodes have sufficient stored energy and operational capability to support neighbouring deficit nodes. Battery state of charge limits

and converter efficiency constraints are considered to determine whether a node is a feasible support source. The algorithm then checks whether there are any feasible support nodes available within the microgrid. If feasible support nodes exist, the system identifies the minimum electrical distance between deficit nodes and potential surplus nodes. This step prioritizes power exchanges between nodes that are electrically closer to each other, reducing distribution losses and improving overall system efficiency. If no feasible local support nodes are available, the system relies on the utility grid to provide the required power.

In this case, the microgrid imports power through the grid interface converter at the PCC. Once the available power sources have been identified, the optimal power flow problem is solved. The OPF algorithm determines the optimal power distribution across the network by minimizing system losses while satisfying operational constraints. Following the optimization step, the algorithm schedules the optimal power dispatch for each node. This determines how much power is generated, stored, exchanged between nodes, or imported/exported from the grid during the current time step. After the power dispatch is determined, the battery state of charge at each node is updated based on the charging or discharging behaviour resulting from the optimization. The results for each hour of operation are stored to enable time-series analysis of the system performance. Finally, the stored simulation results are used to generate plots, tables, and performance metrics that illustrate the behaviour of the microgrid under the selected test case. These outputs provide insight into system efficiency, power sharing behaviour, grid interaction, and overall operational performance. The workflow then concludes, completing the simulation cycle for the selected test case and operating conditions.

3.3.2 Data Acquisition and Test Design

The photovoltaic generation and load demand profiles used in the simulations were obtained from external NREL datasets. Solar irradiance data were taken from the National Solar Radiation Database (NSRDB), which provides serially complete solar and meteorological time-series data, including global horizontal irradiance (GHI), direct normal irradiance (DNI), and diffuse horizontal irradiance (DHI) (NSRDB, n.d.). The hourly load profile was obtained from the NREL Microgrid Load Profile Explorer dataset, which provides hourly electrical load profiles for rural Sub-Saharan African household and commercial microgrid applications (Microgrid Load and LCOE Modelling Results | NLR Data Catalog, n.d.). In the MATLAB workflow, a single day of PV and load data was extracted from the respective files, converted into 24-hour sequences, and normalized to create dimensionless PV and load profiles. These normalized profiles were then used to scale the rated PV generation and nodal load demand in the optimization model. Because both input files are external and user-editable, alternative days, demand scenarios, and solar conditions can be incorporated without modifying the optimization code.

3.3.2.1 Test Case 1: Radial 7-Bus System

The system is modelled as a small DC microgrid consisting of seven buses representing a local community network. One bus represents a central community load (Bus 1), while the remaining buses represent prosumer households equipped with photovoltaic generation and battery storage. The network follows a radial topology, where buses are connected in a tree-like structure without closed loops. Voltage magnitude limits are maintained between 0.95 p.u. and 1.05 p.u. to ensure stable system operation. Each prosumer

household is equipped with a battery storage capacity of 4 p.u.-h, with state-of-charge operating limits between 0.50 and 0.90. The distribution network is modelled using resistive branches with line resistances ranging from 0.04 p.u. to 0.06 p.u., representing typical low-voltage DC distribution lines.



Figure 3: Test Case 1: Radial 7-Bus System

3.3.2.2 Test Case 2: IEEE 14-Bus System

The system is modelled as a modified IEEE 14-bus network adapted to represent a DC microgrid environment. Due to the absence of a standardized benchmark system for islanded DC microgrids, the conventional IEEE 14-bus AC test system is adapted with several modifications to reflect DC operating conditions. The original system data is obtained from MATPOWER and adjusted by removing reactive power components and setting line reactance to zero, since DC systems do not involve reactive power flows. In this configuration, thirteen buses are treated as prosumer households equipped with

photovoltaic generation and battery energy storage, while one bus (Bus 14) represents a central community load node. To better represent DC distribution characteristics, the branch parameters are modified by reducing line resistance by approximately 10%. Additionally, zero-resistance branches representing transformers in the original AC system are replaced with small resistances (0.005 p.u.) to model DC/DC converter interconnections between buses. Voltage magnitude limits are maintained between 0.95 p.u. and 1.05 p.u. to ensure stable network operation. Each prosumer household is also equipped with a battery storage capacity of 4 p.u.-h, with state-of-charge operating limits between 0.50 and 0.90. This modified IEEE 14-bus model provides a more complex network structure than the smaller test system and enables the evaluation of optimal power sharing, voltage regulation, and distribution losses within a larger community-scale DC microgrid.

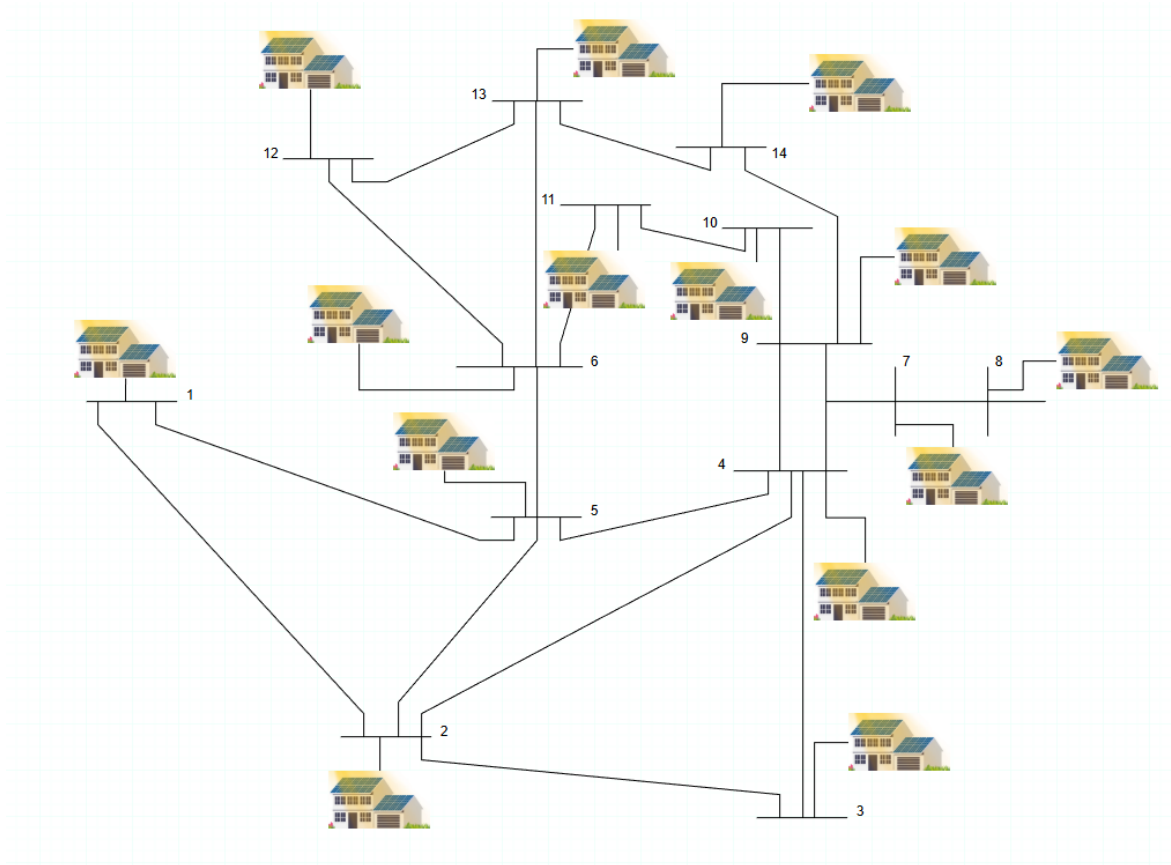


Figure 4: Test Case 2: IEEE 14-Bus System

3.3.2.3 Test Case 3: Clustered 20-Bus System

The system is modelled as a DC community microgrid consisting of twenty buses representing interconnected households. One bus represents a central community load (Bus 20), while the remaining buses represent prosumer households equipped with photovoltaic generation and battery storage. The network is organized into four clusters, each containing five households that are interconnected in a ring topology to allow multiple power-sharing paths between neighboring nodes. Within each cluster, the households can exchange energy locally through the ring connections, improving operational flexibility and resilience to single-line contingencies. The clusters are further interconnected through gateway nodes, forming a higher-level ring that enables power

sharing between clusters across the wider microgrid. Each prosumer household generates electricity through rooftop photovoltaic systems while simultaneously supplying local residential loads. The households are also equipped with battery energy storage systems with a capacity of 4 p.u.-h, allowing excess solar generation to be stored and later discharged when required. Battery state-of-charge limits are maintained between 0.50 and 0.90 to ensure realistic operating conditions. The distribution network is modelled using resistive branches representing low-voltage DC lines. Intra-cluster connections have line resistances of 0.04 p.u., while inter-cluster gateway connections have slightly higher resistances of 0.06 p.u. Voltage magnitude limits are maintained between 0.95 p.u. and 1.05 p.u. to ensure stable network operation. This twenty-bus configuration represents a larger clustered DC microgrid that enables the analysis of power sharing between households both within clusters and across clusters, allowing the evaluation of system performance, network losses, and energy exchange under DGDS architecture.

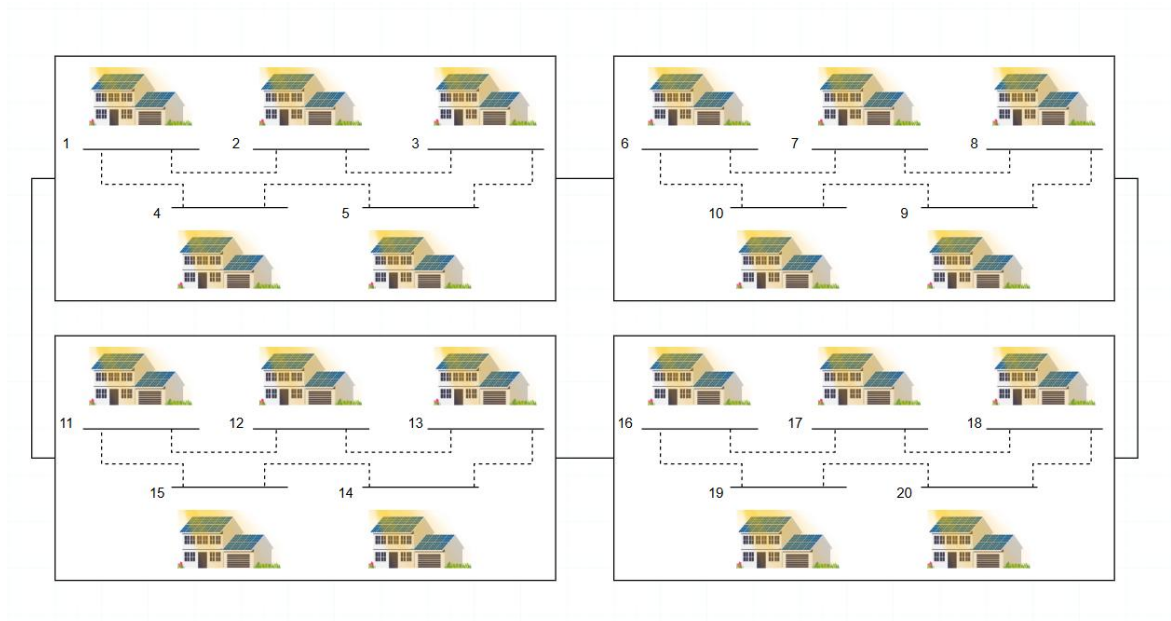


Figure 5: Test Case 3: Clustered 20-Bus System

4 Code Implementation and Structure

This section describes the implementation of the DC microgrid optimization framework and the structure of the code used to support its operation. The system is developed using a modular MATLAB-based approach, where each function performs a specific task within the overall workflow. The implementation is organized to ensure clear separation between system initialization, data handling, optimization, and result processing. This structured design enables efficient execution of the optimal power flow algorithm, supports scalability across different test cases, and ensures consistent data flow between components.

4.1 Program Architecture

The system operates through a structured workflow that integrates system initialization, input data processing, optimization, and result generation. The execution begins when a test case is selected using one of the main driver functions, namely *run_tc1.m*, *run_tc2.m*, or *run_tc3.m*. These functions act as the entry point of the system and are responsible for configuring the operating mode (islanded or grid-connected), initializing the microgrid model, loading input data, and coordinating the overall simulation process. Once a test case is selected, the corresponding system builder function (*build_tc1_7bus.m*, *build_tc2_14bus.m*, or *build_tc3_20bus.m*) is called to construct the system model. These functions define the static properties of the microgrid, including network topology, line parameters, generation capacities, load distribution, battery characteristics, voltage limits, converter loss coefficients, and grid interface settings. The resulting data is stored in a structured variable, *sys*, which serves as the central repository of system parameters. In

the case of Test Case 2, the function *case14_dc_modified.m* is used to adapt the standard IEEE 14-bus system into a DC-compatible format before being incorporated into the system structure. Following system initialization, time-varying input data is incorporated into the model. The function *read_pv_and_load_csv.m* is used to import PV generation and load demand profiles from external CSV and xlsx files. These profiles define the variation of generation and demand over the 24-hour simulation period. The profiles are stored in the system structure and are used to scale generation and load values at each time step.

The optimization process is carried out by the function *solve_opf_24h.m*, which performs a time-series simulation over a 24-hour horizon. This function initializes the results structure, *RES*, and iterates through each hour of the simulation. For each time step, a temporary data structure (*data_t*) is created to store the current load demand and maximum available PV generation. The function then calls *solve_opf_hour_fmincon.m* to solve the optimal power flow problem for that specific hour. Within the hourly solver, the optimization variables are first organized into a single decision vector using the function *opf_index.m*, which ensures consistent indexing of all variables. Lower and upper bounds are then defined for each variable based on system constraints, and an initial guess is generated to improve solver convergence. The nonlinear optimization problem is solved using MATLAB's *fmincon* solver, where the objective function is defined in *opf_objective.m* and the system constraints are defined in *opf_constraints.m*. The objective function minimizes distribution, DC/DC converter, and grid converter losses, while the constraints enforce power balance, voltage limits, battery operation limits, and network physics.

Once a solution is obtained, the function *opf_unpack.m* is used to convert the decision vector into meaningful physical quantities, such as bus voltages, generation outputs, battery power, state of charge, branch power flows, and system losses. These results are

stored in the *RES* structure, and the updated battery state of charge is carried forward to the next time step. This process is repeated for all 24 hours to capture the time-dependent behaviour of the system. After the completion of the optimization process, the system proceeds to the post-processing stage. The function *plot_testcase_figures.m* generates graphical outputs, including PV and load profiles, loss components, power injection schedules, and grid and battery interactions. In addition, the function *generate_tables.m* produces structured tables summarizing per-bus results such as power injection, state of charge, nodal voltage, and converter efficiency. These outputs are saved for further analysis and reporting.

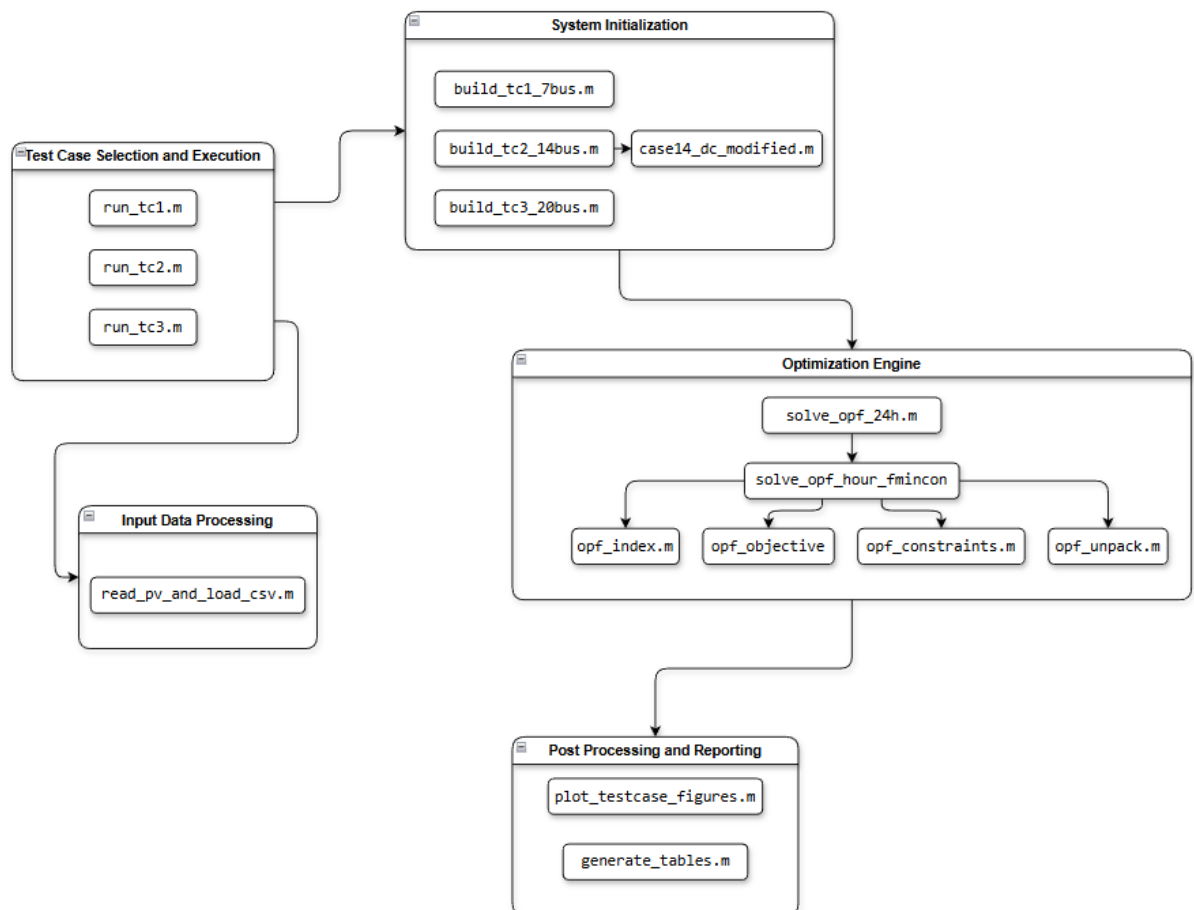


Figure 6: System Architecture Diagram

4.2 Key Functions

4.2.1 Test Case Initialization Functions

The test case initialization functions (*build_tc1_7bus*, *build_tc2_14bus*, and *build_tc3_20bus*) are responsible for constructing the microgrid system model used in the optimization process. These functions define the network topology, bus parameters, generation capacities, load distribution, battery characteristics, converter parameters, and grid interface settings. All system data is stored in a structured variable (*sys*), which serves as the central data container and is passed to all subsequent functions. This allows the system to support multiple microgrid configurations of varying size and complexity.

4.2.2 Main Execution Functions

The main execution functions (*run_tc1*, *run_tc2*, and *run_tc3*) act as the entry point of the system. These functions control the overall workflow by initializing the system, loading or generating time-varying input data, executing the optimization process, and generating output results. They coordinate the interaction between all components and ensure that the simulation is carried out consistently for each test case under both islanded and grid-connected modes.

4.2.3 Optimization Control Function

The optimization control functions manage the execution of the optimal power flow problem across the simulation horizon. The function *solve_opf_24h* performs time-series optimization by iterating over each hour of the simulation and updating the battery state

of charge between time steps. Within this process, the function *solve_opf_hour_fmincon* is used to solve the nonlinear optimization problem for a single time step. It defines the decision variables, variable bounds, and initial solution, and then formulates the objective and constraint functions.

The optimization problem is solved using MATLAB's *fmincon* solver with an interior-point algorithm. This solver handles the nonlinear objective and constraints to compute the optimal operating point while satisfying system limits such as power balance, voltage constraints, and battery operation. This approach enables efficient and reliable solution of the optimal power flow problem at each time step.

```
% --- fmincon
opts = optimoptions('fmincon', ...
    'Algorithm','interior-point', ...
    'Display','off', ...
    'MaxIterations', 4000, ...
    'MaxFunctionEvaluations', 40000, ...
    'ConstraintTolerance', 1e-6, ...
    'OptimalityTolerance', 1e-6);
```

Snippet 6: fmincon solver

4.2.4 Core Optimization Functions

The core optimization functions define the mathematical formulation of the optimal power flow problem. The function *opf_objective* specifies the objective function, which minimizes total system losses, including distribution losses, converter losses, and grid converter losses.

The function *opf_constraints* defines the system constraints, including power balance equations, voltage limits, battery operation constraints, branch flow relationships, and grid interaction limits. These constraints ensure that the optimization solution satisfies both physical laws and operational requirements.

4.2.5 Result Processing Functions

The result processing functions are responsible for converting and presenting the outputs of the optimization. The function *opf_unpack* extracts the solution vector into meaningful physical quantities such as voltages, power flows, state of charge, and system losses. Following this, *plot_testcase_figures* generates graphical outputs for analysis, while *generate_tables* produces structured tables summarizing key results for each bus. These functions enable effective interpretation and evaluation of system performance.

4.3 Data Flow and Interface

The primary data structures used throughout the system are *sys*, *data_t*, and *RES*, which serve as the main interfaces between components.

The *sys* structure contains all static system parameters, including network topology, generation limits, battery characteristics and grid settings. It is created during system initialization and is passed as an input to all major functions, including the optimization routines, objective function, constraint function, and post-processing modules. This ensures that all components operate using a consistent representation of the microgrid.

```
% Pack sys struct
sys = struct();
sys.N = N;
sys.E = E;
sys.from = from;
sys.to = to;
sys.r = r;
sys.Imax = Imax;

sys.v_min = v_min;
sys.v_max = v_max;

sys.PG_peak = PG_peak;
sys.PL_peak = PL_peak;
```



```

sys.has_pv = has_pv;
sys.has_bat = has_bat;

sys.SOC0 = SOC0;
sys.SOC_min = SOC_min;
sys.SOC_max = SOC_max;
sys.Cb = Cb;
sys.PB_min = PB_min;
sys.PB_max = PB_max;

sys.conv = conv;
sys.grid = grid;

sys.inc_from = inc_from;
sys.inc_to = inc_to;

```

Snippet 7: System Structure

The *data_t* structure is used to store time-dependent data for each simulation step. It is created within the 24-hour optimization loop and contains the current time index, the maximum available PV generation, and the load demand for that hour. This structure is passed to the hourly optimization solver, allowing the system to adapt to changing operating conditions over time.

```

for t = 1:T
    data_t = struct();
    data_t.t = t;
    data_t.PG_max = sys.PG_peak .* pv_prof(t);
    data_t.PL = sys.PL_peak .* load_prof(t);

    [sol, ok] = solve_opf_hour_fmincon(sys, data_t, SOC_prev);

```

Snippet 8: 24-hour optimization loop

The *RES* structure is used to store the outputs of the optimization process. It is initialized at the start of the simulation and is updated at each time step with the results obtained from the solver. This includes variables such as voltages, power flows, battery state of charge, and system losses. The *RES* structure is then used by the visualization and reporting functions to generate plots and tables.

```

RES = struct();
RES.mode = "OPF2";
RES.T = T;

RES.v = nan(T, N);

```

```

RES.PG      = nan(T,N);
RES.PB      = nan(T,N);
RES.Pinj    = nan(T,N);
RES.SOC     = nan(T+1,N);
RES.Pij     = nan(T,E);
RES.Pji     = nan(T,E);
RES.lij     = nan(T,E);
RES.Pconv   = nan(T,N);

if use_grid
    RES.PgImp      = nan(T,1);
    RES.PgExp      = nan(T,1);
    RES.Pgrid      = nan(T,1);
    RES.grid_conv_loss = nan(T,1);
end

RES.dist_loss = nan(T,1);
RES.conv_loss = nan(T,1);
RES.total_loss = nan(T,1);

```

Snippet 9: Result Structure

Data is exchanged between functions through clearly defined input and output arguments. For example, the function *solve_opf_24h* receives *sys*, *p_v_prof*, and *load_prof* as inputs and returns the results structure *RES*. Similarly, the function *solve_opf_hour_fmincon* receives *sys*, *data_t*, and the previous state of charge, and returns the solution for the current time step.

4.4 Coding Conventions

To maintain consistency with existing DC microgrid optimal power flow studies, all electrical quantities are expressed in per-unit (p.u.) values. Using per-unit normalization allows power, voltage, and loss variables to be scaled relative to selected base quantities, ensuring that all variables remain within a comparable numerical range. This improves the numerical stability and convergence performance of the optimization solver. Additionally, the use of per-unit values makes the model independent of specific hardware ratings or

system sizes, allowing the framework to represent different microgrid configurations without modifying the underlying formulation.

Time-varying data is organized using matrix representations. System variables are stored as matrices of size $T \times N$, where T represents the number of time steps and N represents the number of buses. Similarly, branch-related variables are stored as $T \times E$ matrices, where E represents the number of branches. This consistent structure simplifies indexing, supports time-series analysis, and improves computational efficiency.

```
RES.v      = nan(T,N) ;  
RES.PG     = nan(T,N) ;  
RES.PB     = nan(T,N) ;  
RES.Pinj   = nan(T,N) ;  
RES.SOC    = nan(T+1,N) ;  
RES.Pij    = nan(T,E) ;  
RES.Pji    = nan(T,E) ;  
RES.lij    = nan(T,E) ;  
RES.Pconv  = nan(T,N) ;
```

Snippet 10: Matrices for Time-varying data

Decision variables are arranged into a single vector for the optimization solver, with their positions defined using a structured indexing approach (*idx*). This ensures consistent mapping between the solver variables and the physical quantities of the system and allows the optimization problem to be formulated in a compact and scalable manner.

5 Results and Analysis

5.1 Overview of Test Cases

The selected test cases are designed to evaluate the performance of the DC microgrid under varying levels of renewable generation and grid interaction. Each test case represents a different operating condition, allowing the system behaviour to be analysed in terms of efficiency, reliability, and energy management. The following questions guide the analysis:

- Can the microgrid operate independently under full photovoltaic (PV) generation in islanded mode? This is investigated using the islanded test case with full PV availability, where all demand must be met using local generation and storage.
- How does grid connection influence system performance when PV generation is sufficient? This is examined by comparing islanded and grid-connected operation under full PV conditions, highlighting the impact of grid support on losses and operational flexibility.
- How does reduced PV availability affect system operation and grid dependence? This is analysed using the grid-connected test case with 50% PV generation, where the system must balance reduced local generation with increased reliance on storage and grid import.
- What is the system behaviour in the absence of PV generation? This is evaluated using the grid-connected test case with no PV generation, representing the most grid-dependent scenario and allowing assessment of system performance under worst-case conditions.

Together, these test cases provide a progressive evaluation of the system, moving from fully self-sufficient operation to complete reliance on the utility grid, and enable a comprehensive analysis of the trade-offs between efficiency, flexibility, and reliability.

5.2 Individual Test Case Results

This section presents the results obtained for the clustered 20-bus DC microgrid system (Test Case 3). The system represents a more complex and interconnected network structure, allowing for the analysis of power sharing, loss behaviour, and operational dynamics under varying generation and grid conditions. For the results of test case 1, see Appendix C and for the results of test case 2, see Appendix D. For in-depth unit and integration testing of all three systems, see the System Testing Document (STD) (myelearning upload).

5.2.1 Islanded Mode with Full PV Generation

Table 1: Hour 24 Operational Results for TC3 (Islanded Mode)

```
===== OFF2 =====
Grid connected mode: 0
Successful hours: 24/24
Total dist loss (p.u.): 0.023932
Total conv loss (p.u.): 0.616246
Total system loss (p.u.): 0.640179
Optimization runtime: 130.6763 seconds
```

===== TC3-20bus-islanded - OPF-2 (Hour 20 snapshot) =====					
Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0	-0.096	86.659	1.049	"Non-Op"
2	0	-0.047	87.68	1.049	"Non-Op"
3	0.002	-0.095	86.143	1.049	"70.640"
4	0	-0.041	88.455	1.049	"Non-Op"
5	0	-0.049	87.793	1.049	"Non-Op"
6	0	-0.06	87.602	1.05	"Non-Op"
7	0	-0.057	87.86	1.05	"Non-Op"
8	0	-0.059	87.4	1.049	"Non-Op"
9	0	-0.085	86.487	1.05	"Non-Op"
10	0	-0.074	86.701	1.05	"Non-Op"
11	0	-0.047	87.272	1.049	"Non-Op"
12	0	-0.091	87.008	1.049	"Non-Op"
13	0.003	-0.049	88.109	1.049	"73.475"
14	0	-0.048	88.047	1.049	"Non-Op"
15	0	-0.096	85.266	1.049	"Non-Op"
16	0.148	-0.205	83.668	1.046	"97.588"
17	0.048	-0.091	87.192	1.048	"96.433"
18	0.035	-0.125	86.41	1.049	"95.736"
19	0.15	-0.237	81.301	1.046	"97.593"
20	-0.399	0	0	1.037	"97.515"

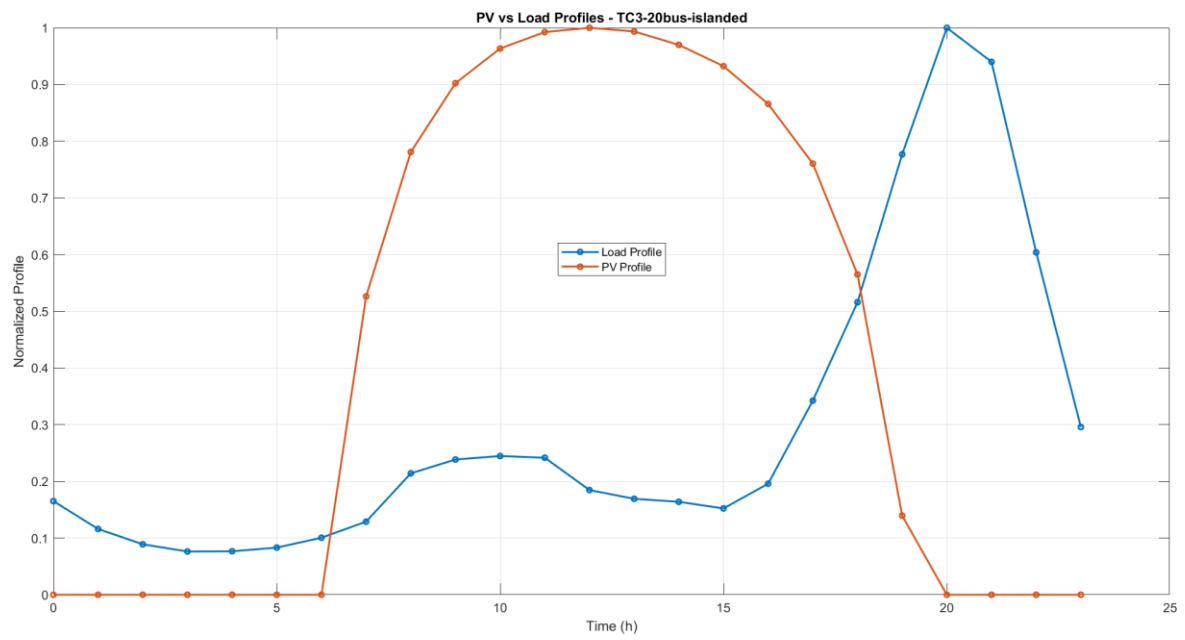


Figure 7: PV and Load Profiles

The figure above presents the PV and load profiles over the 24 hours. PV generation peaks during midday hours, while load demand increases significantly during the evening.

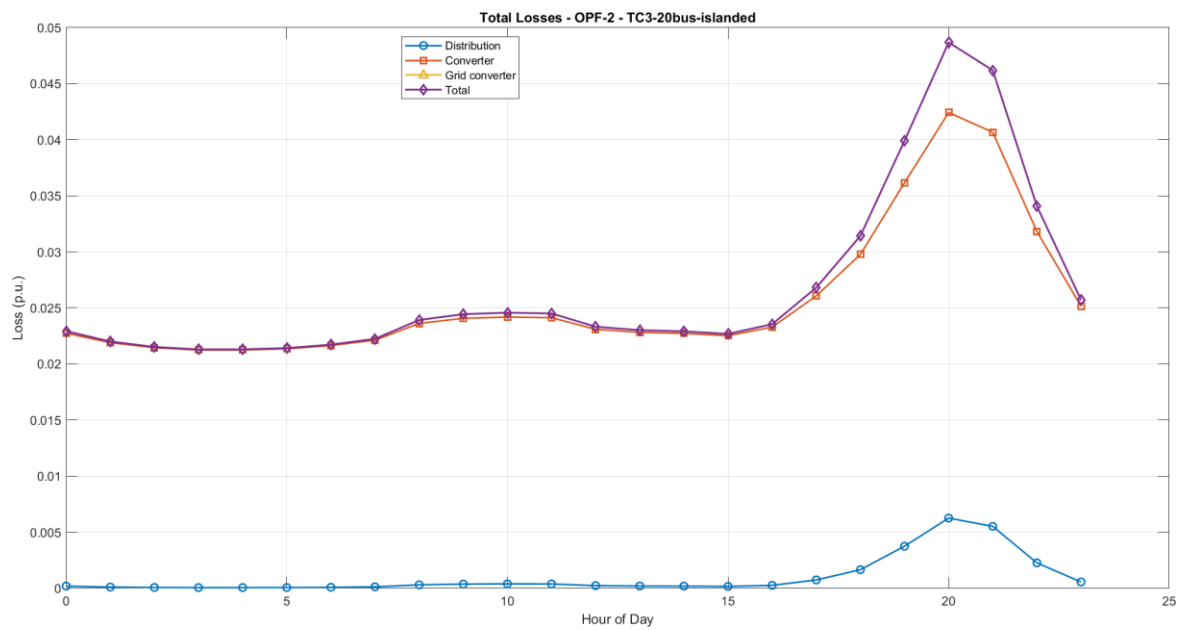


Figure 8: Total System Losses - Islanded Mode

The figure above shows the variation of total system losses over the 24 hours. Losses remain relatively low and stable during daylight hours when PV generation is available. However, a significant increase in losses is observed during the evening period, corresponding to peak load demand and reduced PV generation.

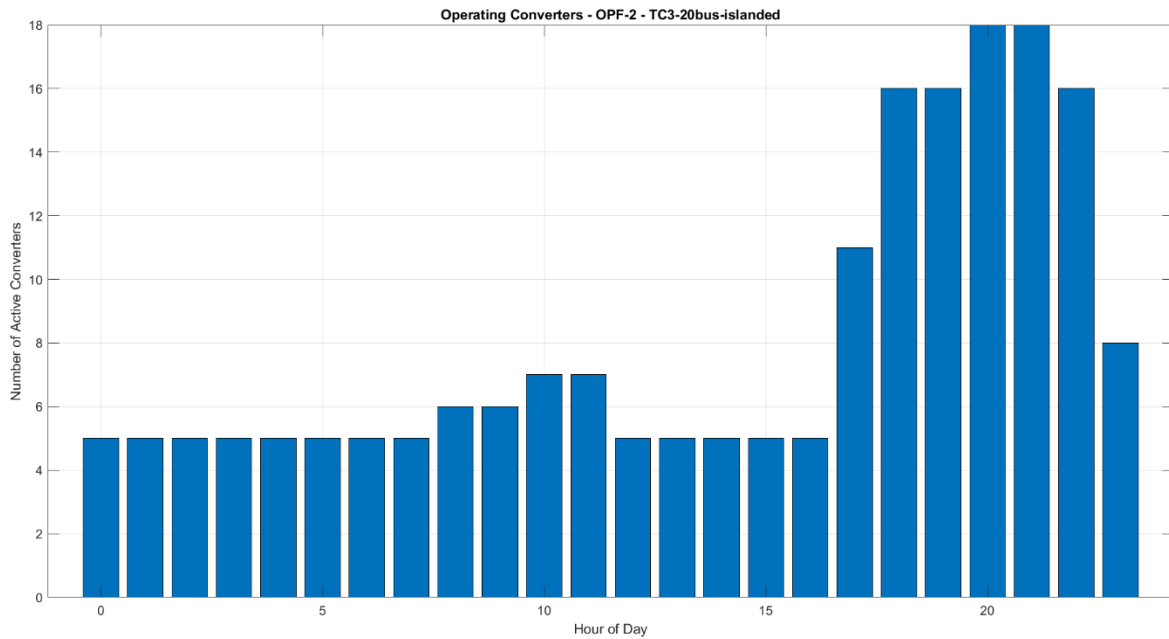


Figure 9: Number of Active Converters - Islanded Mode

The figure above illustrates the number of active converters throughout the day. A relatively low and stable number of converters are active during periods of sufficient PV generation. However, a sharp increase in converter activity is observed during the evening peak, indicating increased power exchange between nodes to meet demand. This increased activity contributes to higher system losses, as more power is processed through converters and network branches.

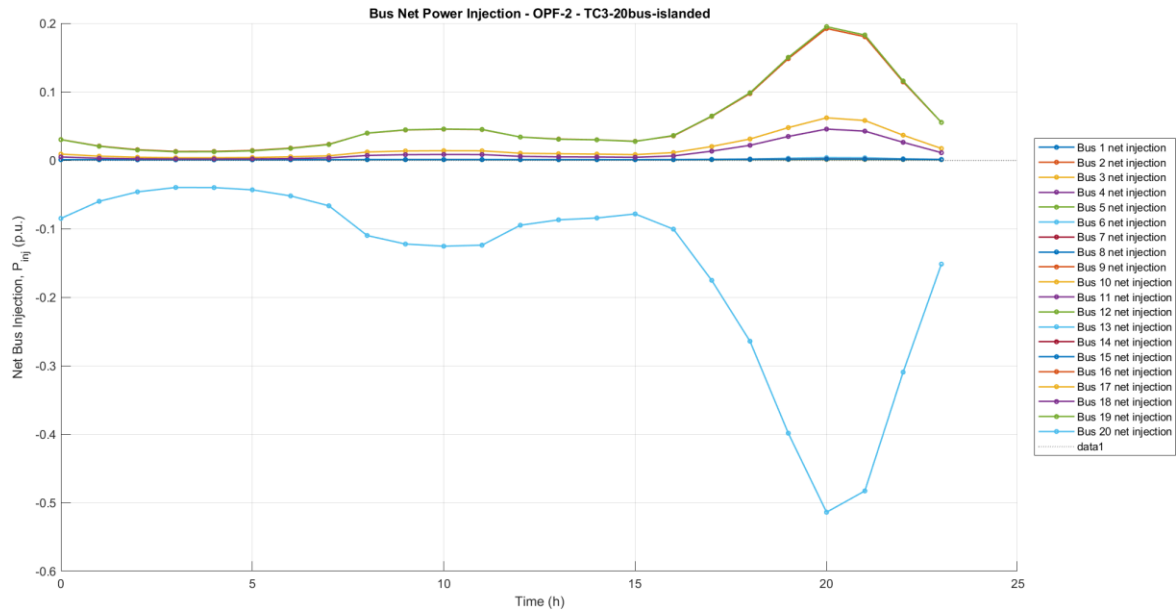


Figure 10: Net Power Injection at each bus - Islanded Mode

The figure above shows the net power injection at each bus. While many buses operate close to zero net injection for large portions of the day, several buses become clear net exporters during the evening peak load demand, whereas the community load bus remains a strong net consumer. This indicates that power is being redistributed throughout the network rather than being consumed only at the point of generation. The figure, therefore, highlights the peer-to-peer sharing behaviour of the clustered microgrid under islanded operation.

5.2.2 Grid-Connected Mode with Full PV Generation

Table 2: Hour 24 Operational Results for TC3 (Grid-connected Mode - 100% PV Generation)

===== OPF2 =====

Grid connected mode: 1
 Successful hours: 24/24
 Total dist loss (p.u.): 0.024041
 Total conv loss (p.u.): 0.616251
 Total grid conv loss (p.u.): 0.048014
 Total system loss (p.u.): 0.688306
 Optimization runtime: 149.3505 seconds

===== TC3-20bus-grid - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.002	-0.097	87.004	1.047	"65.988"
2	0.002	-0.048	87.634	1.047	"67.772"
3	0.003	-0.097	87.043	1.047	"71.799"
4	0.002	-0.041	88.426	1.047	"67.874"
5	0.002	-0.049	87.75	1.047	"66.007"
6	0	-0.061	87.647	1.047	"Non-Op"
7	0	-0.058	87.915	1.047	"Non-Op"
8	0	-0.06	87.424	1.047	"Non-Op"
9	0	-0.087	86.433	1.047	"Non-Op"
10	0	-0.074	86.567	1.047	"Non-Op"
11	0.002	-0.049	87.355	1.047	"67.012"
12	0.002	-0.092	86.926	1.047	"69.238"
13	0.003	-0.049	88.05	1.047	"73.811"
14	0.002	-0.049	88.123	1.047	"68.788"
15	0.002	-0.1	85.542	1.047	"66.573"
16	0.147	-0.201	83.62	1.044	"97.584"
17	0.046	-0.093	87.25	1.046	"96.353"
18	0.03	-0.116	86.669	1.046	"95.300"
19	0.148	-0.233	80.447	1.044	"97.587"
20	-0.399	0	0	1.034	"97.515"

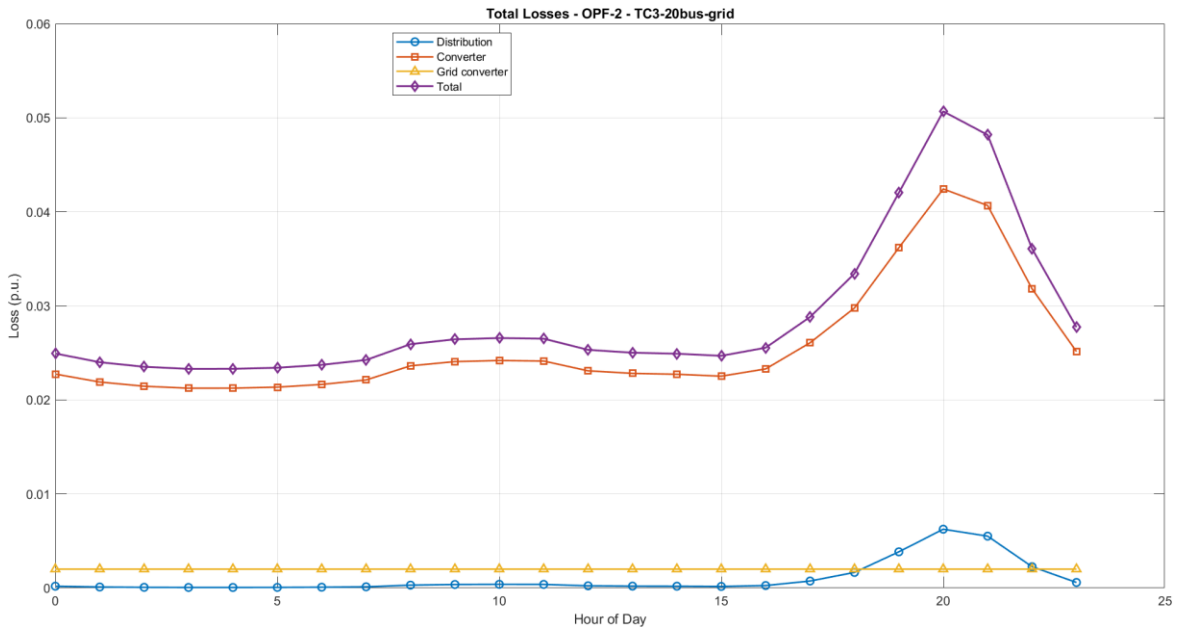


Figure 11: Total System Losses - Grid-Connected Mode

The figure above shows the total system losses under grid-connected operation. Compared to the islanded case, total losses are slightly higher due to the inclusion of grid converter losses. While distribution and local converter losses follow a similar trend, the additional grid interface introduces a constant loss component. Losses still peak during the evening hours, corresponding to high demand and reduced PV generation.

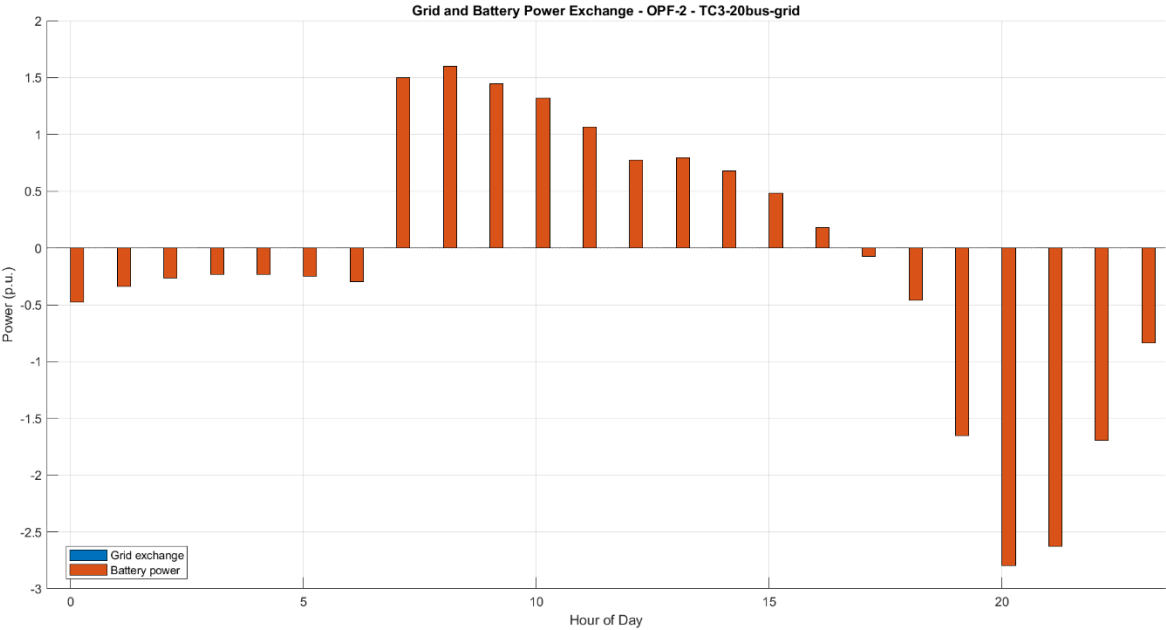


Figure 12: Grid and Battery Power Exchange - Grid-Connected Mode

The figure above illustrates the power exchange between the grid and battery over the 24 hours. During periods of high PV generation, the battery absorbs excess energy, while during peak demand periods, it discharges to support the load. Grid interaction remains limited, indicating that the system primarily relies on local generation and storage, with the grid providing supplementary support when required.

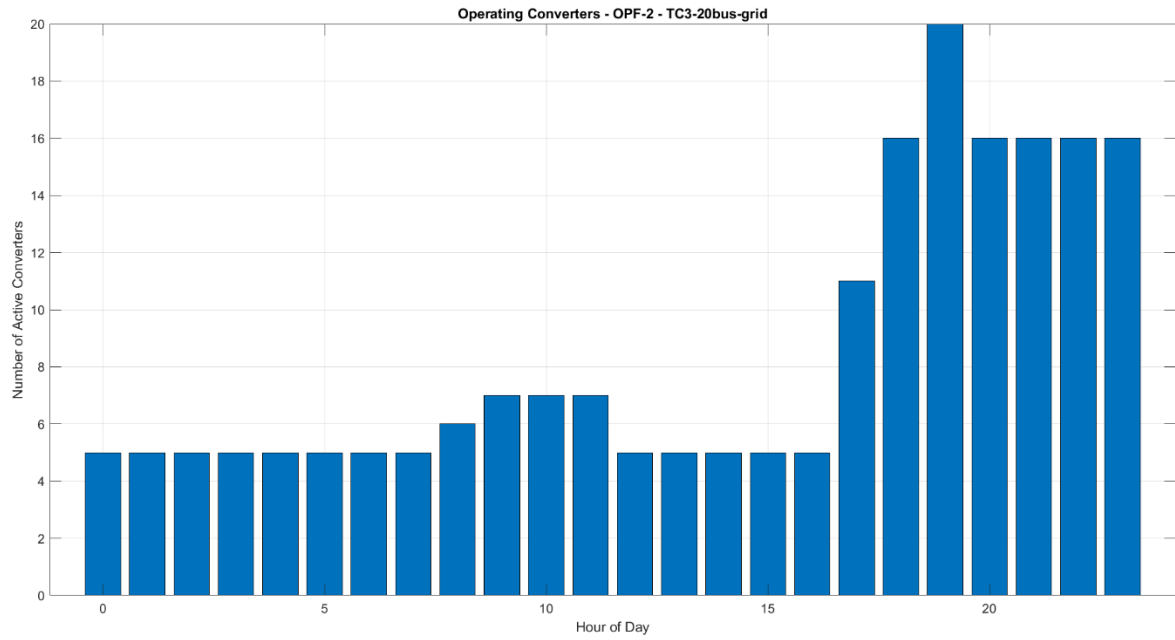


Figure 13: Number of Active Converters - Grid-Connected Mode

Similar to the islanded case, converter activity increases significantly during peak demand periods. However, the presence of a grid connection allows for more flexible power routing, contributing to slightly higher system activity. This increased utilization is associated with the additional losses observed in the system.

5.2.3 Grid-Connected Mode with 50% PV Generation

Table 3: Hour 24 Operational Results for TC3 (Grid-connected Mode - 50% PV Generation)

===== OPF2 =====

Grid connected mode: 1
 Successful hours: 24/24
 Total dist loss (p.u.): 0.026556
 Total conv loss (p.u.): 0.621032
 Total grid conv loss (p.u.): 0.048012
 Total system loss (p.u.): 0.695600
 Optimization runtime: 160.3928 seconds

===== TC3-20bus-grid - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0	-0.109	85.004	1.049	"Non-Op"
2	0	-0.059	84.896	1.049	"Non-Op"
3	0.002	-0.108	83.527	1.049	"70.636"
4	0	-0.051	88.415	1.049	"Non-Op"
5	0	-0.061	85.733	1.049	"Non-Op"
6	0	-0.07	85.784	1.05	"Non-Op"
7	0	-0.068	86.737	1.05	"Non-Op"
8	0	-0.07	84.479	1.049	"Non-Op"
9	0	-0.098	82.211	1.05	"Non-Op"
10	0	-0.086	82.147	1.05	"Non-Op"
11	0	-0.059	84.73	1.049	"Non-Op"
12	0	-0.107	86.15	1.049	"Non-Op"
13	0.003	-0.059	87.435	1.049	"71.931"
14	0	-0.058	87.017	1.049	"Non-Op"
15	0	-0.109	80.138	1.049	"Non-Op"
16	0.148	-0.217	77.196	1.046	"97.588"
17	0.048	-0.108	86.739	1.048	"96.441"
18	0.035	-0.14	85.568	1.049	"95.727"
19	0.15	-0.255	73.642	1.046	"97.593"
20	-0.399	0	0	1.037	"97.515"

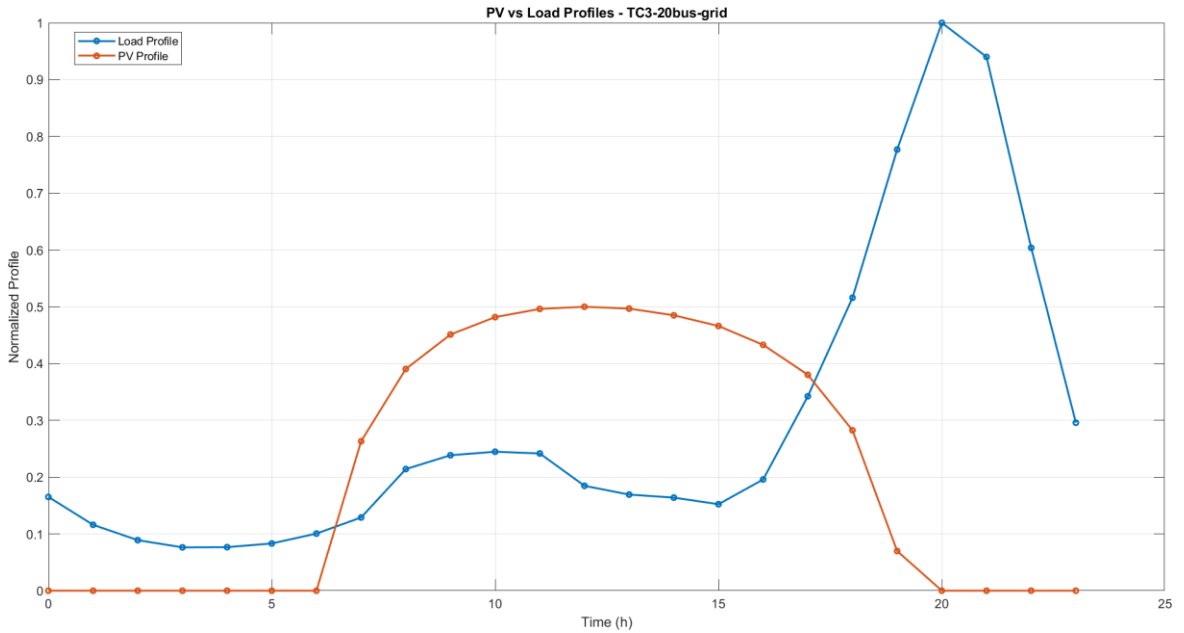


Figure 14: PV and Load Profiles with 50% PV Generation

The figure above shows the PV and load profiles for the 50% PV case. Compared to the full PV scenario, the available PV generation is significantly reduced while the load profile remains unchanged.

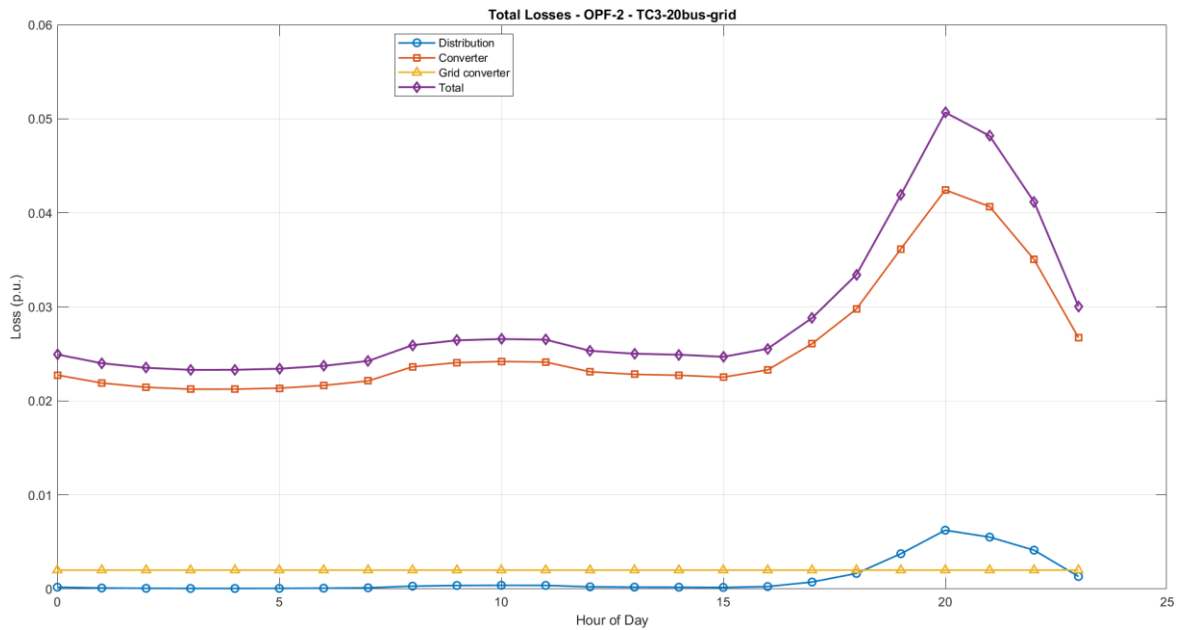


Figure 15: Total System Losses - Grid-Connected Mode with 50% PV Generation

The figure above illustrates the total system losses for the grid-connected case with 50% PV generation. Compared to the full PV condition, the system exhibits higher total losses due to the reduced contribution of local renewable generation. The evening period shows the largest loss values, as the system must increasingly rely on battery discharge to satisfy demand.

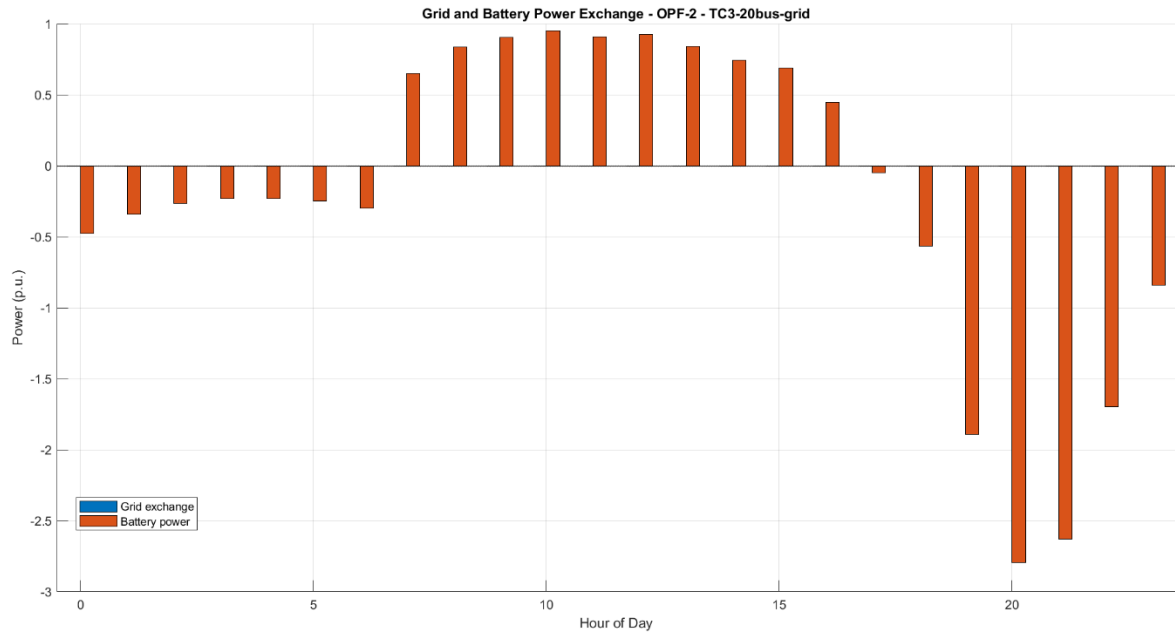


Figure 16: Grid and Battery Power Exchange - Grid-Connected Mode with 50% PV Generation

The figure above shows the grid and battery power exchange for the 50% PV scenario. Although PV generation is reduced, the system does not rely on grid import to satisfy demand. Instead, the battery absorbs available excess energy during the daytime and discharges during periods of higher demand, particularly in the evening. This indicates that the storage system is sufficiently sized to maintain self-sufficient operation even under reduced renewable availability.

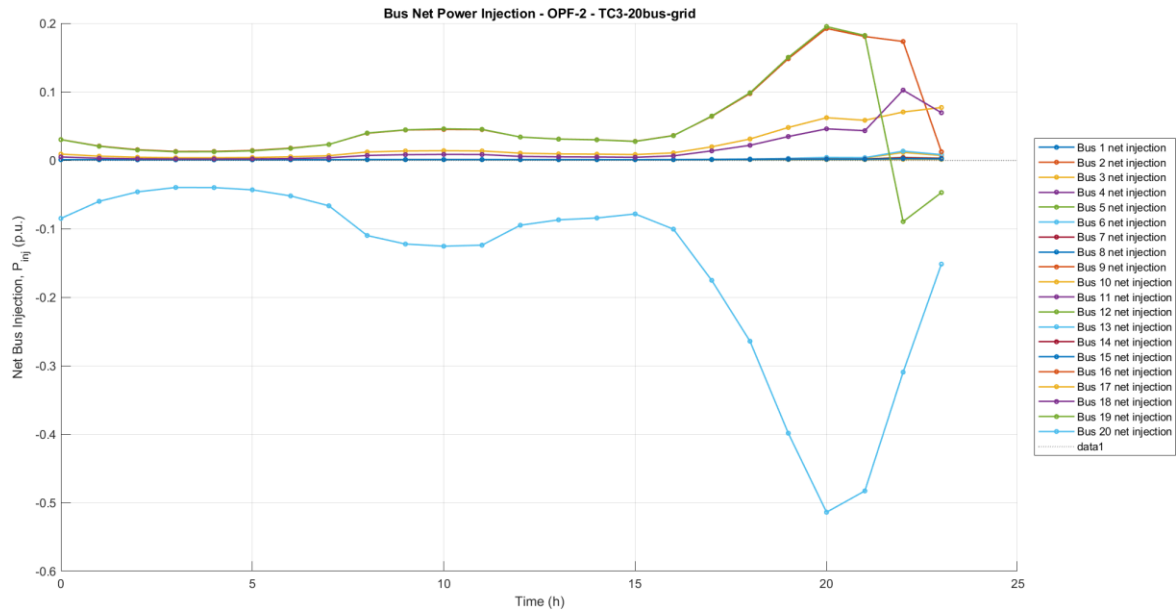


Figure 17: Net Power Injection at each bus - Grid-Connected Mode with 50% PV Generation

The figure above shows the net power injection at each bus for the grid-connected 20-bus system with 50% PV generation. The behaviour is similar to that observed in the islanded mode case, with many buses operating close to zero net injection for large portions of the day. However, slightly greater variation is observed during peak demand hours, indicating increased peer-to-peer power sharing across the network.

5.2.4 Grid-Connected Mode with No PV Generation

Table 4: Hour 24 Operational Results for TC3 (Grid-connected Mode - 0% PV Generation)

===== OPF2 =====

Grid connected mode: 1
 Successful hours: 24/24
 Total dist loss (p.u.): 0.148074
 Total conv loss (p.u.): 0.697362
 Total grid conv loss (p.u.): 0.051117
 Total system loss (p.u.): 0.896553
 Optimization runtime: 121.8849 seconds

===== TC3-20bus-grid - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.003	-0.125	66.411	1.05	"75.139"
2	0.007	-0.08	63.273	1.05	"86.861"
3	0.028	-0.145	64.685	1.049	"95.107"
4	0.007	-0.072	76.434	1.05	"86.493"
5	0.003	-0.079	63.33	1.05	"74.689"
6	0	-0.083	67.908	1.05	"Non-Op"
7	0.002	-0.083	69.282	1.05	"66.484"
8	0.007	-0.088	65.518	1.05	"85.950"
9	0.002	-0.114	59.642	1.05	"66.484"
10	0	-0.101	58.717	1.05	"Non-Op"
11	0.003	-0.077	61.857	1.05	"75.038"
12	0.008	-0.132	67.364	1.05	"88.249"
13	0.032	-0.101	71.22	1.049	"95.497"
14	0.007	-0.078	68.591	1.05	"86.666"
15	0.003	-0.123	59.596	1.05	"74.078"
16	-0.084	0	50.004	1.033	"97.218"
17	0.281	-0.361	69.021	1.045	"97.639"
18	0.22	-0.346	63.089	1.046	"97.661"
19	-0.123	0	50	1.032	"97.503"
20	-0.399	0	0	1.023	"97.515"

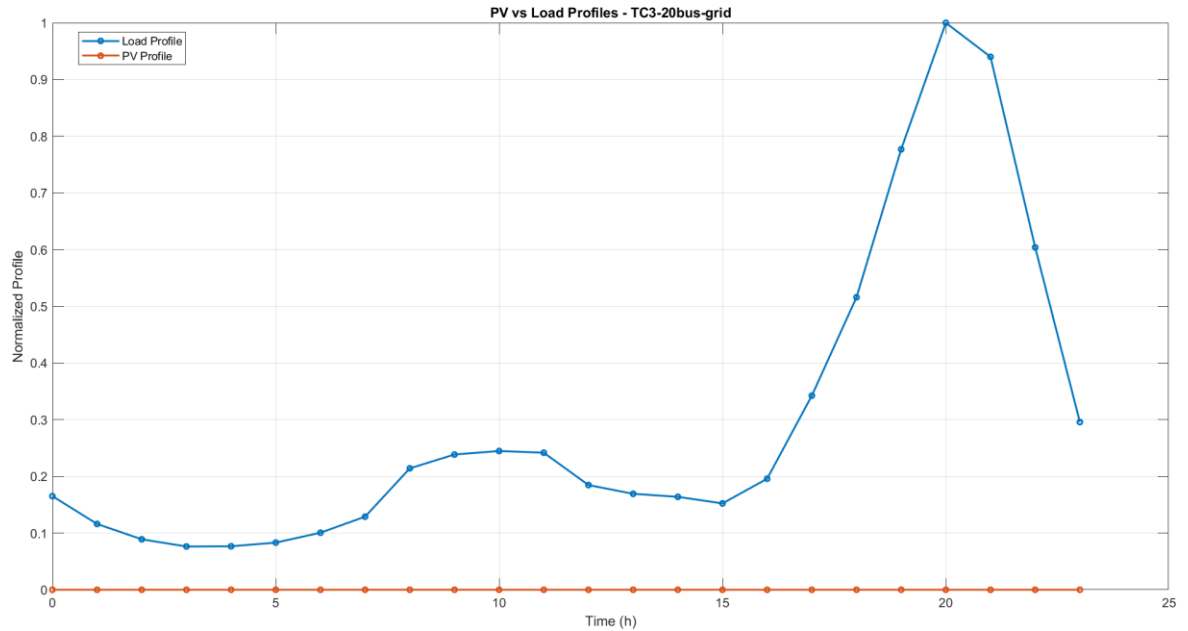


Figure 18: PV and Load Profiles with No PV Generation

The figure above shows the PV and load profiles for the no-PV case. As expected, PV generation remains zero throughout the entire 24-hour period, while load demand continues to follow the normal daily pattern with a pronounced evening peak. This creates the most constrained operating condition, since no local renewable generation is available to contribute to demand or support battery charging.

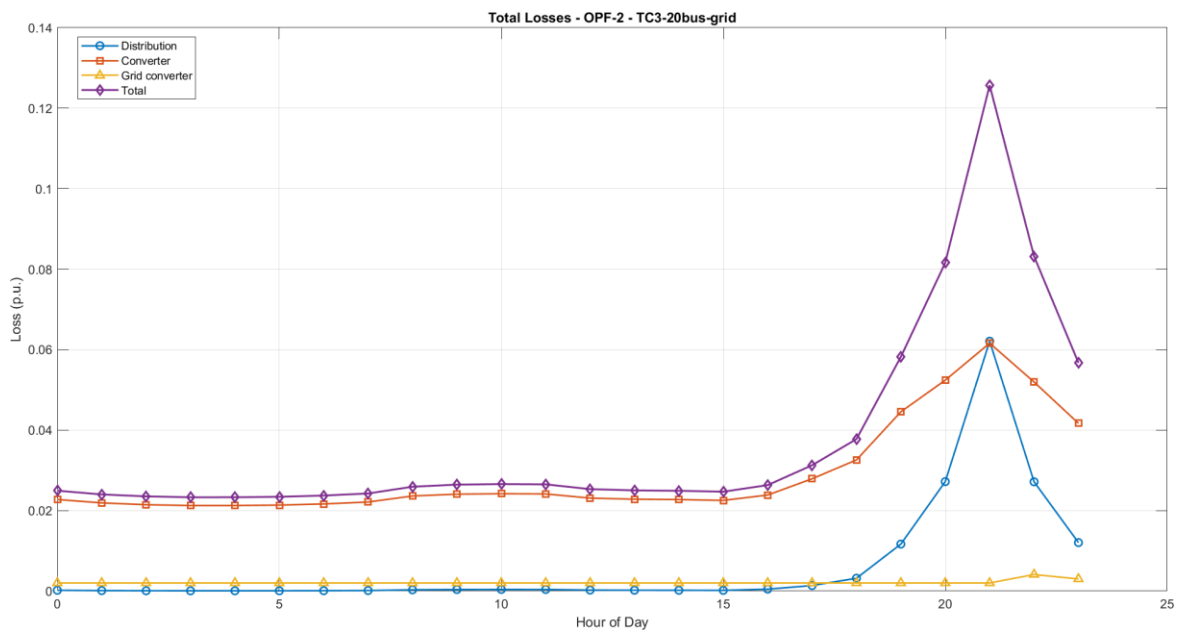


Figure 19: Total System Losses - Grid-Connected Mode with No PV Generation

The above figure illustrates the total system losses under grid-connected operation with no PV generation. This case produces the highest overall losses among all operating conditions considered. In particular, distribution losses rise sharply during the evening peak, indicating heavier power transfer across the network. DC-DC converter and grid converter losses also remain significant, reflecting the increased dependence on stored energy and grid support.

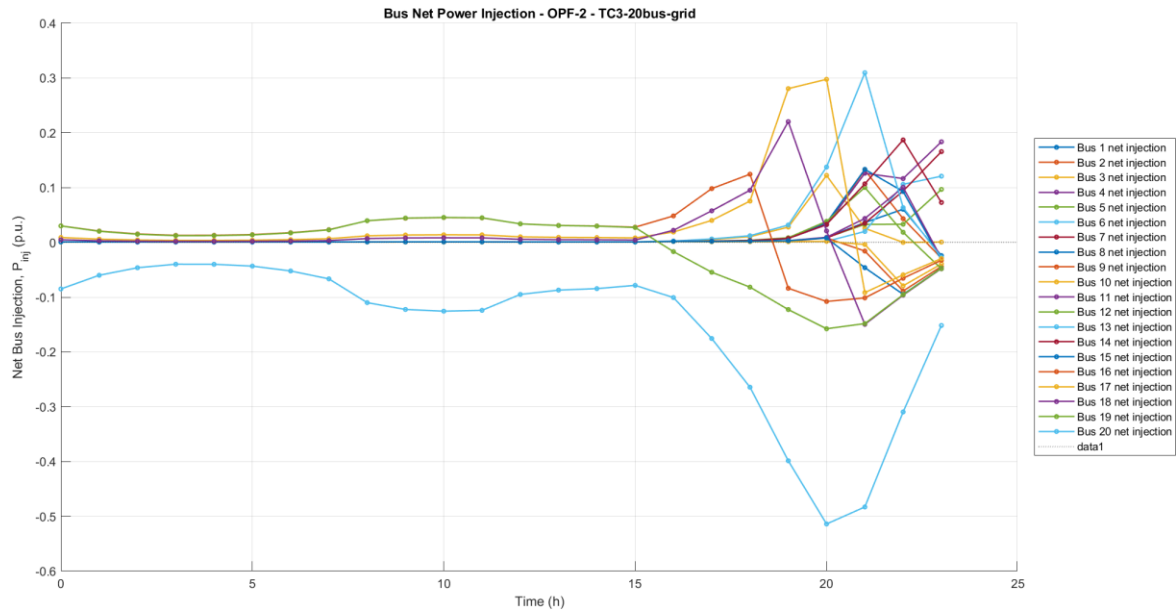


Figure 20: Net Power Injection at each bus - Grid-Connected Mode with No PV Generation

The figure above shows the net power injection at each bus for the grid-connected 20-bus system with no PV generation. The behaviour remains similar to the previous cases during the early hours, with most buses operating close to zero net injection. However, significantly larger variations are observed during peak demand hours, indicating increased peer-to-peer power sharing across the network as the system relies heavily on battery support and grid interaction.

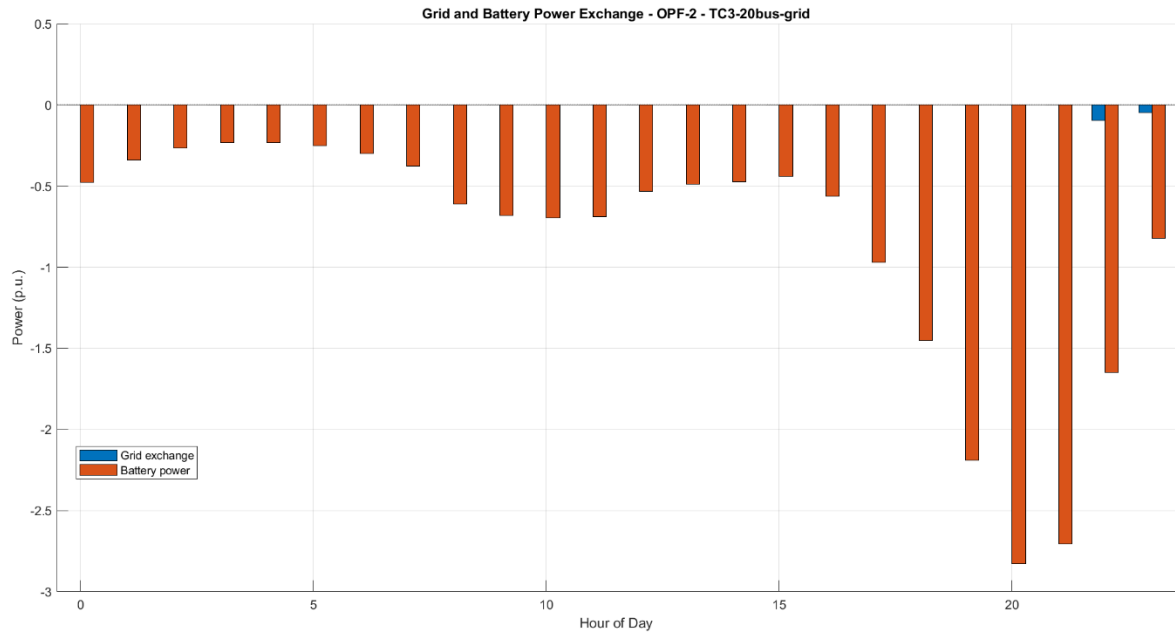


Figure 21: Grid and Battery Power Exchange - Grid-Connected Mode with No PV Generation

The figure above shows the grid and battery power exchange for the no-PV case. In the absence of photovoltaic generation, the battery is forced to discharge throughout the day to support the load. During the evening peak, the battery discharge becomes significantly larger, and grid support begins to appear as the system reaches the limits of what storage alone can provide. This confirms that, unlike the 50% PV case, the no-PV scenario results in genuine grid dependence.

5.3 Comparative Analysis

5.3.1 Effect of Grid Connection

Comparing the full PV cases, the islanded system achieves lower total system losses due to the absence of grid converter losses. In contrast, the grid-connected case introduces additional losses associated with the AC/DC interface at the PCC. Although this increases total losses, the grid-connected configuration provides greater operational flexibility,

allowing the system to balance supply and demand under varying load and generation conditions.

5.3.2 Effect of PV Reduction

As PV generation decreases from full capacity to 50%, the system exhibits a greater reliance on internal energy storage. In this case, the batteries compensate for the reduced generation and are sufficient to meet the load demand without requiring grid import. This highlights the capability of distributed storage to maintain system autonomy even under reduced renewable availability, while also demonstrating increased stress on battery resources.

5.3.3 No PV Scenario (Worst Case)

In the absence of PV generation, the system becomes heavily dependent on non-renewable sources to meet demand. Battery discharge increases significantly, and grid support becomes necessary to sustain operation. This scenario results in the highest total system losses due to increased power flow through converters and higher network loading, emphasizing the importance of renewable generation in maintaining system efficiency.

5.3.4 Battery Utilization

Battery operation varies significantly across the different scenarios. Under full PV conditions, batteries primarily act as energy buffers, storing excess generation during the day and supplying power during peak demand periods. As PV availability decreases, batteries shift to a more critical role in maintaining system balance, supplying a larger

portion of the demand. In the absence of PV, battery usage becomes intensive, leading to lower state of charge levels and increased dependence on external support.

5.3.5 Loss Trade-offs

A clear trade-off is observed between system efficiency and operational flexibility. Islanded operation minimizes total losses by avoiding grid converter losses, making it more efficient under sufficient local generation. However, grid-connected operation enhances reliability and system resilience at the cost of increased converter losses. Additionally, reduced PV availability leads to higher losses due to increased reliance on battery discharge and power redistribution within the network.

6 Discussion

This chapter analyzes and evaluates the results obtained from the three microgrid test systems, namely the 7-bus radial system, the modified IEEE 14-bus system, and the clustered 20-bus system, under varying operating conditions. The results are examined in light of established DC microgrid theory and existing research on optimal power flow and energy management. Across all three systems, similar trends were observed under the different operating scenarios, indicating that system behaviour is primarily influenced by the interaction between generation, storage, and grid support rather than the specific network topology. Under full PV conditions, both islanded and grid-connected modes resulted in relatively low total system losses. This is consistent with theoretical expectations, as local generation reduces the need for extensive power transfer and limits the amount of power processed through converters. As PV availability was reduced, the role of battery storage became more prominent. In the 50% PV scenario, all systems were able to meet demand without requiring grid import, relying instead on stored energy to balance supply and demand. This behaviour aligns with established microgrid operation principles, where batteries act as buffers to compensate for variability in renewable generation. It also demonstrates that sufficient storage capacity can maintain system autonomy under moderate reductions in renewable output.

In the absence of PV generation, the systems exhibited their most constrained operating condition. Battery discharge increased significantly, and grid support became necessary to maintain operation. This resulted in higher total system losses, particularly during peak demand periods. This behaviour is consistent with theoretical models, where reduced local generation leads to increased reliance on external supply and higher levels of power

transfer within the network. The inclusion of grid connection introduced a clear trade-off across all systems. While grid-connected operation improved flexibility and provided an additional source of support, it also resulted in increased total losses due to the presence of grid converter losses. This observation is consistent with previous studies, which identify power electronic conversion as a key contributor to inefficiency in DC microgrids. As a result, grid connection enhances reliability but may reduce overall efficiency depending on operating conditions. An additional observation from the results is the variation in converter activity across the different system configurations. In the 7-bus and 14-bus systems, a larger number of converters were active throughout the day, whereas the 20-bus clustered system showed more selective converter operation. This can be attributed to the smaller network size and reduced path diversity in the 7-bus and 14-bus systems, which require more nodes to actively participate in power transfer to maintain system balance. In contrast, the clustered 20-bus system provides greater flexibility for localized power sharing, allowing fewer converters to be active at any given time. Converter losses were observed to contribute significantly to total system losses, particularly during periods of high demand and increased power transfer. While distribution losses were also present, the results suggest that power electronic conversion plays an important role in determining overall system efficiency, although the exact contribution varies depending on system configuration and operating conditions.

The results also demonstrate that system behaviour is strongly time-dependent. Across all test cases, higher losses and increased system activity were observed during evening periods when load demand was highest and PV generation was unavailable. This reflects the mismatch between generation and demand, which is a well-known challenge in renewable-based microgrid systems. No major anomalies were observed in the results, as

all trends followed expected theoretical behaviour. Minor variations between systems are attributed to differences in network size and configuration, but these did not significantly affect the overall trends. This consistency strengthens the validity of the results and supports the general applicability of the findings.

7 Conclusion

This project developed and implemented a MATLAB-based optimal power flow framework to analyse power sharing in DC microgrids for rural electrification. The model integrated photovoltaic generation, battery energy storage, peer-to-peer energy exchange, and grid interaction, while accounting for distribution, converter, and grid interface losses. The framework was successfully applied to three different systems: a 7-bus radial network, a modified IEEE 14-bus system and a clustered 20-bus microgrid, demonstrating its scalability and adaptability. The results showed consistent behaviour across all systems, confirming that microgrid performance is primarily influenced by renewable generation availability, distributed storage capacity, and grid support. Under full PV conditions, systems operated efficiently with low losses, while reduced PV increased reliance on battery storage. In the absence of PV, grid support became necessary, resulting in the highest losses. These findings highlight the importance of renewable integration and storage in maintaining efficient and reliable microgrid operation, while also demonstrating the trade-off between flexibility and efficiency introduced by grid connection.

Despite these contributions, the study is limited to steady-state simulation and does not account for dynamic behaviour, detailed converter modelling, or long-term variability in generation and demand. The grid was represented using a simplified model, and the analysis was restricted to a 24-hour horizon, which may not fully capture real-world operating conditions. Future work can extend this framework by incorporating dynamic system modelling, more detailed converter and grid representations, and longer-term simulations. Additionally, economic optimization and real-time or decentralized control strategies could be explored to enhance practical applicability. Overall, this project

demonstrates that coordinated optimization of generation, storage, and grid interaction is essential for designing efficient, reliable, and scalable DC microgrid systems suitable for rural electrification.

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Appendices

Appendix A: Technical Risk Analysis

Table 5: Technical Risk Assessment and Mitigation Worksheet

Technical Threat	Risk Score	Mitigation Strategy	Details of Mitigation
Non-convergence of optimization solver (fmincon)	0.7	Reduce	Improve initial guesses, adjust solver tolerances, and increase iteration limits to enhance convergence reliability
Infeasible solutions due to constraint conflicts	0.7	Reduce	Carefully define bounds and constraints, and include feasibility checks within the solver
Unrealistic input data (PV/load profiles)	0.5	Reduce	Use validated datasets (NREL, NSRDB) and normalize profiles to reflect realistic operating conditions

(Adapted from Hong, R. 2013. "Risk Assessment." Tufts University. Accessed July 2019 at: <https://sites.tufts.edu/eeseniordesignhandbook/2013/risk-management/>)

Table 6: Technical Risk Assessment Scores

Probability of Occurrence	Significant	0.3	0.7	0.9
	Moderate	0.2	0.5	0.7
	Low	0.1	0.3	0.6
		Low	Moderate	Significant
		Severity of Impact		

(Adapted from Hylton, P. 2006. "Technical Risk Management as the Connectivity in a Capstone Design Course." *Journal of Engineering Technology* 23 (1):48.)

Appendix B: Environmental Health Safety Risk Assessment

PART I: Version Control

Version Number	Date of Revision
1.0	26/03/2026

PART II: Project Identification

Project Student: Aaron Alves

Project Title: Optimal Power Sharing in DC Grid-Connected Microgrids for Rural Electrification

Project Supervisor: Dr. Arvind Singh

Project Objectives:

1. Review microgrid topologies/ features in rural electrification settings, relevant to the optimization problem
2. Review optimization techniques applied to the power sharing problem
 - Provide a justification of an optimization method(s) to be implemented
3. Formulate the objective function and constraints for the optimization problem
4. Create a function set to read a system description file (to be defined), carry out the optimization routine and generate results
 - Code must be modular and well documented
 - A testing and validation document should be provided
5. Document at least 3 test cases for microgrids of different size
 - Discuss results and limitations

Provide an overview of the project's methodology:

The methodology of this project involves developing an optimal power flow (OPF) framework to analyze power sharing in DC microgrids using a distributed generation and storage architecture. The system is modelled using a branch flow formulation, and an optimization problem is defined to minimize total system losses while satisfying electrical and operational constraints. The model is implemented in MATLAB and solved using the fmincon solver over a 24-hour time horizon with time-varying PV and load profiles. The framework is applied to multiple microgrid configurations under both islanded and grid-connected modes, and the results are analyzed to evaluate system performance in terms of power distribution, storage utilization, and overall efficiency.

Describe the tasks involved in carrying out the practical work (attach additional pages/documentation as needed):

- Collection and processing of real-world solar irradiance and load demand data
- Design and configuration of the microgrid system
- Testing and validation of system performance through simulation, including analysing results, troubleshooting errors, and refining system parameters to ensure reliable operation under different condition

Risk Assessment and Risk Mitigation Worksheet

Risk Priority	The Risk What can happen and How it can happen	Time frame Duration for which the risk exists	Severity Rating	Probability /Likelihood Rating	Risk Level Rating	Risk Treatment/Controls to be implemented	Severity Rating after treatment/ controls	Probability /Likelihood rating after treatment/ controls	Reduced Risk Level rating after treatment/ controls
1	Non-convergence of optimization solver (fmincon) leading to incomplete results	Simulation	Marginal	Likely	H	Improve initial guesses, adjust solver tolerances, increase iterations	Negligible	Occasional	L
2	Inaccurate modelling of converter losses affecting results validity	Modelling	Marginal	Occasional	M	Validate model using literature and perform sensitivity analysis	Negligible	Seldom	L
3	Errors in PV/load input data leading to unrealistic simulations	Data Processing	Marginal	Likely	H	Use verified datasets (NREL/NSRDB) and cross-check data	Negligible	Occasional	L
4	Improper constraint formulation causing infeasible or incorrect solutions	Modelling	Critical	Occasional	H	Carefully validate constraints and test edge cases	Marginal	Seldom	L
5	Numerical instability due to poor scaling of variables	Simulation	Marginal	Occasional	M	Use per-unit (p.u.) normalization for all variables	Negligible	Seldom	L

Appendix C: Individual Test Case Results (Test Case 1)

Islanded Mode with Full PV Generation

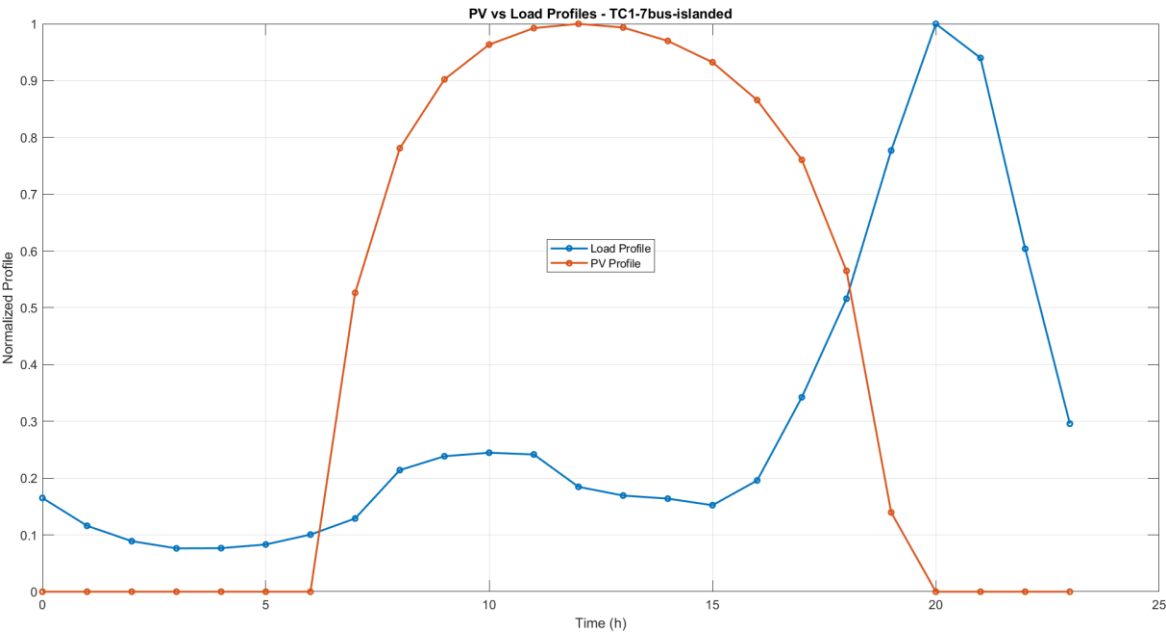
Table 7: Hour 24 Operational Results for TC1 (Islanded Mode)

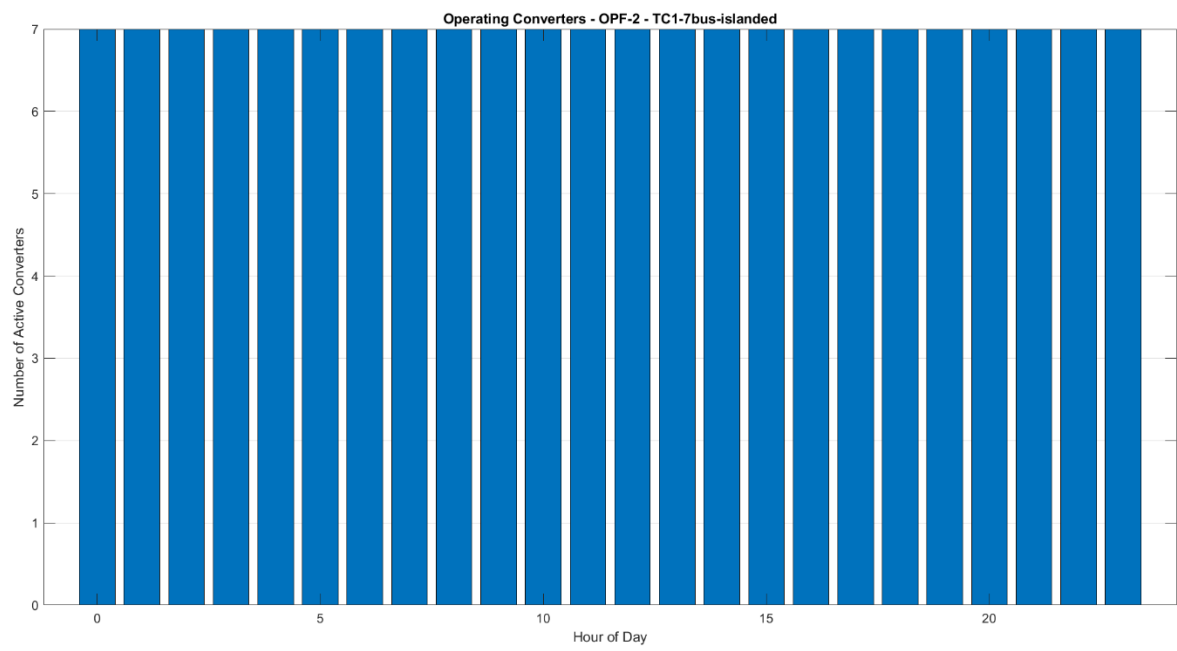
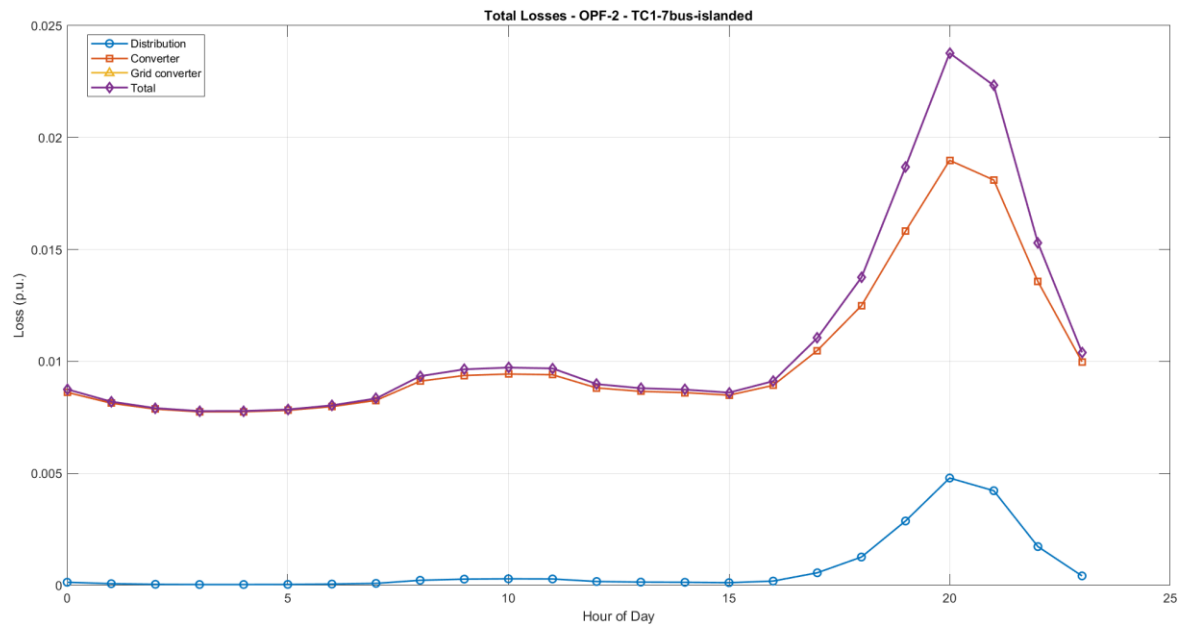
===== TC1 OPF2 =====

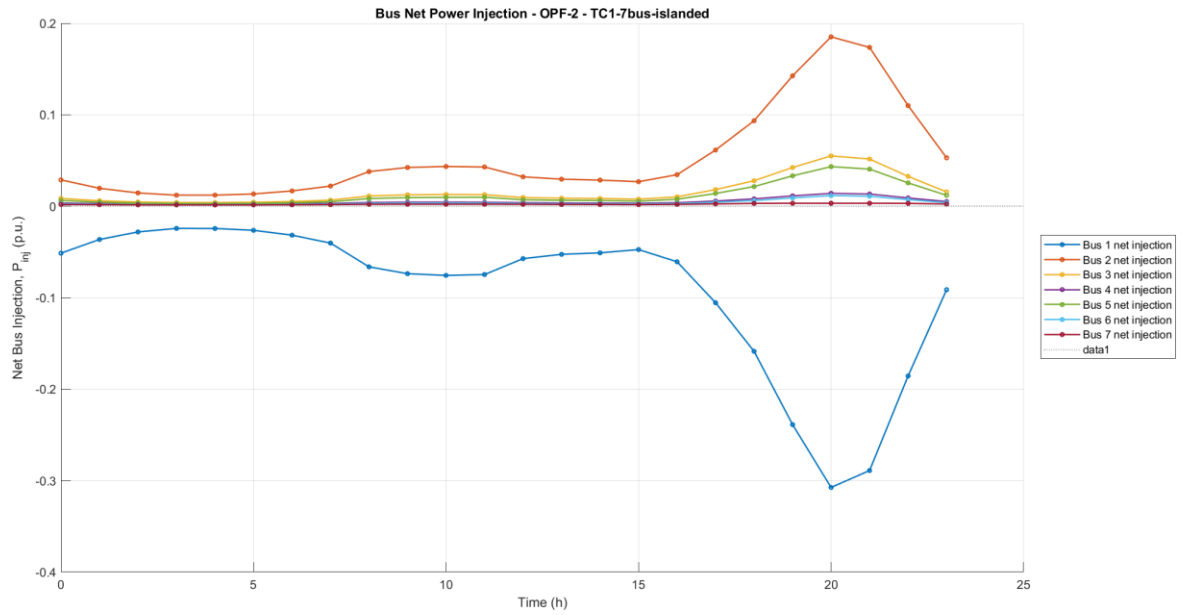
Grid connected mode: 0
Successful hours: 24/24
Total dist loss (p.u.): 0.018168
Total conv loss (p.u.): 0.244426
Total system loss (p.u.): 0.262594
Optimization runtime: 26.0157 seconds

===== TC1-7bus-islanded - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	-0.239	0	0	1.036	"97.660"
2	0.143	-0.187	83.668	1.047	"97.573"
3	0.042	-0.102	86.535	1.049	"96.202"
4	0.011	-0.047	87.304	1.05	"90.586"
5	0.033	-0.089	86.968	1.049	"95.621"
6	0.009	-0.052	87.653	1.05	"88.844"
7	0.003	-0.068	87.309	1.05	"75.494"







Grid-connected Mode with Full PV Generation

Table 8: Hour 24 Operational Results for TC1 (Grid-connected Mode - 100% PV Generation)

```

===== TC1 OPF2 =====
Grid connected mode: 1
Successful hours: 24/24
Total dist loss (p.u.): 0.018194
Total conv loss (p.u.): 0.244430
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.310638
Optimization runtime: 26.5543 seconds

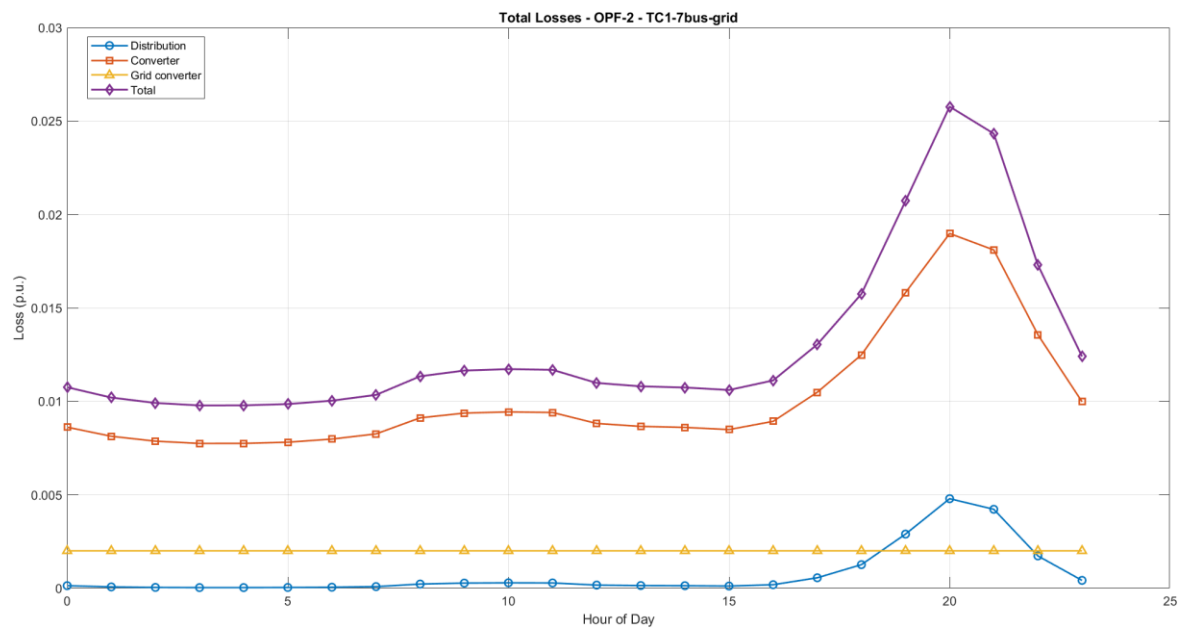
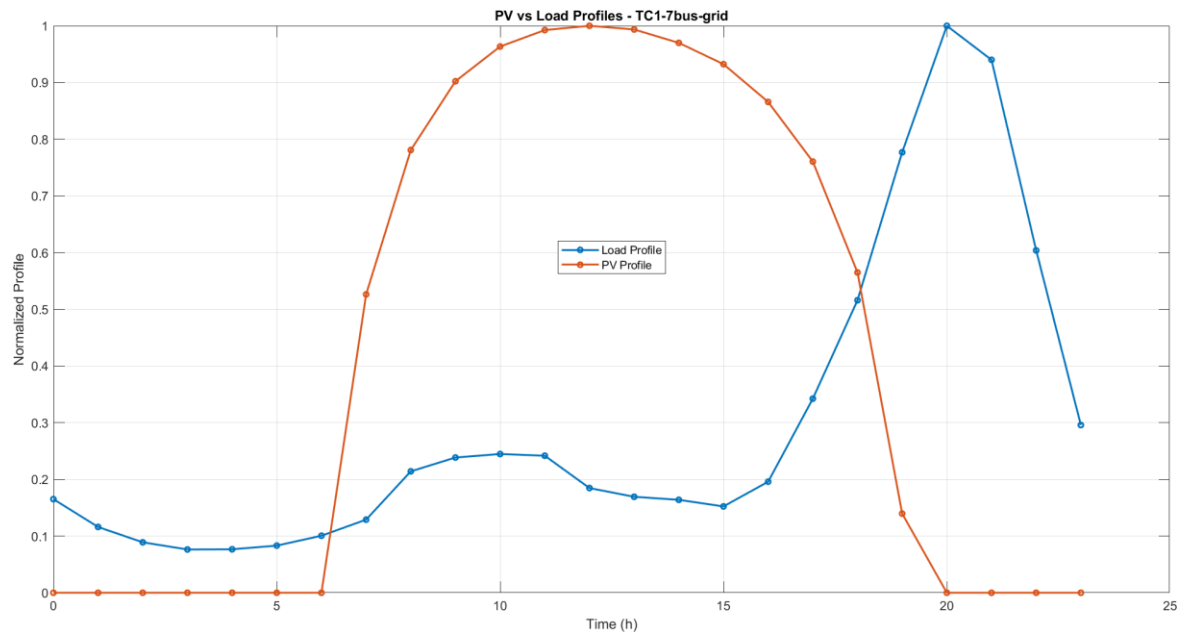
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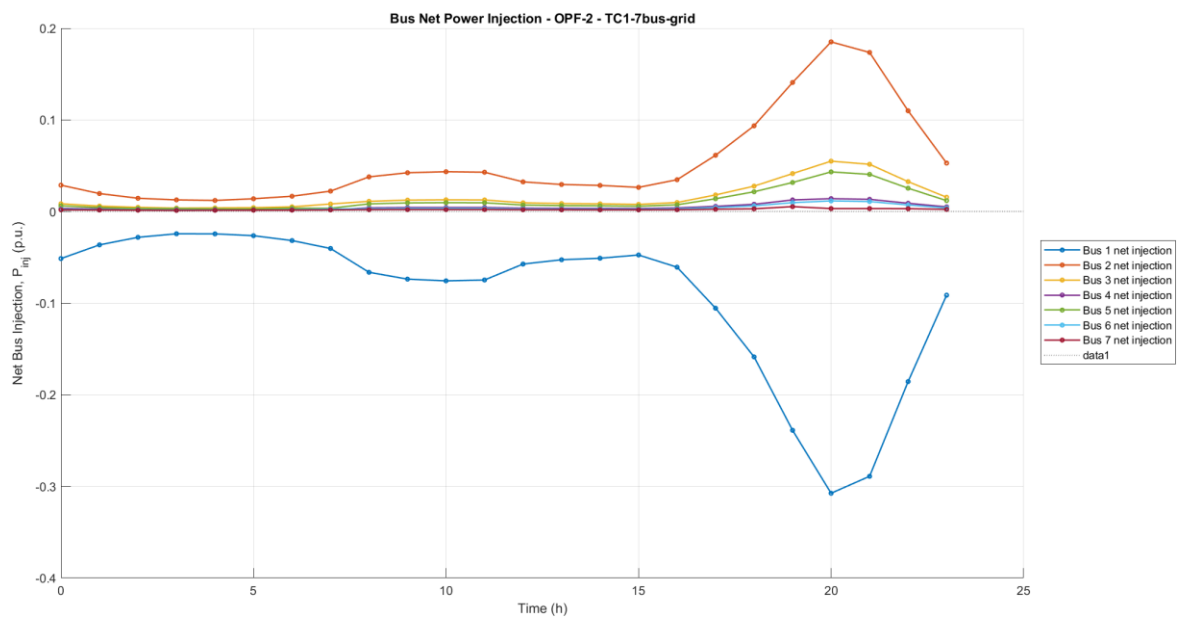
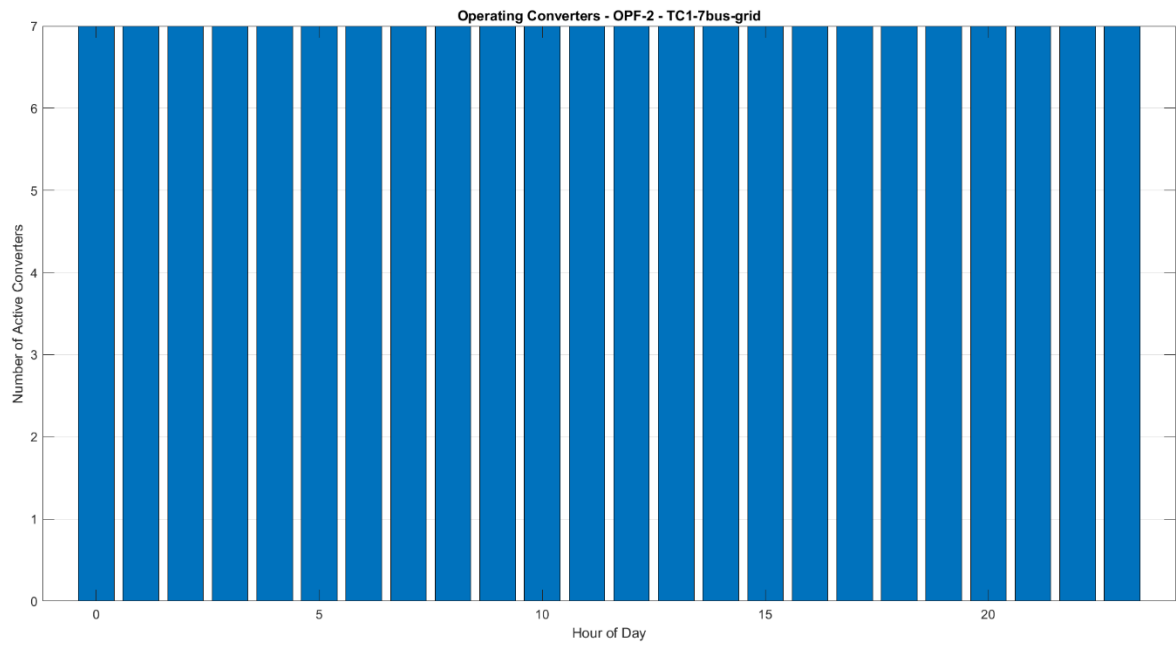
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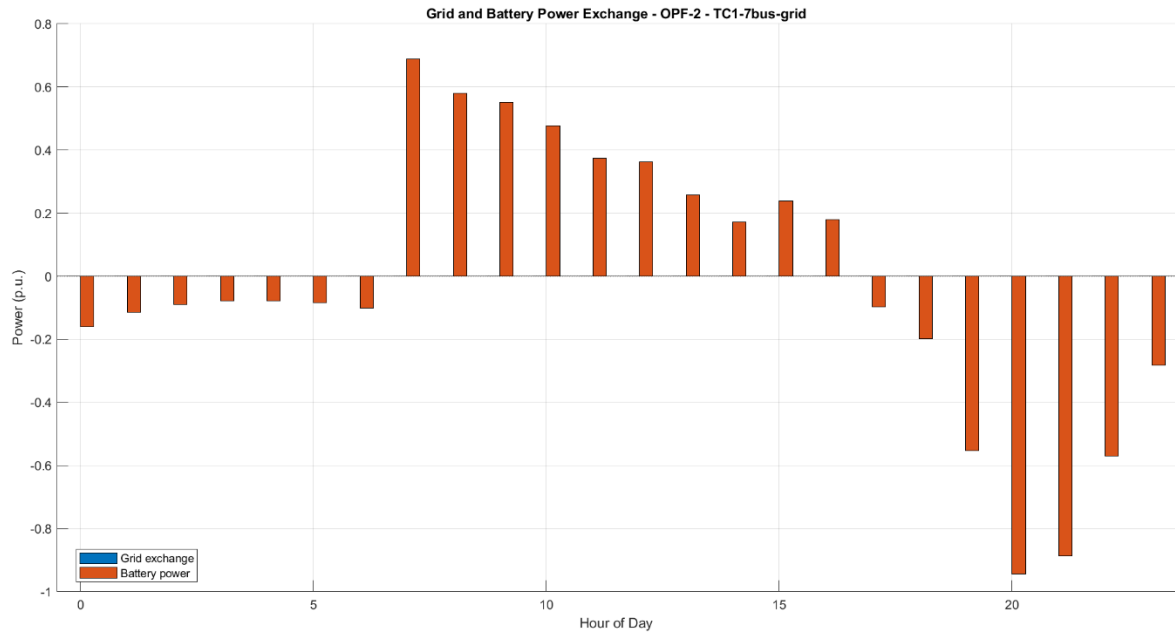
===== TC1-7bus-grid - OPF-2 (Hour 20 snapshot) =====

```

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	-0.239	0	0	1.035	"97.659"
2	0.141	-0.188	85.944	1.046	"97.567"
3	0.041	-0.102	86.284	1.048	"96.150"
4	0.013	-0.05	87.869	1.049	"91.375"
5	0.032	-0.09	87.582	1.048	"95.487"
6	0.01	-0.051	87.66	1.049	"89.397"
7	0.005	-0.071	87.596	1.049	"83.159"







Grid-connected Mode with 50% PV Generation

Table 9: Hour 24 Operational Results for TC1 (Grid-connected Mode - 50% PV Generation)

===== TC1 OPF2 =====

Grid connected mode: 1

Successful hours: 24/24

Total dist loss (p.u.): 0.018168

Total conv loss (p.u.): 0.244427

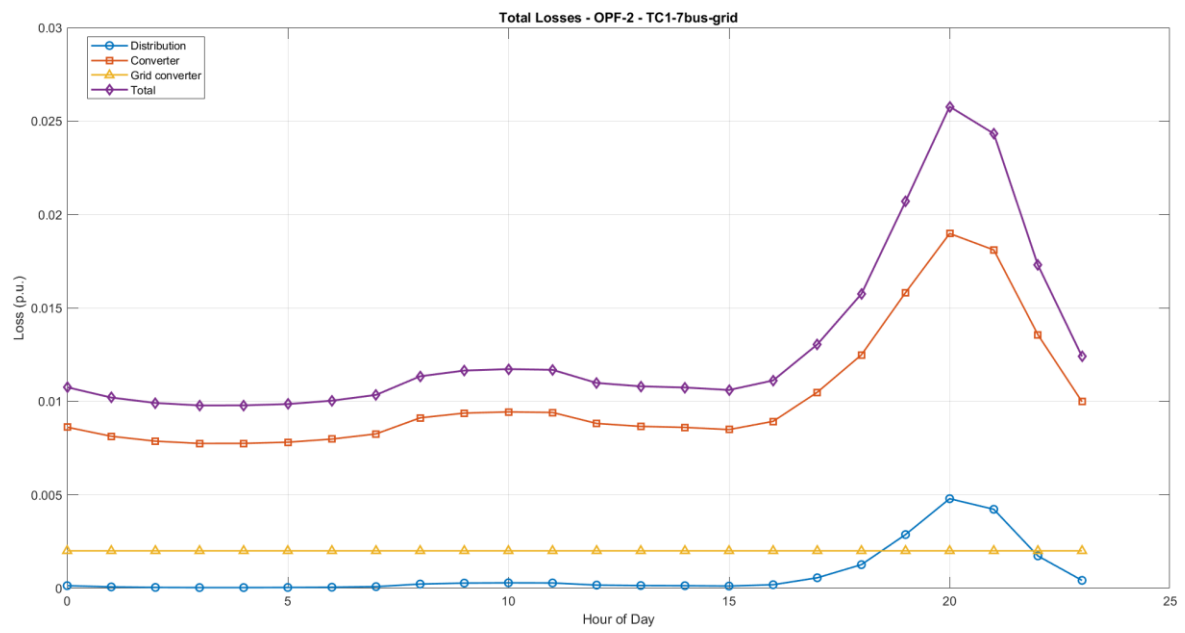
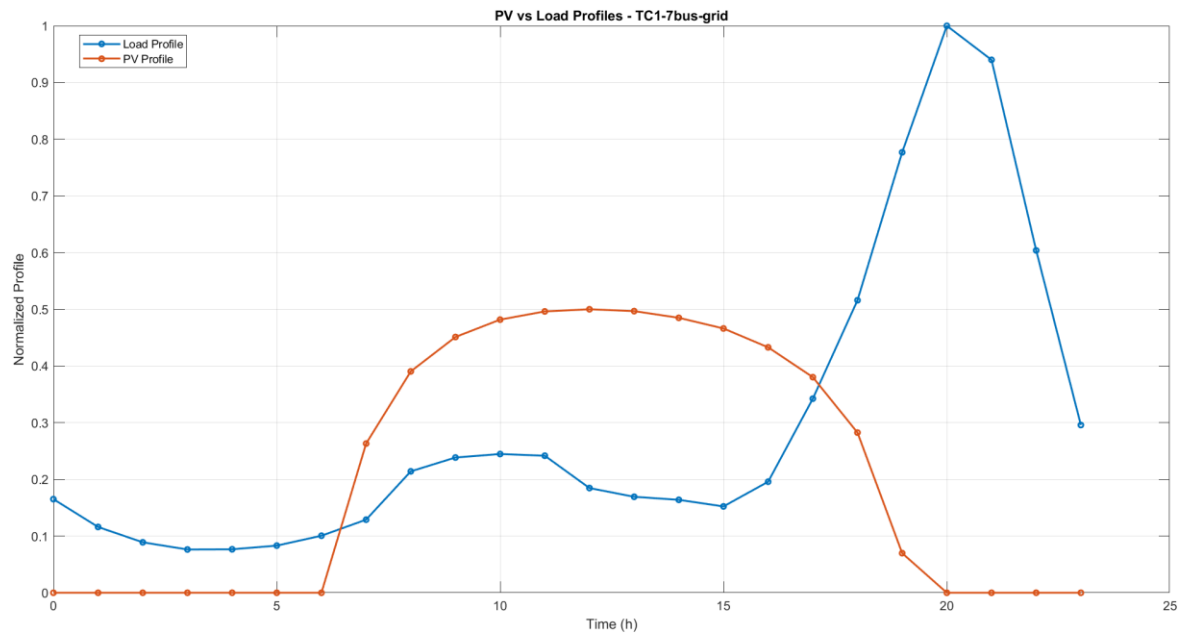
Total grid conv loss (p.u.): 0.048012

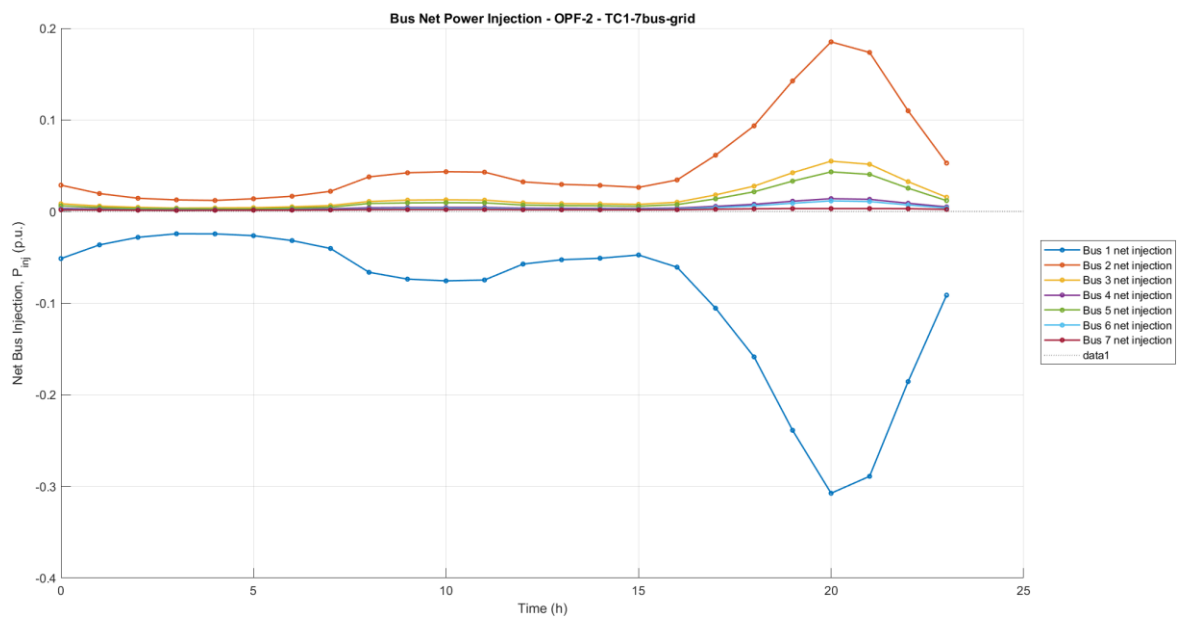
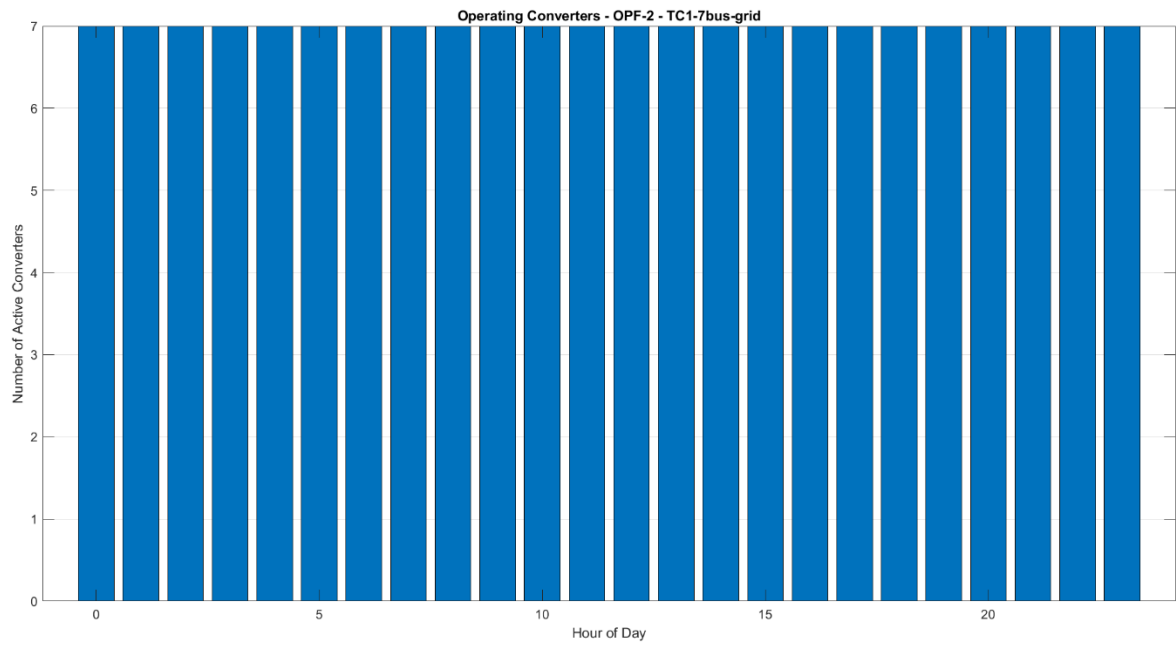
Total system loss (p.u.): 0.310607

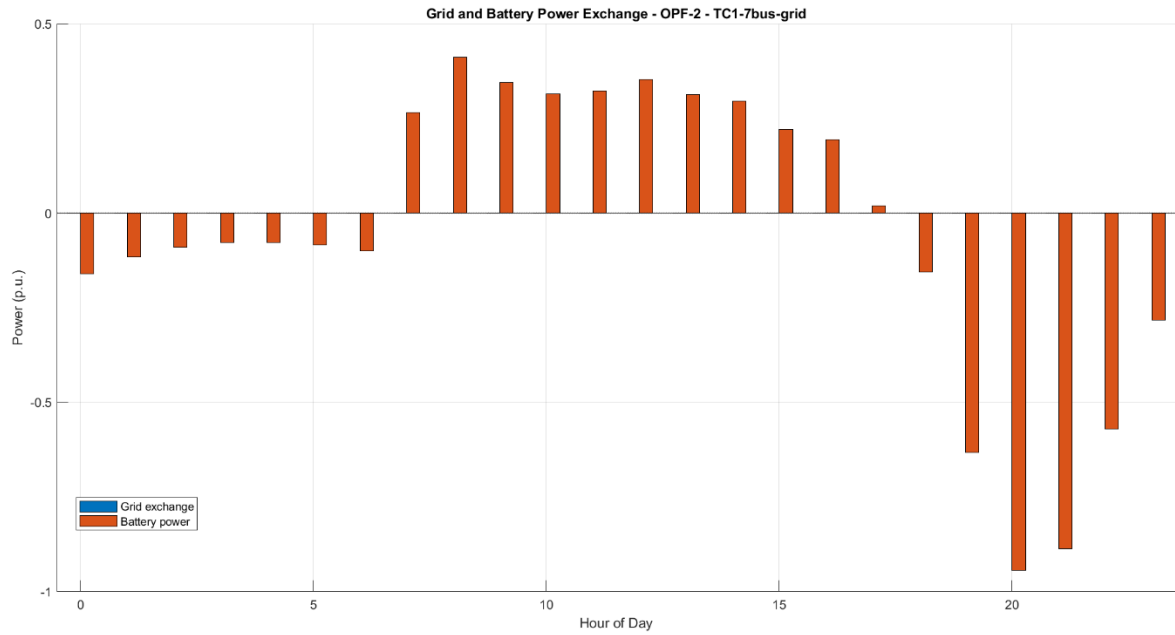
Optimization runtime: 25.9871 seconds

===== TC1-7bus-grid - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	-0.239	0	0	1.036	"97.660"
2	0.143	-0.207	82.737	1.047	"97.573"
3	0.042	-0.119	84.769	1.049	"96.203"
4	0.011	-0.059	82.399	1.05	"90.577"
5	0.033	-0.104	85.554	1.049	"95.620"
6	0.009	-0.062	84.405	1.05	"88.824"
7	0.003	-0.081	85.937	1.05	"75.550"







Grid-connected Mode with No PV Generation

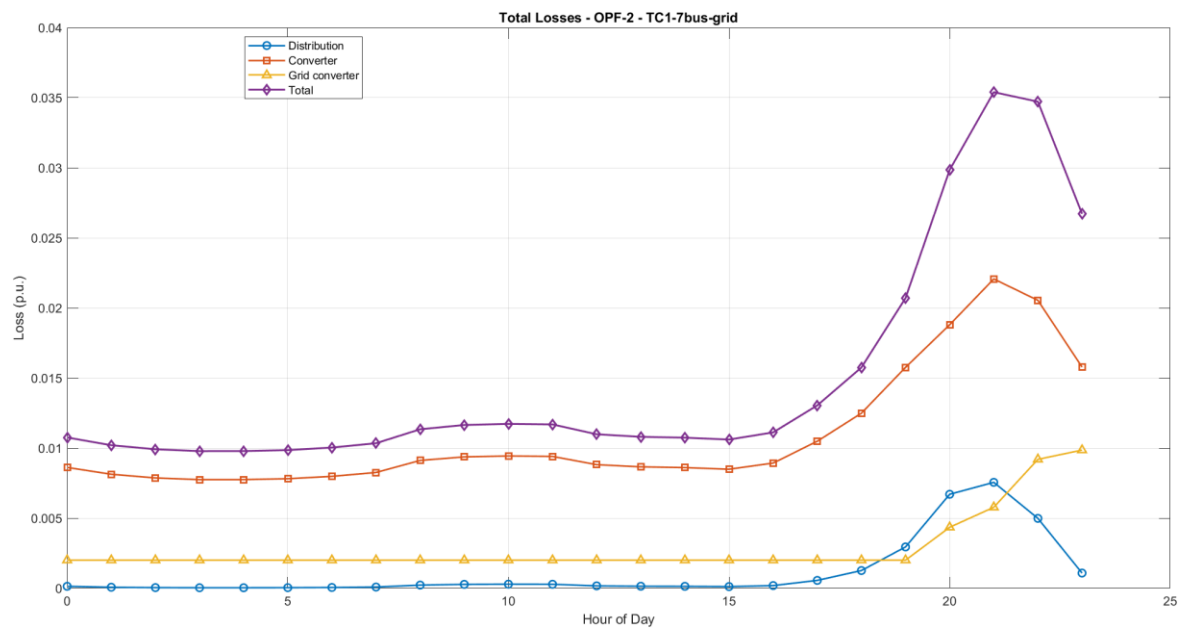
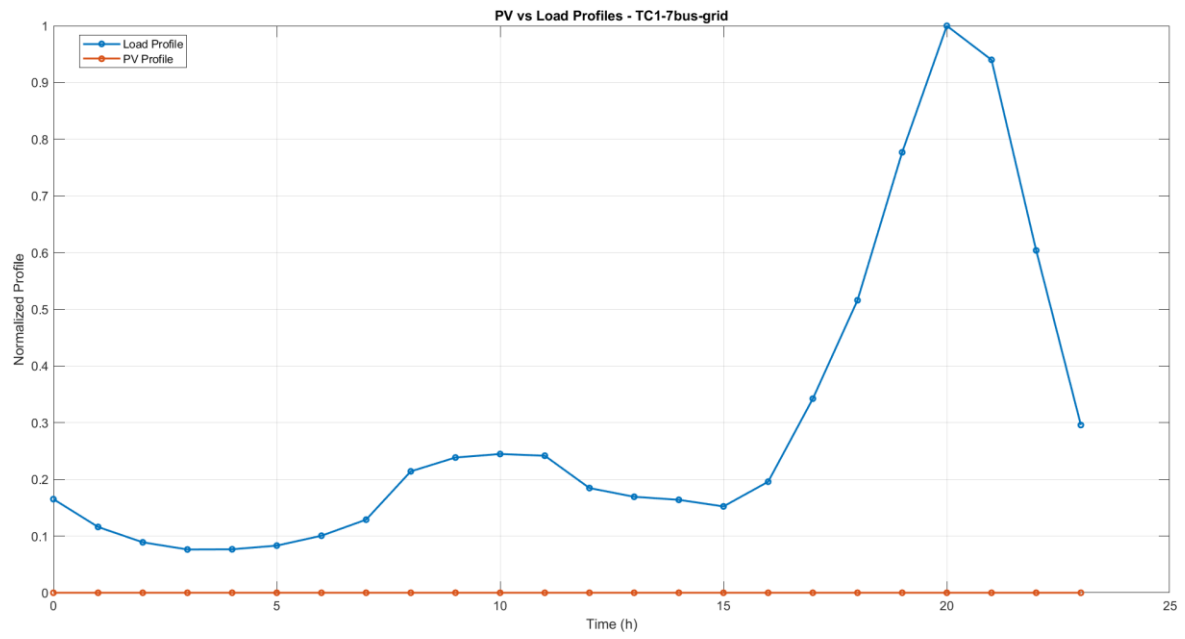
Table 10: Hour 24 Operational Results for TC1 (Grid-connected Mode - 0% PV Generation)

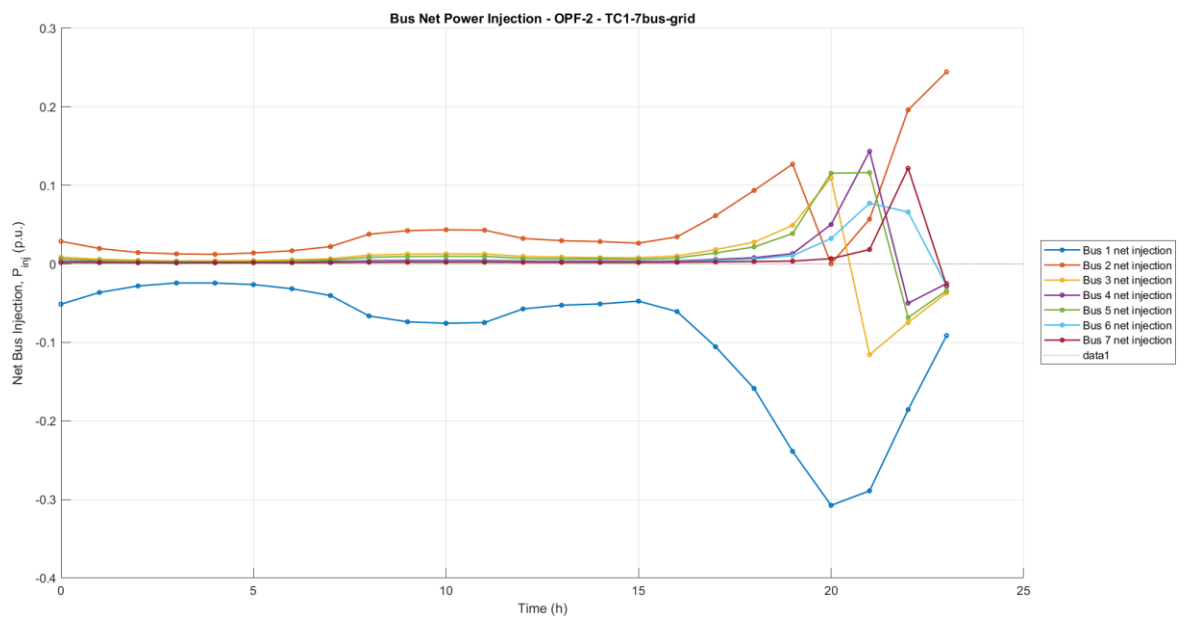
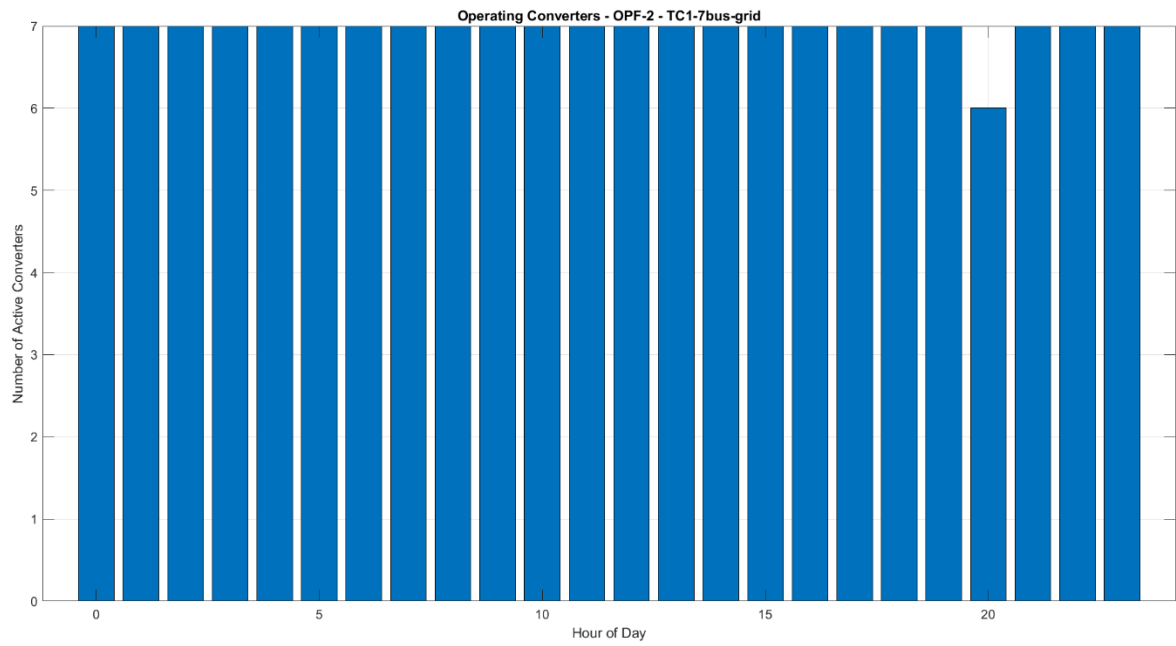
```

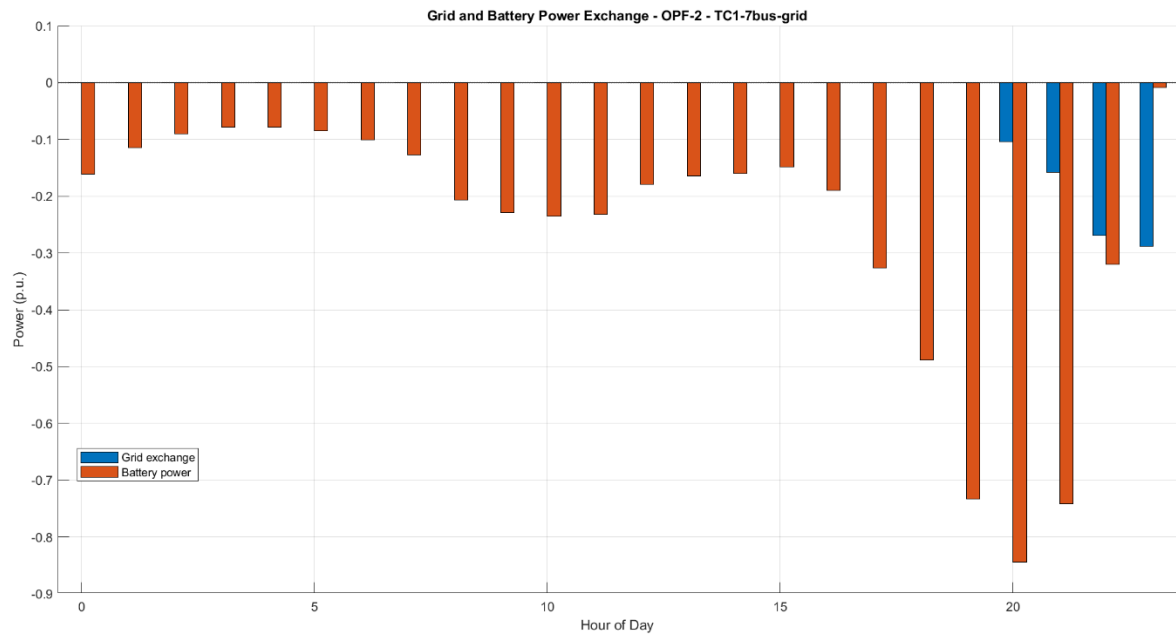
===== TC1 OPF2 =====
Grid connected mode: 1
Successful hours: 24/24
Total dist loss (p.u.): 0.027402
Total conv loss (p.u.): 0.260920
Total grid conv loss (p.u.): 0.069183
Total system loss (p.u.): 0.357504
Optimization runtime: 22.2960 seconds

```

===== TC1-7bus-grid - OPF-2 (Hour 20 snapshot) =====					
Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	-0.239	0	0	1.035	"97.660"
2	0.127	-0.208	55.208	1.047	"97.521"
3	0.049	-0.144	59.415	1.049	"96.488"
4	0.013	-0.076	60.762	1.05	"91.552"
5	0.039	-0.126	64.422	1.049	"95.995"
6	0.011	-0.082	62.304	1.05	"90.098"
7	0.004	-0.098	64.107	1.05	"77.144"







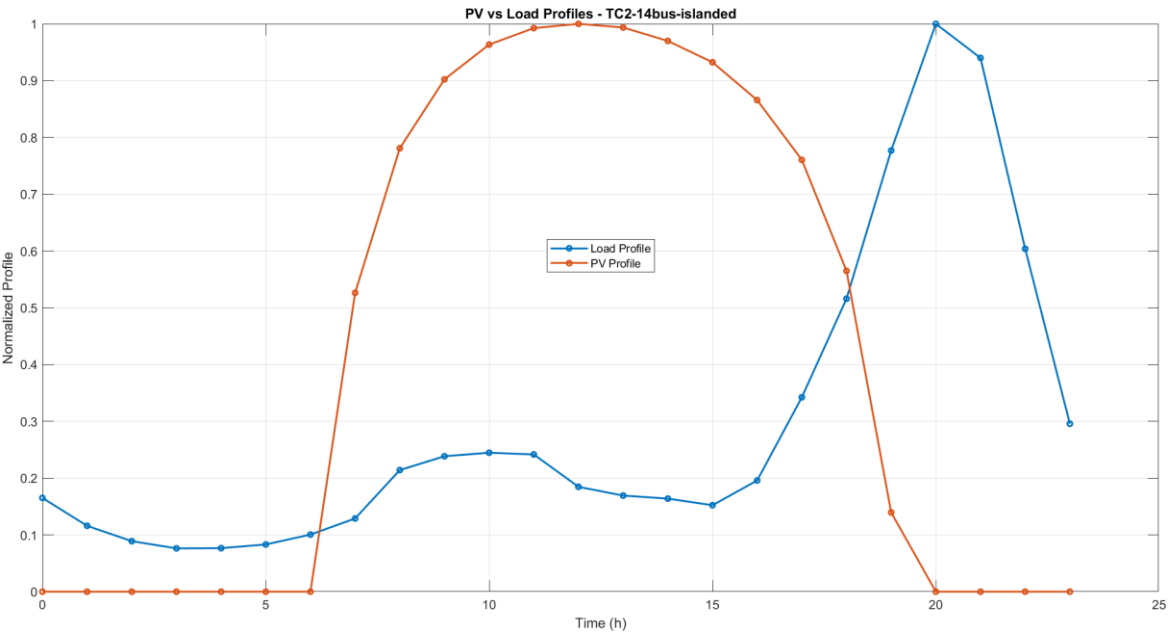
Appendix D: Individual Test Case Results (Test Case 2)

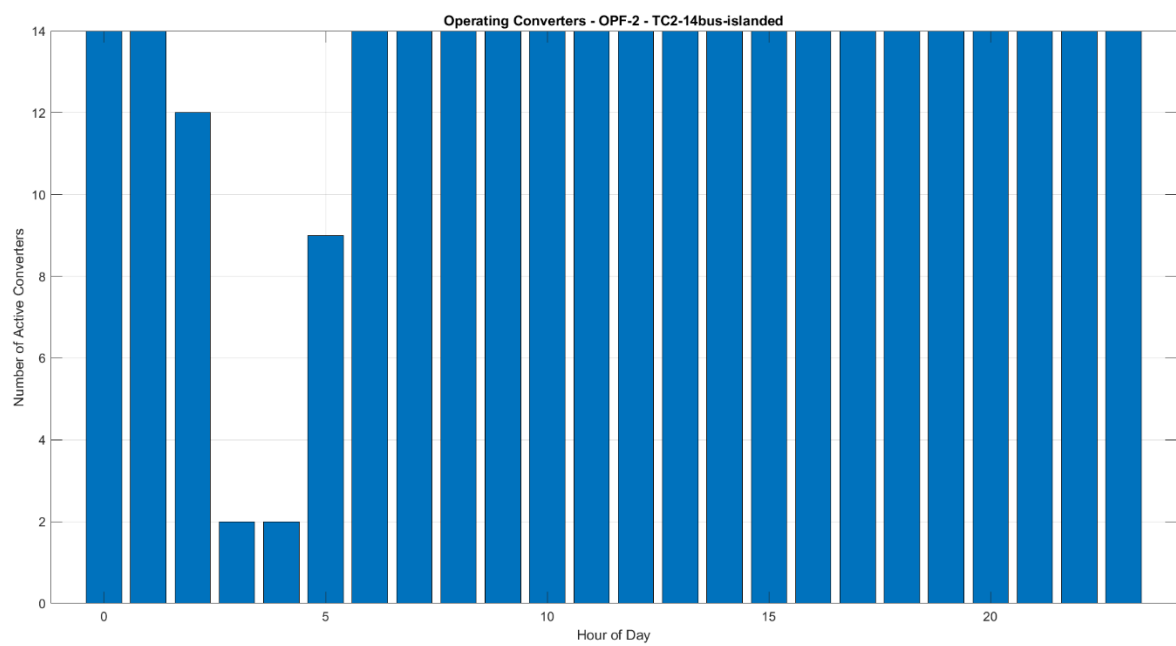
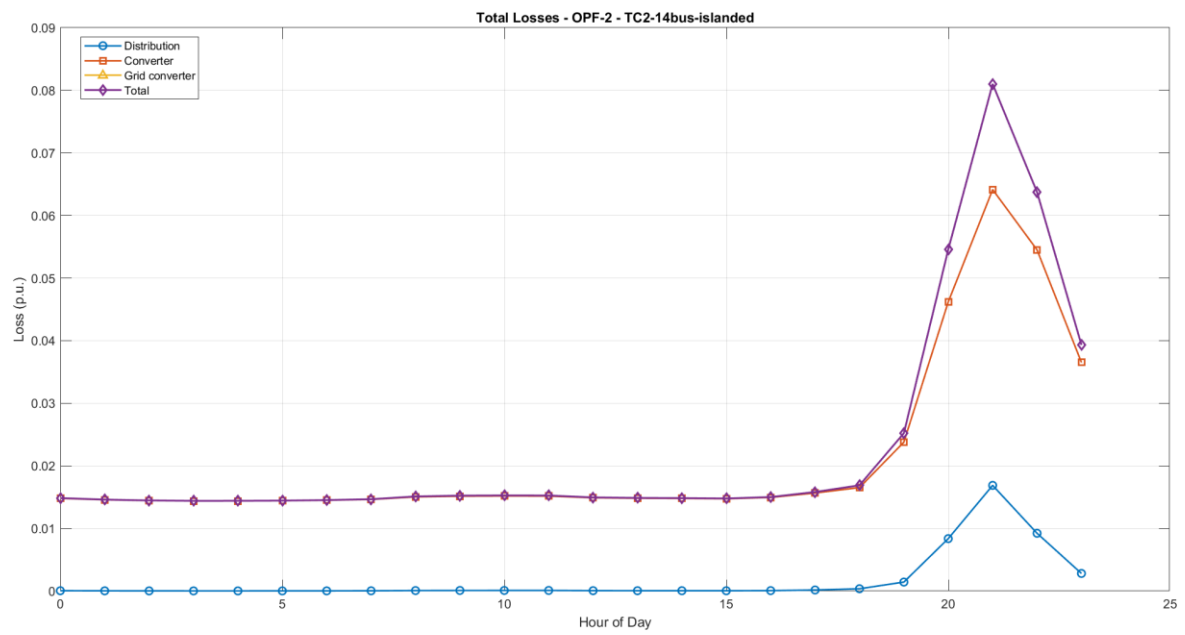
Islanded Mode with Full PV Generation

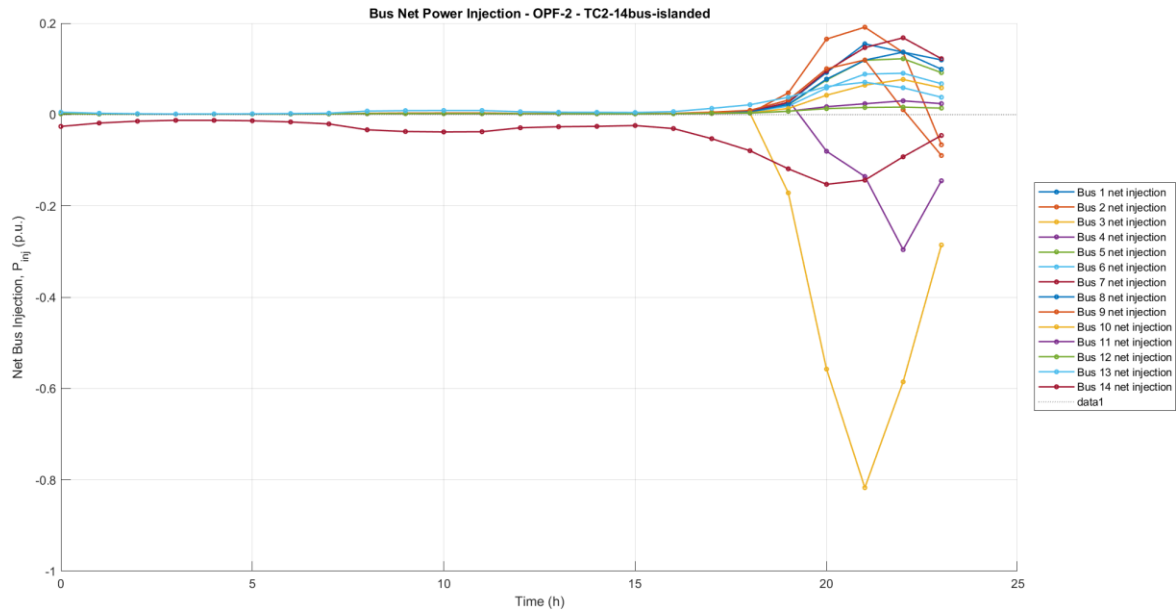
Table 11: Hour 24 Operational Results for TC2 (Islanded Mode)

===== TC2 OPF2 =====
Grid connected mode: 0
Successful hours: 24/24
Total dist loss (p.u.): 0.040081
Total conv loss (p.u.): 0.508212
Total system loss (p.u.): 0.548293
Optimization runtime: 72.1294 seconds
|
===== TC2-14bus-islanded - OPF-2 (Hour 20 snapshot) =====

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.026	-0.024	83.72	1.049	"94.914"
2	0.047	-0.199	81.387	1.048	"96.425"
3	-0.172	-0.4	72.371	1.045	"97.631"
4	0.032	-0.349	76.646	1.049	"95.505"
5	0.022	-0.075	77.88	1.049	"94.169"
6	0.018	-0.097	83.767	1.049	"93.473"
7	0.027	-0.025	83.921	1.049	"94.996"
8	0.022	-0.02	87.187	1.049	"94.300"
9	0.032	-0.234	80.554	1.049	"95.495"
10	0.013	-0.077	77.351	1.049	"91.704"
11	0.007	-0.032	81.897	1.049	"86.505"
12	0.007	-0.05	82.695	1.049	"86.801"
13	0.038	-0.133	78.359	1.049	"95.951"
14	-0.119	0	0	1.042	"97.485"







Grid-connected Mode with Full PV Generation

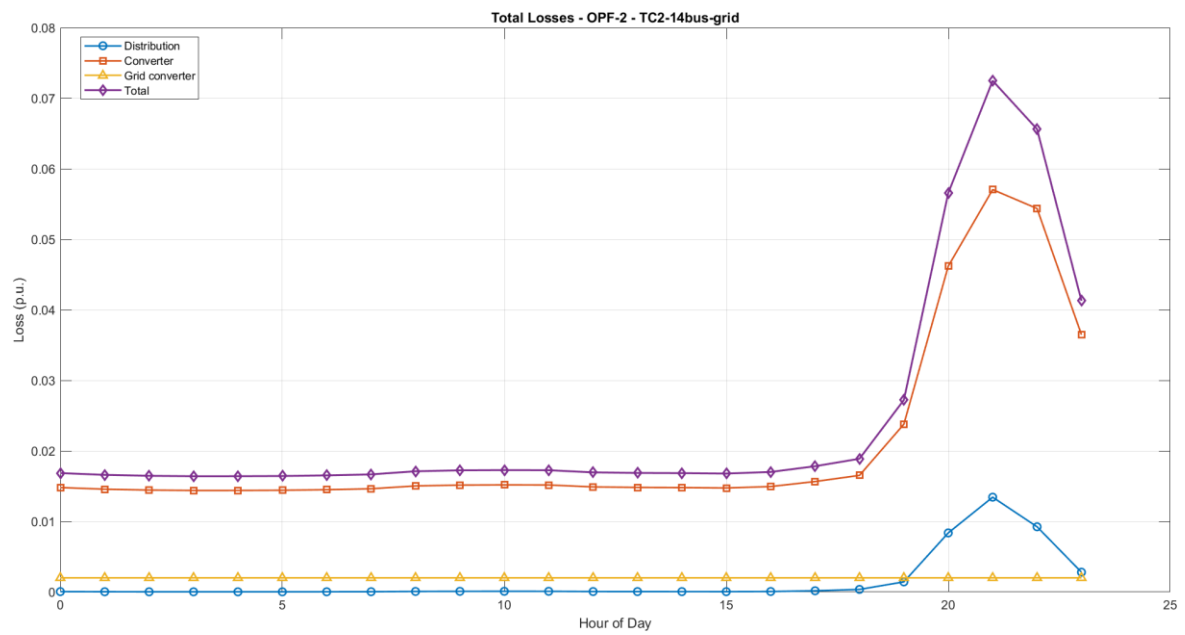
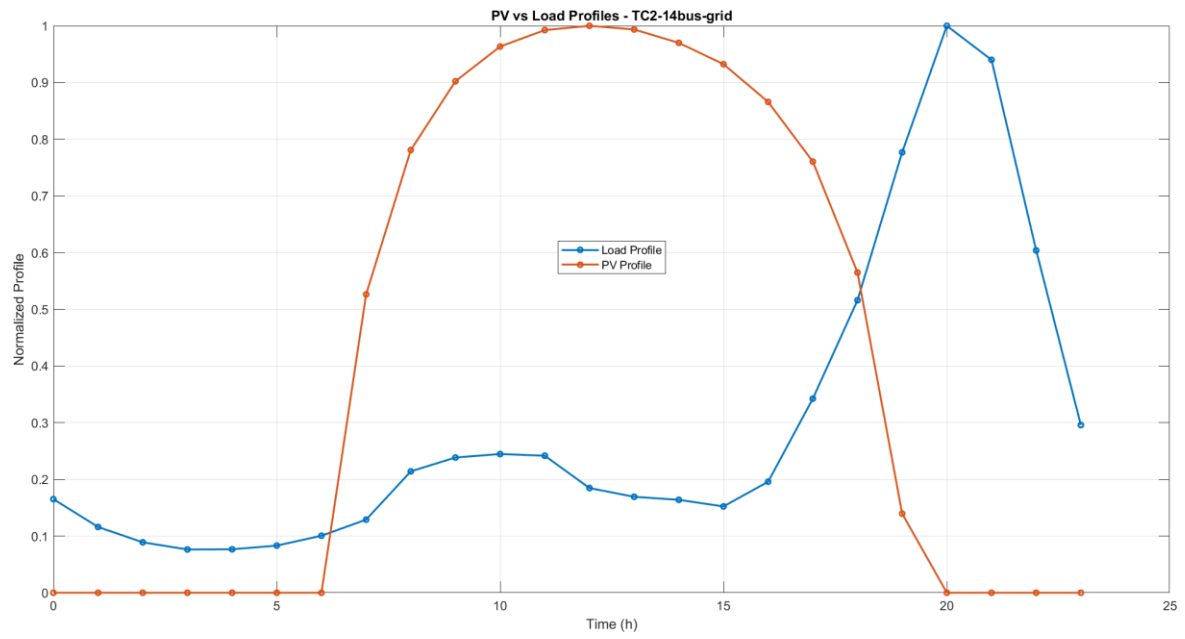
Table 12: Hour 24 Operational Results for TC2 (Grid-connected Mode - 100% PV Generation)

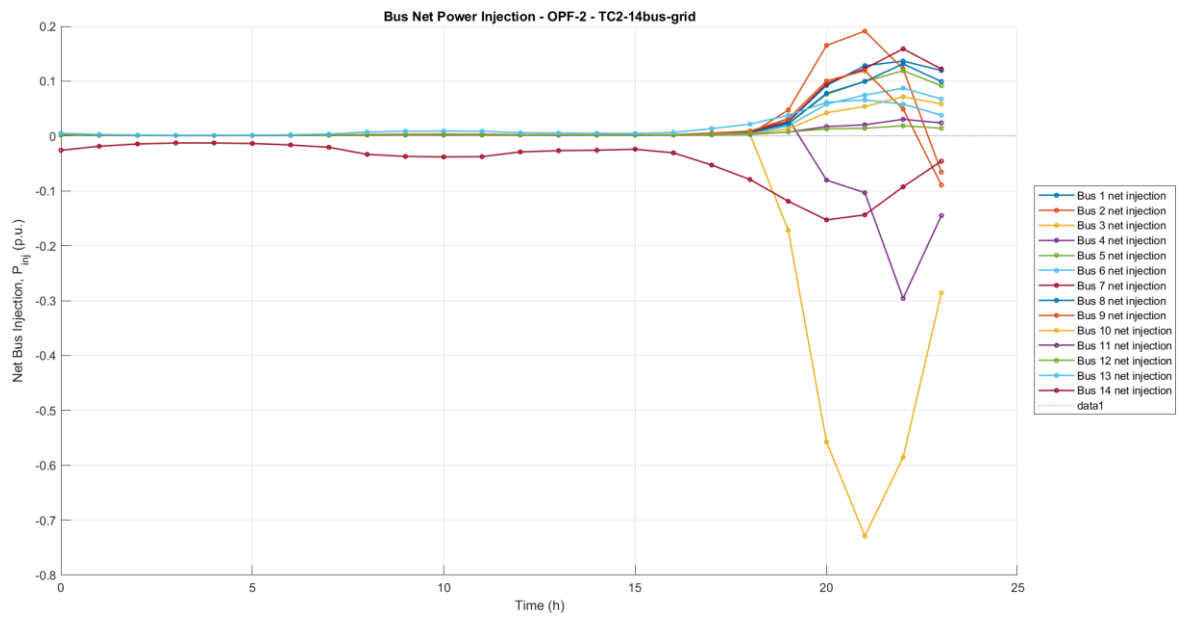
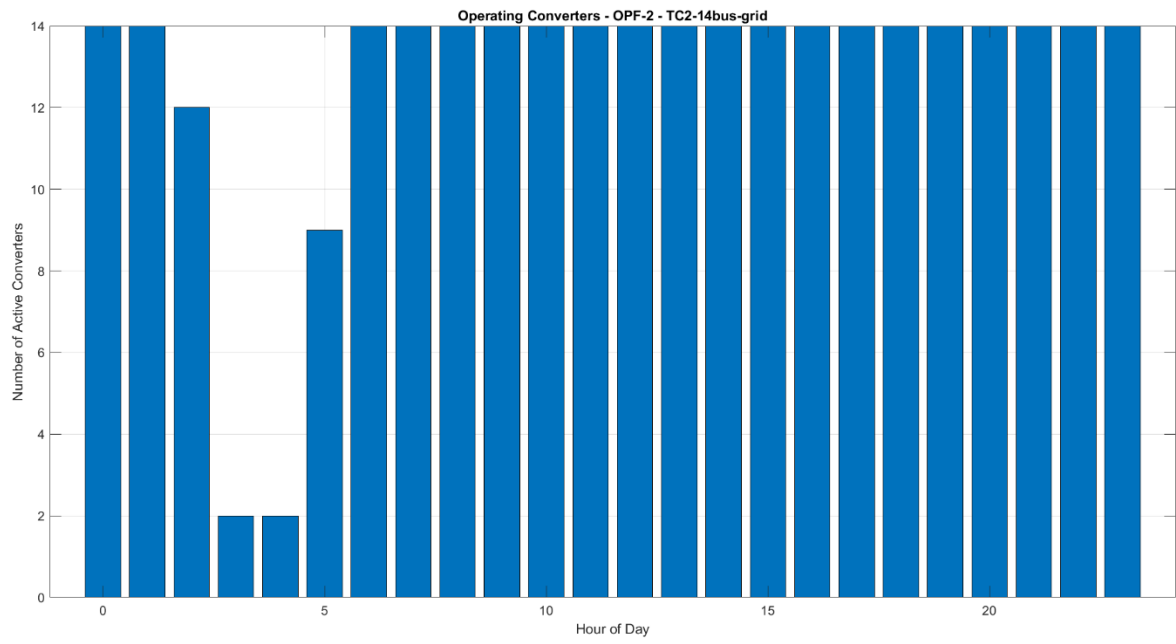
```

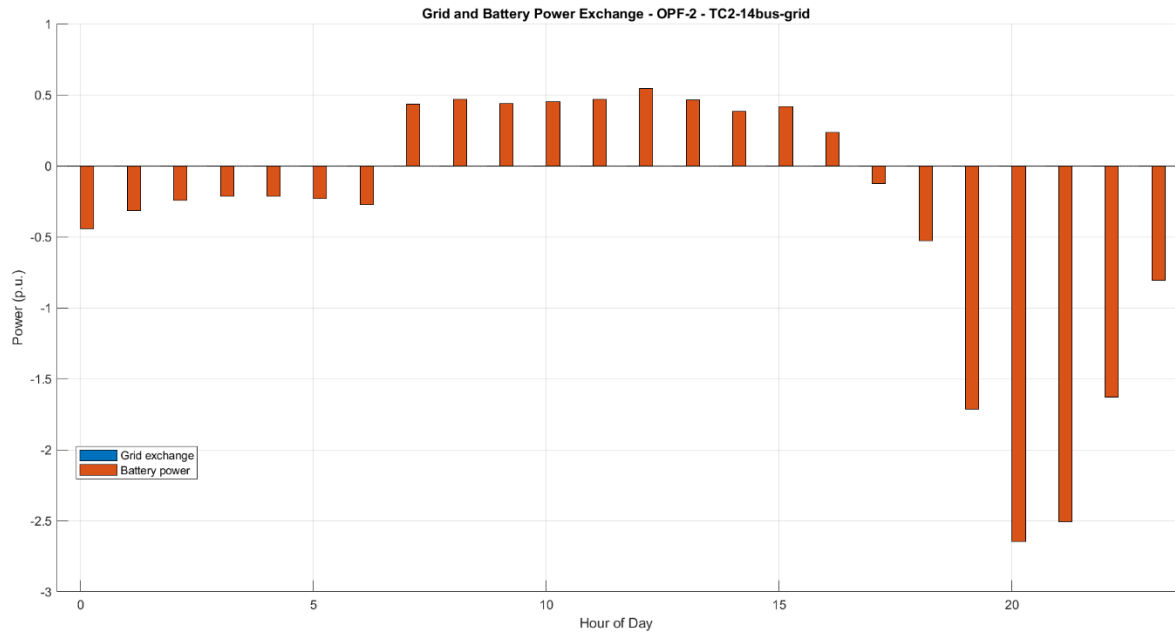
===== TC2 OPF2 =====
Grid connected mode: 1
Successful hours: 24/24
Total dist loss (p.u.): 0.036651
Total conv loss (p.u.): 0.500959
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.585624
Optimization runtime: 80.7177 seconds

```

===== TC2-14bus-grid - OPF-2 (Hour 20 snapshot) =====					
Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.026	-0.024	82.108	1.049	"94.902"
2	0.047	-0.202	81.124	1.048	"96.429"
3	-0.172	-0.4	74.477	1.045	"97.631"
4	0.031	-0.345	77.356	1.049	"95.469"
5	0.022	-0.076	75.982	1.049	"94.253"
6	0.018	-0.097	72.691	1.049	"93.499"
7	0.027	-0.025	84.056	1.049	"94.965"
8	0.023	-0.021	79.359	1.049	"94.393"
9	0.032	-0.233	81.502	1.049	"95.485"
10	0.013	-0.076	85.65	1.049	"91.398"
11	0.007	-0.032	75.677	1.049	"86.761"
12	0.007	-0.05	78.892	1.049	"86.885"
13	0.038	-0.134	85.036	1.049	"95.954"
14	-0.119	0	0	1.042	"97.485"







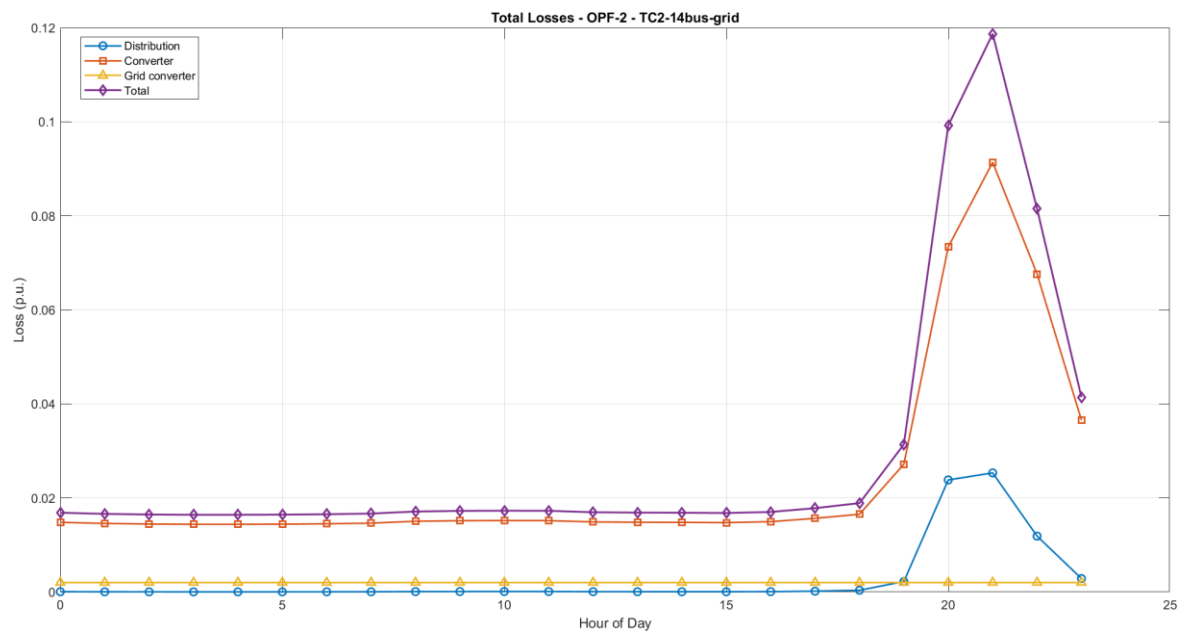
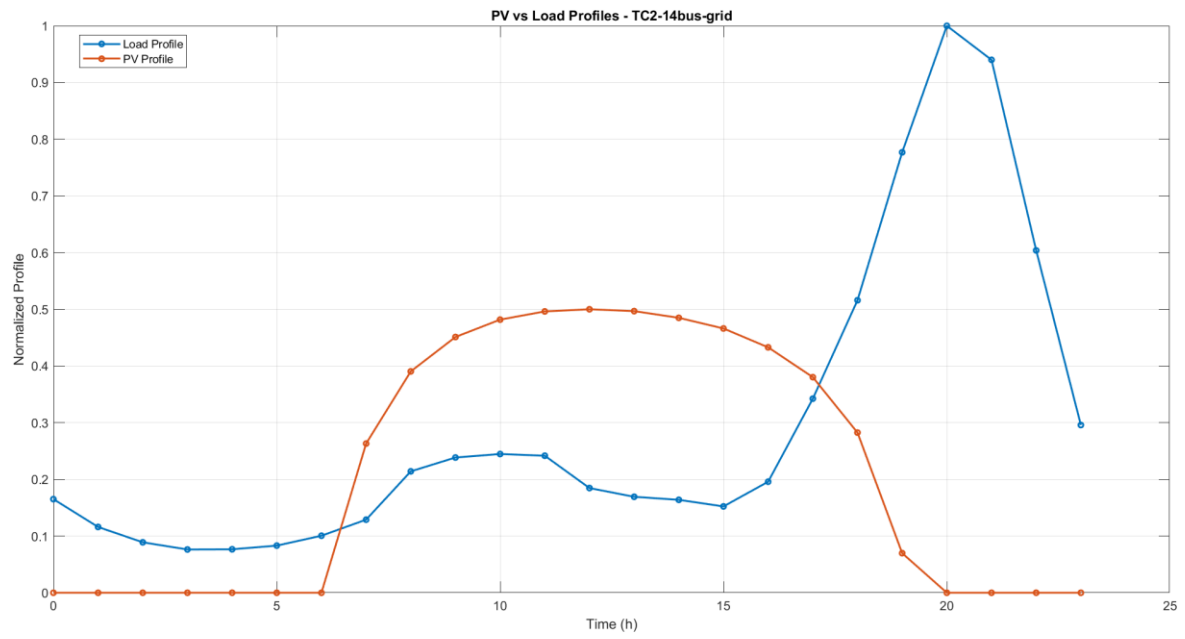
Grid-connected Mode with 50% PV Generation

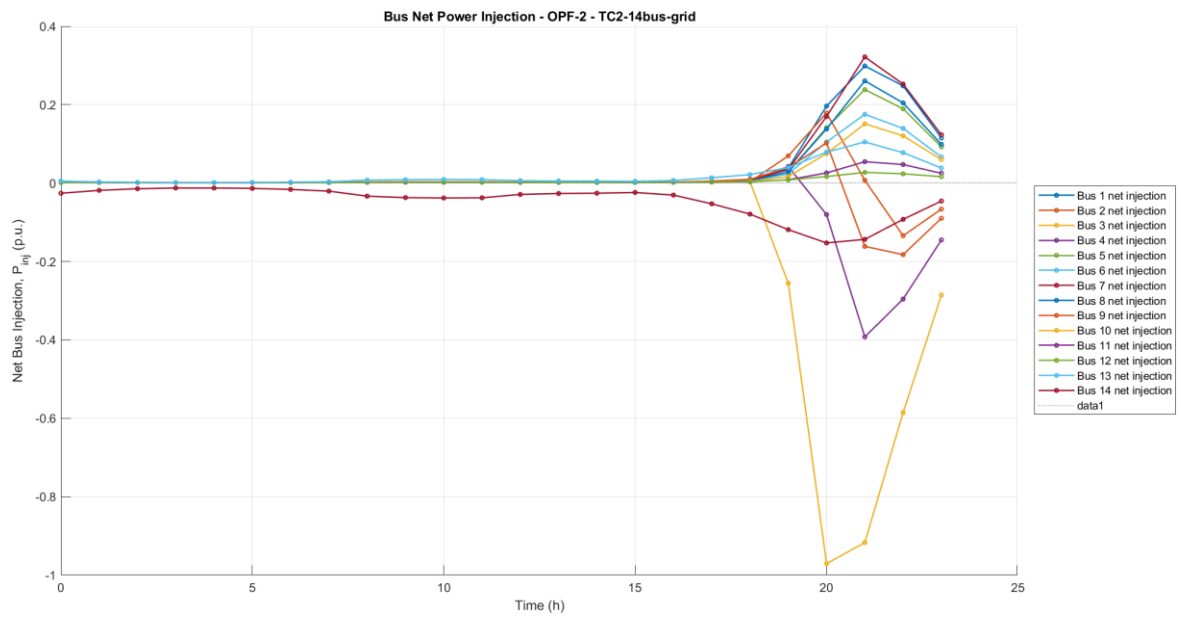
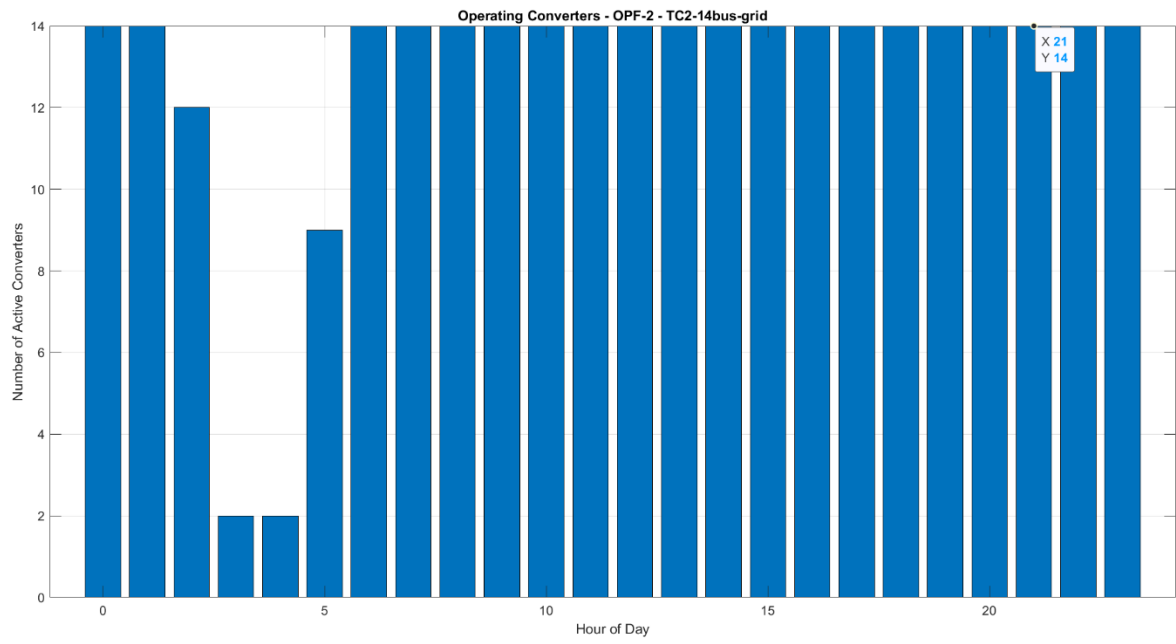
Table 13: Hour 24 Operational Results for TC2 (Grid-connected Mode - 50% PV Generation)

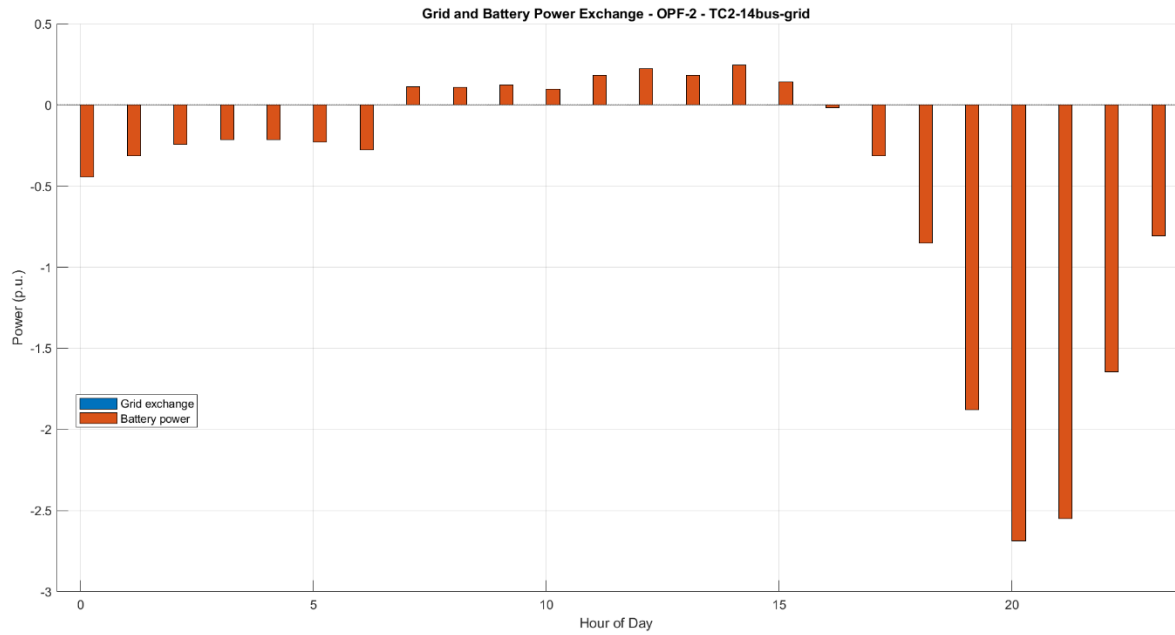
```
===== TC2 OPF2 =====
Grid connected mode: 1
Successful hours: 24/24
Total dist loss (p.u.): 0.067461
Total conv loss (p.u.): 0.579183
Total grid conv loss (p.u.): 0.048014
Total system loss (p.u.): 0.694658
Optimization runtime: 79.3152 seconds
```

```
===== TC2-14bus-grid - OPF-2 (Hour 20 snapshot) =====
```

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.038	-0.038	73.169	1.049	"95.973"
2	0.069	-0.23	71.06	1.048	"97.010"
3	-0.256	-0.4	60.139	1.043	"97.654"
4	0.042	-0.384	71.268	1.049	"96.199"
5	0.029	-0.086	80.115	1.049	"95.180"
6	0.024	-0.107	76.268	1.049	"94.515"
7	0.034	-0.034	84.638	1.049	"95.721"
8	0.028	-0.028	83.665	1.049	"95.164"
9	0.04	-0.257	69.407	1.049	"96.054"
10	0.016	-0.084	81.083	1.049	"92.920"
11	0.008	-0.035	76.962	1.05	"88.228"
12	0.008	-0.054	80.115	1.05	"87.797"
13	0.04	-0.14	78.315	1.049	"96.055"
14	-0.119	0	0	1.042	"97.485"







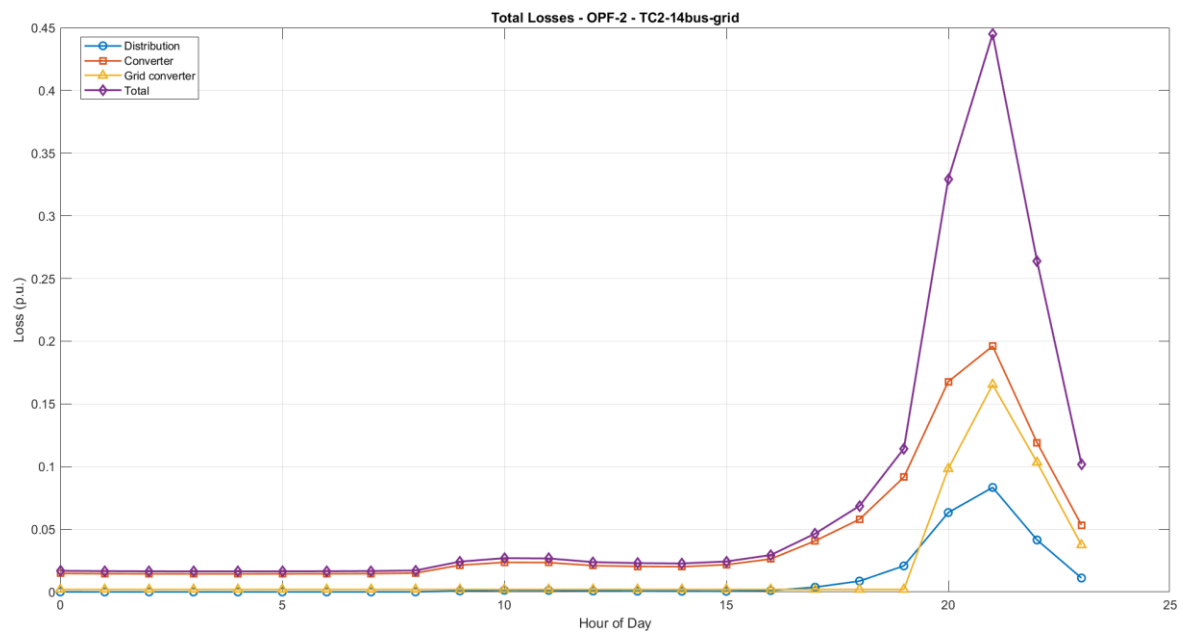
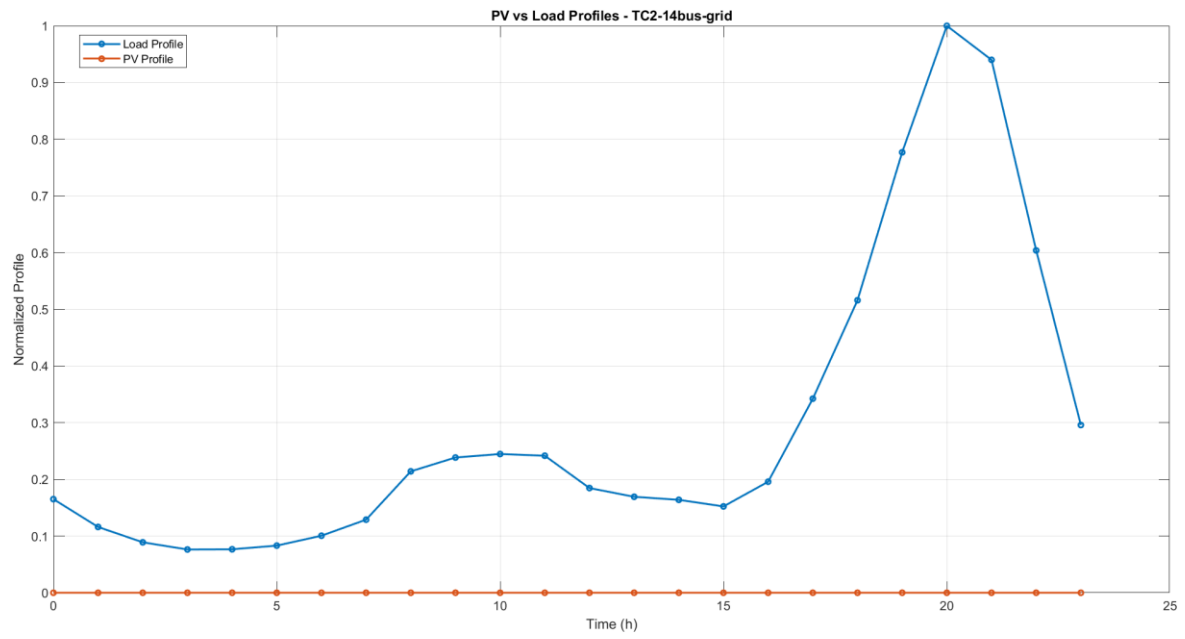
Grid-connected Mode with No PV Generation

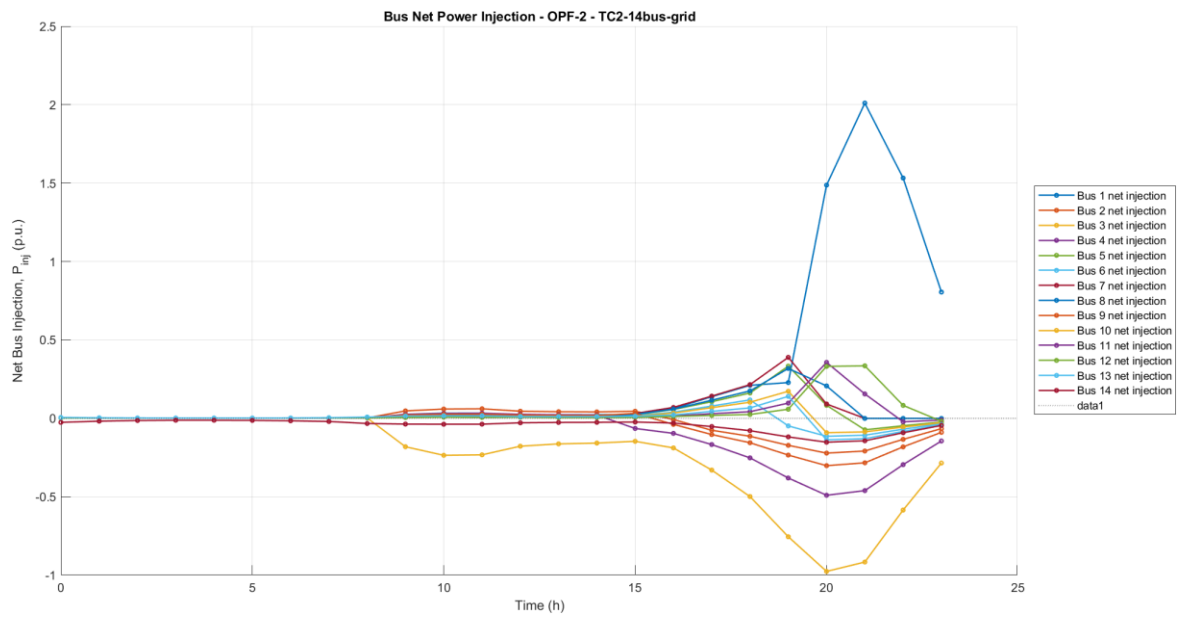
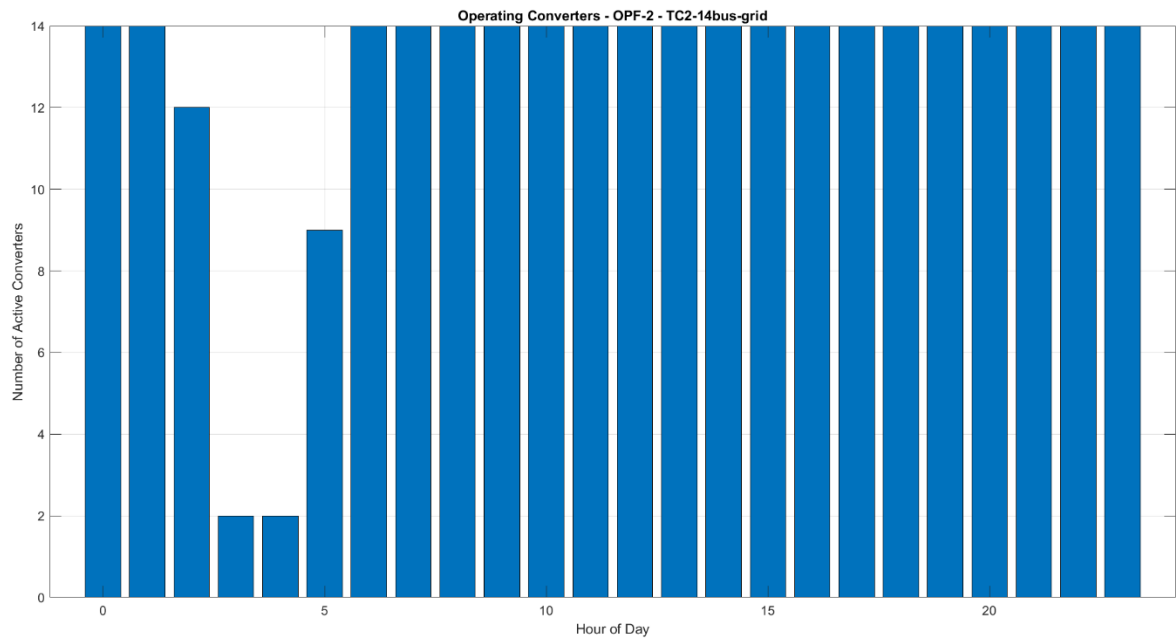
Table 14: Hour 24 Operational Results for TC2 (Grid-connected Mode - 0% PV Generation)

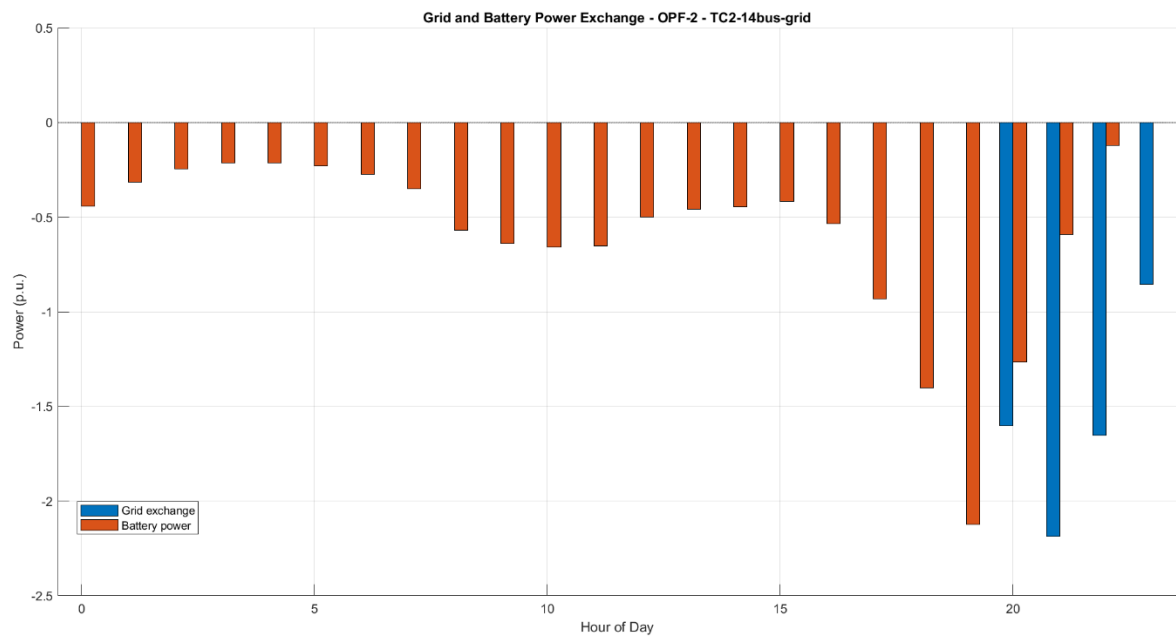
```
===== TC2 OPF2 =====
Grid connected mode: 1
Successful hours: 24/24
Total dist loss (p.u.): 0.240013
Total conv loss (p.u.): 1.035565
Total grid conv loss (p.u.): 0.444473
Total system loss (p.u.): 1.720051
Optimization runtime: 75.3695 seconds
```

```
===== TC2-14bus-grid - OPF-2 (Hour 20 snapshot) =====
```

Bus No.	Pi (p.u.)	PBi (p.u.)	SOCi (%)	NodalVoltage (p.u.)	Converter Efficiency (%)
1	0.228	-0.233	55.827	1.042	"97.661"
2	-0.173	0	50.001	1.037	"97.633"
3	-0.756	0	50.001	1.023	"96.952"
4	-0.381	0	50.001	1.042	"97.538"
5	0.333	-0.4	64.021	1.045	"97.593"
6	-0.049	-0.04	51.001	1.045	"96.480"
7	0.388	-0.398	62.292	1.044	"97.528"
8	0.318	-0.325	63.416	1.046	"97.608"
9	-0.235	0	50.001	1.043	"97.660"
10	0.172	-0.246	56.165	1.048	"97.632"
11	0.097	-0.126	67.957	1.05	"97.344"
12	0.058	-0.107	75.708	1.05	"96.758"
13	0.14	-0.248	56.22	1.049	"97.565"
14	-0.119	0	0	1.039	"97.485"







Appendix E: Use of Generative AI

Generative AI was used throughout this project as a support tool to aid understanding, improve clarity of explanations and assist with structuring documentation. Specifically, AI was used to:

- clarify theoretical concepts related to DC microgrids and optimal power flow (OPF)
- assist in structuring the system architecture and workflow descriptions
- refine written explanations to improve technical clarity and readability
- support debugging and restructuring of MATLAB code

All AI-generated outputs were critically evaluated before use. The responses were not used verbatim; instead, they were modified, verified, and adapted to align with the specific implementation and mathematical formulation of the project. Engineering judgement was applied to ensure correctness and consistency with the system model.

Generative AI was not used to replace independent analysis or decision-making, but rather as a supplementary tool to enhance the development and presentation of the work.

Appendix F: GitHub for Project

Link to GitHub: <https://github.com/speedygonzales4/Optimal-Power-Sharing-in-DC-Grid-Connected-Microgrids-for-Rural-Electrification.git>